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**FINAL REPORT**

# Towards Proactive Emergency Response:

## Anticip-42, a Human-in-the-Loop AI Intervention-Support System for Firefighters


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# DECLARATION

I certify that I am the author of this project and that any assistance I received in its preparation is fully acknowledged and disclosed in this project. I have also cited any source from which I used data, ideas, or words, either quoted or paraphrased. Further, this report meets all the rules of quotation and referencing in use at MedTech, as well as adheres to the fraud policies listed in the MedTech honor code.

No portion of the work referred to in this study has been submitted in support of an application for another degree or qualification to this or any other university or institution of learning.

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# ABSTRACT

**Introduction and Background:** Emergency-response officers must analyze changing information and anticipate possible developments while retaining responsibility for operational decisions. This study investigates how human-in-the-loop AI can support the anticipation process at SDIS 42 without replacing professional judgment.

**Methodology:** A Design Science Research approach was followed through four iterations. Operational requirements were translated into BPMN and UML models, followed by a Streamlit feasibility prototype and an integrated React–FastAPI proof of concept. The final system combined local language models, stage-specific Retrieval-Augmented Generation, persistent storage, and external contextual services. It was evaluated using warehouse-fire and methane-release scenarios. Six local models were compared in the main test using structural measures, task executability, runtime, and an LLM-as-judge assessment.

**Results:** Anticip-42 implemented a progressive workflow covering possible situations, COS objectives, maneuver ideas, and detailed tasks. Human review and validation were maintained between every generation stage. Under the tested conditions, Qwen3 30B provided the strongest balance among workflow completeness, semantic quality, task executability, reliability, and runtime. The evaluation also showed that structurally complete outputs could still contain unrealistic or incomplete operational content.

**Conclusion:** The study demonstrates the technical feasibility of combining local AI, document retrieval, contextual information, persistent operation state, and human validation in a modular decision-support workflow. Anticip-42 remains a proof of concept and requires evaluation on the final SDIS infrastructure, with the complete authorized corpus and qualified operational users.

**Keywords:** Human-in-the-Loop AI, Emergency Response, Firefighter Anticipation, Decision Support, Large Language Models, Retrieval-Augmented Generation

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# LIST OF ABBREVIATIONS

- **AI:** Artificial Intelligence
- **API:** Application Programming Interface
- **ARIA:** Analyse, Recherche et Information sur les Accidents
- **BARPI:** Bureau d'analyse des risques et pollutions industriels
- **BPMN:** Business Process Model and Notation
- **CASDIS:** Conseil d'administration du Service départemental d'incendie et de secours
- **CODIS:** Centre opérationnel départemental d'incendie et de secours
- **COS:** Commandant des opérations de secours
- **CPU:** Central Processing Unit
- **CRM:** Centre de regroupement des moyens
- **CTA:** Centre de traitement de l'alerte
- **DGSCGC:** Direction générale de la sécurité civile et de la gestion des crises
- **DOS:** Directeur des opérations de secours
- **DSR:** Design Science Research
- **DSRM:** Design Science Research Methodology
- **DTA:** Différentes tâches à accomplir
- **ENS:** Événement non souhaité
- **ENSOSP:** École nationale supérieure des officiers de sapeurs-pompiers
- **EPS:** Évolution prévisible du sinistre
- **GDO:** Guide de doctrine opérationnelle
- **GPU:** Graphics Processing Unit
- **GTO:** Guide de techniques opérationnelles
- **HADRIA:** Hub d'application, de développement et de recherche en intelligence artificielle de la sécurité civile
- **HITL:** Human-in-the-Loop

- **IA:** Intelligence artificielle
- **IAP:** Incident Action Plan
- **IM:** Idée de manoeuvre
- **INERIS:** Institut national de l'environnement industriel et des risques
- **INRS:** Institut national de recherche et de sécurité
- **LLM:** Large Language Model
- **MD:** Moyens disponibles
- **OSM:** OpenStreetMap
- **PC:** Poste de commandement
- **PCASDIS:** Président du conseil d'administration du Service départemental d'incendie et de secours
- **PEX:** Partage d'expérience
- **POC:** Proof of Concept
- **RAG:** Retrieval-Augmented Generation
- **RAM:** Random Access Memory
- **RETEX:** Retour d'expérience
- **RGPD:** Règlement général sur la protection des données
- **RQ:** Research Question
- **SDIS:** Service départemental d'incendie et de secours
- **SDIS 42:** Service départemental d'incendie et de secours de la Loire
- **SE:** Situation envisageable
- **SITAC:** Situation tactique
- **UML:** Unified Modeling Language
- **VM:** Virtual Machine

# 1 INTRODUCTION

"Plans are worthless, but planning is everything." This observation by Dwight D. Eisenhower captures the purpose of operational anticipation: preparing for an uncertain future without assuming that events will follow a fixed plan [1]. During an emergency, the situation continues to evolve while officers receive information, coordinate resources, and prepare for what may happen next. This study explores how artificial intelligence can support the anticipation officer at SDIS 42 ("Service départemental d'incendie et de secours de la Loire") by organizing operational information and helping prepare structured proposals without replacing human decision-making.

This chapter introduces the problem addressed by the study and the context in which the project was conducted. It presents the host and partner institutions, the purpose and objectives of the research, the research questions, the adopted approach, the study boundaries, its main contributions, and the overall structure of the report.

## 1.1 Institutional Setting and Project Collaboration

This study was conducted at the DISP laboratory at Université Lumière Lyon 2, in partnership with the *Service départemental d'incendie et de secours de la Loire* (SDIS42). The two institutions contributed complementary forms of expertise to the project.

DISP conducts research on decision and information systems, with particular attention to modeling, data management, artificial intelligence, and decision support in complex organizations [2], [3]. This research environment provided the methodological and technical basis for examining how Large Language Models and Retrieval-Augmented Generation could be integrated into an operational decision-support system.

The academic supervision focused on the research approach, software architecture, experimental choices, and evaluation of the proof of concept while SDIS42 provided the operational setting in which the research problem was defined. The project was conducted in collaboration with the *"Groupement Analyse et Anticipation des Risques"*, which contributes to the analysis of departmental risks and the preparation of emergency response [4]. The operational supervision was provided by Capitaine Yoan, who explained the anticipation process, supplied local operational documents and definitions, and reviewed the interpretation of fire-service terminology and working practices.

The collaboration was organized through regular meetings, document reviews, and observation of command-post simulations. These exchanges were used to understand how information circulates during an intervention, identify the difficulties experienced by operational officers, and progressively refine the proposed system. The simulations were particularly useful because they

made it possible to observe the anticipation workflow in closer conditions. This institutional collaboration directly shaped the technical orientation of the project.

## 1.2 Project Context

Every emergency intervention begins with an incomplete picture. A caller may describe smoke, an accident, an unusual smell, or a person in danger, but this first account rarely provides everything the emergency services need to know. In the Loire department, calls made through the emergency numbers 18 and 112 are received by the *"Centre de traitement de l'alerte"* (CTA). Its first task is to transform this initial information into an operational response: identify the event, locate it, evaluate its apparent urgency, and dispatch the appropriate resources [5], [6].

Once the first crews arrive, the incident is no longer simply an emergency call. It becomes an evolving operational situation. Initial information may be confirmed, corrected, or contradicted. New information may appear, additional victims may be discovered, access routes may become unusable, and the resources initially dispatched may no longer be sufficient. At departmental level, the *"Centre opérationnel départemental d'incendie et de secours"* (CODIS) follows this evolution, coordinates reinforcements, maintains an overview of operational activity, and informs the relevant authorities [4], [6].

For complex interventions, the command structure progressively expands. On the incident ground, the *"Commandant des opérations de secours"* (COS) commands the emergency resources under the authority of the *"Directeur des opérations de secours"* (DOS). Depending on the scale and complexity of the operation, a command post may be established to organize information, coordinate operational sectors, monitor resources, and prepare decisions [6].

The anticipation function is not activated for every emergency call. It appears mainly during major emergency operations requiring a *"poste de commandement de site"*. Such operations may include a major flood, a large forest fire, an industrial or chemical incident, a mass-casualty event, or a terrorist attack. The SDIS 42 operational regulation identifies the *"officier anticipation"* as part of the command structure established at site-command level [6].

These events represent only a small proportion of the interventions managed during a year. This creates a particular difficulty: anticipation is most important during situations that officers encounter least frequently in real operations. Firefighters therefore rely on regular training to maintain their operational knowledge. Based on observations made during the project, SDIS 42 regularly conducted exercises as part of its weekly training. Larger exercises were also organized at specific times of the year, sometimes with other emergency services, to simulate events such as wildfires, floods, industrial accidents, and incidents involving many casualties.[1]

[1] Based on project meetings with Capitaine Yoan and observations of SDIS 42 training exercises conducted in 2026.

When the anticipation function is activated, the officer must look beyond the current response. The objective is not to predict the future with certainty, but to identify potential escalations of the situation early enough for the command structure to prepare. For example, if a forest fire may reach an exposed settlement, if floodwater may isolate a neighborhood, or if a chemical release may change direction with the wind. The necessary personnel, vehicles, specialist teams, evacuation resources, and logistical support must be considered before the situation occurs.

This work gives the COS time to reserve resources rather than waiting until the situation has already deteriorated. Anticipation is consequently not an attempt to foresee one certain future. It is a way of avoiding being surprised by a plausible one.This reasoning depends on information from several sources: field reports, the *"situation tactique"* (SITAC), weather and geographical data, resource availability, technical expertise, operational doctrine, and knowledge gained from previous incidents and simulations. Although each source may provide useful information, bringing them together quickly and consistently remains demanding, particularly when the situation is changing and decisions cannot be postponed.

This context led to the collaboration between the DISP laboratory and SDIS 42. The project investigates how artificial intelligence can assist the anticipation officer in organizing these different sources and preparing structured, traceable proposals. The study focuses on a proof-of-concept dashboard combining locally deployed Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), internal SDIS documents, and external contextual services. The system is conceived as a proactive intervention system: it helps stay one step ahead, but the interpretation, validation, allocation of resources, and operational decision remain human responsibilities.

## 1.3 Problem Statement

Large-scale emergency operations create a paradox. They require the highest level of preparation and anticipation, but they provide officers with the fewest opportunities to learn through repetition.

### 1.3.1 Rare Operations in a Changing Risk Environment

Most firefighter activity consists of everyday emergency operations that do not require a site-level command post. In 2025, SDIS 42 carried out 48,475 operations, more than 80% of which concerned emergency assistance to people [7]. By contrast, the anticipation function is mainly mobilized during complex, large-scale, long-lasting, or high-consequence operations requiring a *"poste de commandement de site"* [6].These operations may involve a major flood, an extensive forest fire, an industrial or chemical accident or a terrorist attack. They remain uncommon in comparison with daily emergency activity, but their consequences can be considerable. They may

involve several operational sectors, specialist teams, partner organizations, critical infrastructure, and resources requested from outside the department.

Wildfires illustrate how the operational context is changing. The Loire was historically less exposed to major forest fires than Mediterranean departments. However, the departmental authorities now identify it as one of France's *nouveaux territoires de feux*. Several notable fires have already occurred in the department, including events affecting 50 hectares at Saint-Marcellin-en-Forez, 105 hectares at Châteauneuf, 23 hectares at Véranne, and 35 hectares at La Valla-en-Gier [8].Climate change does not mean that every year will produce a major fire. Nevertheless, higher temperatures, drier vegetation and changing rainfall patterns create conditions that favor the propagation of forest fires. Météo-France expects the fire risk to extend into territories that were previously less affected and projects that the fire season could become one to two months longer in some regions [9].

Recent activity in the Loire demonstrates that this is no longer only a long-term projection. By 7 July 2026, SDIS 42 had already responded to 55 forest and natural-area fires since the beginning of June, affecting a total of 217 hectares [10]. These figures do not establish a long-term trend by themselves, but they provide a concrete example of the operational pressure that an emerging fire season can create.This changing environment raises a knowledge problem because large-scale incidents have historically been relatively uncommon in the department; officers cannot rely on a large local history of similar operations. Each event may also combine circumstances that have not previously occurred together.

Training and simulation partially compensate for this limited exposure. For example, before the 2026 summer season, SDIS 42 organized a zonal wildfire exercise involving more than 90 firefighters and 25 vehicles. The exercise reproduced reconnaissance, water supply, protection of exposed areas, coordination, command, and the use of drones and aerial resources [11]. Such exercises allow officers to practice roles and procedures that they may rarely perform during everyday interventions.However, a simulation cannot reproduce every combination of constraints. During a real incident, the anticipation officer must still reason from incomplete information and draw on doctrine, training, feedback from previous operations, technical expertise, and the evolving situation on the ground.

### 1.3.2 Maintaining Information Continuity at the Command Post

The second problem appears when the command post is activated. During a major operation, information does not arrive as one complete and stable report. It arrives progressively through multiple resources: radio messages, sector reports, observations, maps, telephone calls, expert advice, resource updates, weather information, and exchanges with the CODIS and part-

ner organizations. An operational command-post simulation attended during this project made this difficulty particularly visible.[2] Three principal difficulties were identified.

- **Limited information traceability.** Information was frequently communicated orally or recorded across different notes and operational supports. It could therefore become difficult to determine when an element had been received, who had provided it, whether it had been confirmed, and whether it was still current.
- **Rapid and continuous updates.** The operational picture changed quickly as new reports arrived. An observation could confirm an existing assumption, contradict earlier information, or introduce a new event. Maintaining an up-to-date understanding while simultaneously preparing for future situations was therefore demanding.
- **Dependence on human memory and oral restitution.** Officers had to remember and verbally reproduce numerous details, including exposed elements, resources, actions in progress, pending requests, assumptions and possible developments. Under time pressure, important information could be forgotten, repeated, or separated from its original context.

These difficulties are closely connected. When information is not reliably recorded, each update increases the memory burden placed on the command team. When the pace of updates increases, officers have less time to check earlier reports. When information must be reconstructed orally, the link between an observation, its source, and the planning proposal derived from it may become difficult to preserve.

For the anticipation officer, this problem is particularly important. The officer must reason about future situations while remaining connected to the current operational reality. A proposed *"situation envisageable"* may become irrelevant if the information on which it was based has changed. A manoeuvre may appear feasible until a resource becomes unavailable. A task may no longer be appropriate after a change in wind, access, water supply, or the condition of exposed people.

### 1.3.3 Resulting Research Problem

The research problem therefore combines two forms of scarcity. First, major operations provide limited opportunities to accumulate direct local experience. Second, during an actual operation, the time and attention available for collecting, checking and structuring information are also limited.The problem is not simply a lack of information. The command post may receive a large quantity of information, but this information can be fragmented, rapidly outdated, difficult to

[2]Observations recorded during an operational command-post simulation attended as part of this study and discussed with Capitaine Yoan, SDIS 42, in 2026. The methodology chapter describes how these observations were used to identify system requirements.

trace, and demanding to remember. Nor can the problem be solved by predicting one supposedly certain future. Operational anticipation requires several plausible situations to be considered while being related to objectives, maneuver, tasks, and resource requirements.

This study consequently addresses the need for a proactive intervention system that can preserve the history and source of operational information, highlight significant updates, retrieve relevant knowledge from SDIS documents, and help the anticipation officer prepare structured and traceable proposals. Such support must reduce the information-management burden without removing professional judgment. The officer remains responsible for interpreting and validating the proposals, and the COS retains responsibility for the operational decision.

## 1.4 Purpose of the study

In response to the research problem described above, this study aims to design and evaluate Anticip-42, a proof-of-concept dashboard that uses locally deployed Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) to support the anticipation officer during complex fire and rescue operations. Anticip-42 is intended to bring together operational information entered by officers, contextual data obtained from external services, and knowledge retrieved from local SDIS documents. The purpose of this combination is to help the anticipation officer organize rapidly changing information, preserve its traceability, and prepare structured planning proposals for the command structure.

The system is not intended to predict one certain future, replace real actors, or automate operational command. Its outputs remain proposals that must be reviewed, corrected, and validated by the appropriate officers. Ultimately, the study seeks to determine whether a locally deployed LLM-and-RAG approach can provide useful support for operational anticipation while preserving human responsibility, data control, and the established chain of command.

## 1.5 Research Questions

This study examines how a proactive intervention system based on human-in-the-loop artificial intelligence can support operational decision-making during emergency response. It is organized around one main research question and four supporting research questions. Together, the questions follow the progression of the study from requirements identification and system modeling to human-in-the-loop AI design and technical implementation.

## Main Research Question

> **Main RQ:** How can a proactive intervention system for firefighters be designed and implemented using human-in-the-loop AI to support operational decision-making during emergency response?

**Rationale.** Emergency-response operations require officers to interpret incomplete and rapidly evolving information while preparing for possible developments, operational objectives, maneuvers, and detailed tasks. Artificial intelligence may help structure this reasoning and generate timely proposals, but operational authority and accountability must remain with human officers. The main research question therefore investigates the complete design and implementation of a proactive intervention system that combines AI assistance with explicit human supervision, validation, and control.

## Supporting Research Question 1

> **SQ1:** What requirements shape the design of a proactive intervention system for emergency response?

**Rationale.** A proactive intervention system cannot be designed from technological assumptions alone. Its design must reflect the operational process, participating actors, information flows, decision responsibilities, time constraints, data sensitivity, traceability needs, and the points at which officers must review and validate AI-generated proposals. SQ1 therefore identifies and organizes the functional and non-functional requirements of the system. These requirements establish the foundation for the subsequent system models, architecture, implementation, and evaluation.

## Supporting Research Question 2

> **SQ2:** How can system modeling be used to translate the identified requirements into a comprehensive architectural specification?

**Rationale.** The requirements identified through SQ1 must be translated into a clear description of how Anticip-42 should work and how its different parts should work together. System modeling supports this step by showing the operational workflow, the roles of users and system components, and the information exchanged between them.

Although existing decision-support architectures address emergency response, they do not fully represent progressive anticipation processes. SQ2 therefore examines how system models can be used to derive an architectural specification for a progressive anticipation system.

## Supporting Research Question 3

> **SQ3:** How can a dynamic human-in-the-loop AI system be designed to enable proactive intervention while maintaining effective collaboration between human operators and AI components?

**Rationale.** Emergency-response information and operational priorities change continuously. Consequently, the system must support repeated generation, review, correction, validation, and updating of the proposals rather than produce a single static recommendation. SQ3 focuses on the collaboration between human operators and AI components: what the AI proposes, where officers intervene, how their feedback affects later outputs, and how validated information is propagated through the operational workflow. It therefore examines how Anticip-42 can provide proactive assistance without becoming an autonomous operational decision-maker.

## Supporting Research Question 4

> **SQ4:** What services and system interfaces are required to implement the proposed proactive intervention system and support interoperability among its components?

**Rationale.** Anticip-42 brings together several parts, including its user interface, backend, local language models, document-retrieval mechanism, and external sources of contextual information. These parts need clearly defined ways of communicating and exchanging data. SQ4 therefore examines the services and interfaces needed to connect them, achieve interoperability, and implement the architecture developed through SQ2.

## 1.6 Study Objectives

Based on the research questions presented in the previous section, this study aims to design, implement, and evaluate Anticip-42, a proof-of-concept proactive intervention system that uses human-in-the-loop artificial intelligence to support operational decision-making during complex fire and rescue operations. The system is intended to assist the anticipation officer in analyzing the current situation, considering possible developments, and preparing operational proposals while keeping decision-making authority with human officers.

The study is structured around four specific objectives. Each objective corresponds directly to one supporting research question and represents one stage in the development of Anticip-42:

1. **Objective 1: Identify the requirements of a proactive intervention system for emergency response.**

This objective is to identify and specify the functional and non-functional requirements of the proposed system. These requirements cover the actors and responsibilities involved, operational information flows, time constraints, traceability needs, data-protection considerations, and the points at which human validation is required.

2. **Objective 2: Translate the identified requirements into a comprehensive architectural specification through system modeling.**

   This objective is to develop a set of system models representing different aspects of Anticip-42, including its operational processes, system boundaries, actors, components, interactions, information exchanges, data structures, responsibilities, and dependencies. Together, these models provide a coherent architectural specification that can guide the implementation and evaluation of the system.

3. **Objective 3: Design a dynamic human-in-the-loop AI model for proactive intervention.**

   This objective is to design and prototype a dynamic workflow in which AI-generated proposals can be progressively reviewed, corrected, validated, and updated by human operators as an emergency intervention evolves. The model must support effective collaboration between officers and AI components while ensuring that operational decisions remain under human control.

4. **Objective 4: Define and implement the services and interfaces required for system interoperability.**

   This objective is to identify, specify, and implement the services, application programming interfaces, data contracts, and information exchanges required to integrate the user interface, backend, locally deployed language models, retrieval-augmented generation pipeline, documentary sources, and external contextual services. The resulting implementation should support modularity, component replaceability, and interoperability.

## 1.7 Approach and Boundaries of the Study

The following subsections define the scope, limitations, delimitations, and assumptions under which the study was conducted.

### 1.7.1 Scope

This study covers the analysis, design, implementation, and controlled evaluation of Anticip-42, a proof-of-concept system intended to support the anticipation officer during complex fire and

rescue operations. It includes the identification of system requirements, the modeling of the operational process and system architecture, the design of a progressive human-in-the-loop AI workflow, and the implementation of the services and interfaces needed to connect the prototype's components.

Understanding the operational domain required the study to draw on two types of documentation. The first consisted of publicly available sources recommended by Capitaine Yoan.[3] These sources included the national *"Guides de doctrine opérationnelle"* (GDO) and *"Guides de techniques opérationnelles"* (GTO) [12]; Partage d'expérience (PEX) and Retour d'expérience (RETEX) publications available through the "École nationale supérieure des officiers de sapeurs-pompiers" (ENSOSP) portal [13]; industrial-accident records from the "Analyse, Recherche et Information sur les Accidents" (ARIA) database [14]; INERIS Omega technical reports [15]; and references concerning dangerous chemical reactions [16].

The GDO and GTO describe fire-service doctrine, whereas the other sources provide supporting knowledge about operational experience, risk analysis, industrial accidents, fires, explosions, hazardous-material dispersion, and dangerous chemical reactions. Together, these sources provided a general foundation for understanding the operational domain addressed by the project.

The second type consisted of selected non-public working materials made available through a restricted shared repository. These materials provided additional examples of operational terminology, equipment, risks, and document structures. Only the documents explicitly made available for this phase of the project were used. The repository did not represent the complete operational documentation of SDIS 42.

The technical scope includes a dashboard, backend services, locally deployed language models, a retrieval-augmented generation (RAG) pipeline using the selected documents, data storage, and external services providing contextual information such as weather and geographic data. The study examines how these parts can exchange information and support the preparation of structured proposals that officers can review, modify, and validate.

The prototype is evaluated in a controlled environment using representative scenarios, authorized documents, predefined inputs, and saved system outputs. The evaluation considers whether the identified requirements are covered, whether retrieved documents are relevant, whether generated proposals are supported by the available sources, and whether the integrated components function as intended. The study evaluates the technical feasibility of the proposed system; it does not claim that Anticip-42 improves operational outcomes during real interventions.

[3]The initial list of resources and their access conditions was communicated by Capitaine Yoan during project supervision in 2026.

### 1.7.2 Limitations

The first limitation concerns access to internal documentation. SDIS 42 is a public emergency-response organization whose operational documents may contain sensitive information concerning risks, sites, resources, and response procedures. Access to complete collections therefore requires formal authorization and an appropriate agreement between the partner institutions.

This capstone represents an initial phase of a broader research initiative expected to continue beyond the present study. During the capstone period, the institutional agreement covering wider access was still being discussed. Capitaine Yoan was able to provide access to selected working documents, but complete collections of departmental experience reports and other internal operational resources remained subject to additional authorization. The project could therefore not use a complete or continuously updated SDIS 42 knowledge base.

The distinction between public and internal material affects the RAG evaluation. The public sources provide broad doctrine and transferable operational knowledge, but they do not necessarily capture every local procedure or organizational practice. The restricted documents provide greater local specificity, but only a limited selection was available. The prototype results consequently reflect the corpus available during this phase, not the full operational knowledge of SDIS 42.

A second limitation results from the operational setting itself. Interventions that require a dedicated anticipation function are relatively uncommon, and the prototype was not tested during a live emergency. The evaluation relies instead on controlled scenarios, technical experiments, and feedback from a limited number of domain participants. These conditions are suitable for evaluating an initial proof of concept, but they cannot fully reproduce the conditions of a real command post, including time pressure, stress, incomplete information, and operational consequences.

The LLM component also has inherent limitations. A model may produce incomplete, inconsistent, or unsupported content when the operational input is ambiguous or when the retrieval process does not provide sufficient context. RAG, structured prompts, automated checks, and human review can reduce these risks, but they cannot guarantee that every output is operationally appropriate.

Local model execution introduces a further technical constraint. LLMs require significant computing resources, and their response time depends on model size, quantization, available memory, processor capacity, and the presence or absence of GPU acceleration. The measurements obtained in this study are specific to the university virtual machine with 96 GB of RAM, four allocated virtual processors, and an Intel Xeon Silver 4410Y CPU running at 2.00 GHz. Production workloads, larger models, multiple simultaneous users, and alternative hardware environments were not

evaluated.

Finally, the system depends on external services whose availability, coverage, latency, and data quality are outside the researcher's control. A service that responds correctly during testing may later return incomplete information, change its interface, or become temporarily unavailable. These dependencies must be considered when interpreting the prototype results.

### 1.7.3 Delimitations

The study focuses specifically on operational anticipation rather than on the command post as a whole. This focus allows the anticipation process to be examined in sufficient depth while still considering the wider command-post environment when it directly affects that process.A human-in-the-loop approach was also chosen as a central design principle. Artificial intelligence is therefore considered as a means of helping officers prepare and revise operational proposals, while professional judgment and decision-making authority remain with the appropriate human actors.

The study follows a design-and-demonstration approach. The technical components were selected because they address the identified requirements and make it possible to demonstrate the proposed workflow. The aim is therefore to examine whether these components can work together within a coherent architecture, rather than to compare every available technology or determine which configuration performs best.

Finally, the evaluation focuses on whether the proposed concepts can be implemented in a functioning proof of concept. Measuring their effects on operational performance, working practices, or long-term adoption would require a different type of study and is therefore outside the chosen evaluation scope.

### 1.7.4 Assumptions

The study assumes that information entered by the user is sufficiently accurate and current to support the generation process. The prototype does not independently confirm every field observation. Incomplete or incorrect input may therefore produce proposals that are incomplete or inappropriate.

It is assumed that the authorized corpus contains enough relevant doctrine, terminology, operational experience, and risk knowledge to investigate document retrieval and grounded generation. This assumption does not imply that the corpus is complete or that retrieval automatically guarantees the correctness of an output.

The system is also designed on the assumption that a qualified officer remains available to review, modify, reject, and validate every generated proposal. No output is treated as correct

solely because it was generated by an LLM or associated with retrieved documentation.The evaluation assumes that operational anticipation does not have one universally correct answer. Different proposals may be defensible depending on the available information, uncertainty, priorities, and professional judgment. Output quality must therefore be assessed through several complementary criteria rather than by comparison with one fixed reference answer.

Finally, the selected scenarios and contextual services are assumed to be sufficient to demonstrate the principal interactions of the proof of concept. They are not assumed to represent every intervention type or every condition that may arise during operational use.

## 1.8 Contributions

The contributions of this study follow the progression of the research questions, from requirements identification and system modeling to human-in-the-loop AI design and technical integration. They include both the artifacts produced during the development of Anticip-42 and the knowledge gained through their controlled evaluation. Since Anticip-42 remains a proof of concept, these contributions provide evidence of technical feasibility and a foundation for subsequent research; they do not establish operational effectiveness.

### 1.8.1 Technical Contributions

The technical contributions correspond to the four supporting research questions.

1. **A requirements specification for Anticip-42.**

   The study identifies the functional and non-functional requirements that shape the design of a proactive intervention system.

2. **A system modeling and architectural specification.**

   The study translates the identified requirements into models of the operational workflow, user interactions, system components, information exchanges, and data structures. Together, these models form the architectural specification of Anticip-42.

3. **A progressive human-in-the-loop AI workflow.**

   The prototype supports the progressive generation, review, modification, and validation of proposals. Information accepted by an officer is preserved and used during the following stages of the anticipation process.

4. **An integrated proof-of-concept implementation.**

   The implementation connects the dashboard, backend, local language models, document-retrieval mechanism, data storage, and external contextual services through defined services and interfaces.

### 1.8.2 Scientific Contributions

The scientific contribution of this study is exploratory and mainly design-oriented. It does not demonstrate that Anticip-42 improves decisions during real interventions. Instead, it provides the following context-specific findings:

1. **Confirmation of the importance of operational requirements.**

   The study confirms that an emergency-response AI system cannot be designed from technical requirements alone. Operational roles, information exchanges, time constraints, and validation responsibilities must also be considered. This supports existing human-centered approaches to decision-support system design.

2. **A stage-by-stage approach to human–AI collaboration.**

   The study adds a design approach in which human control is included throughout the workflow rather than only at the end. Possible situations, objectives, maneuver ideas, and DTA are generated separately, and each result can be reviewed, corrected, or rejected before it is used in the following stage.

3. **Exploratory evidence from an integrated local prototype.**

   The proof of concept shows that local language models, document retrieval, persistent storage, and external context APIs can be connected through a shared workflow. However, the evaluation also shows that this integration does not guarantee reliable operational output. Its usefulness remains dependent on retrieval quality, response structure, execution time, service availability, and human review.

These findings provide a basis for further evaluation with qualified SDIS personnel and realistic operational conditions. No conclusion is made about the effectiveness or safety of the system during live emergency response.

## 1.9 Structure of the Report

This report follows the development of the project from the operational problem identified at SDIS 42 to the design, implementation, and controlled evaluation of Anticip-42. It is organized into seven main sections.

- **Introduction:** Section 1 introduces the operational context, the institutions involved, and the problem addressed by the project. It then presents the purpose of the study, its objectives, the research questions, the study boundaries, and its main contributions.
- **Background and Literature Review:** Section 2 provides the background needed to understand the project. It describes SDIS 42, the main actors and information exchanges within the command post, and the operational anticipation process. It then reviews previous research on emergency-response decision support, human-AI collaboration, system modeling, large language models, retrieval-augmented generation, and service integration before presenting the research gap addressed by the study.
- **Methodology:** Section 3 explains how the study was conducted using Design Science Research. It presents the research process, the relationship between the research questions and the iterations, the study materials, the requirements-identification activities, the technical environment, and the evaluation procedures.
- **Design and Development Iterations:** Section 4 retraces the development of Anticip-42 through four iterations. The first investigates the operational process and identifies the system requirements. The second translates the process into a progressive, human-supervised workflow. The third develops the modular architecture and integrates local language models, document retrieval, data storage, and contextual services. The fourth brings the previous results together and evaluates the integrated proof of concept. Each iteration presents the problem addressed, the artifact produced, the evaluation performed, and the knowledge gained.
- **Results:** Section 5 presents the results obtained from the development and evaluation of the proof of concept. It brings together the requirements, system models, human-in-the-loop workflow, services, interfaces, and evaluation findings to answer the four supporting research questions and the main research question.
- **Discussion:** Section 6 interprets the findings in relation to the research questions and previous studies. It discusses the technical and research contributions, considers the relevance of the findings for SDIS 42, and examines the study's limitations and threats to validity.
- **Conclusion:** Section 7 returns to the research problem and summarizes what the study was able to demonstrate. It presents the main conclusions, recommendations, and possible directions for the next phase of the project.

# 2 BACKGROUND AND LITERATURE REVIEW

Supporting anticipation during emergency response requires an understanding of both the operational environment and the technologies that can assist decision-making under uncertainty. This chapter establishes the background needed to examine the design of a proactive intervention system for this context.

The chapter first introduces proactive intervention systems for emergency response. It then presents the French fire and rescue context, the organization of SDIS 42, and the anticipation process within the site command post. The technical foundations of AI-assisted decision support are subsequently examined, followed by existing research on system requirements, architectures, human–AI collaboration, and related emergency-response applications. The chapter concludes by identifying the research gap addressed by Anticip-42.

## 2.1 Proactive Intervention Systems for Emergency Response

A proactive intervention system for emergency response supports emergency staff in looking beyond the current situation and preparing for possible developments. It brings together available operational information, relevant knowledge, contextual data, and decision-support capabilities to help staff identify how an incident could evolve, assess the possible consequences, and consider appropriate actions in advance. Its purpose is therefore not only to present what is currently happening, but also to support timely and informed preparation for what may happen next.

Many emergency information systems primarily support situational awareness by collecting reports, displaying maps, monitoring resources, and sharing information. These functions help emergency staff understand the current incident and coordinate the ongoing response. A proactive intervention system extends this support by helping staff examine plausible developments and prepare alternative responses before the situation changes or becomes more difficult to manage.

Proactive intervention does not mean predicting a single certain outcome or allowing a system to make autonomous operational decisions. Emergency situations develop under uncertainty, and available information may be incomplete, conflicting, or subject to change. Possible developments and proposed responses must therefore be updated as new information becomes available and remain subject to human interpretation and validation. This understanding is consistent with situation-awareness theory, which includes the projection of a situation's possible future state [17], and with dynamic emergency-response systems that revise information and plans as an incident evolves [18].

Figure 1 illustrates the relationship between understanding the current situation and anticipating how it may develop. In Endsley's model, situation awareness progresses from the perception

of relevant elements to comprehension of their meaning and then to the projection of their future status. This final level is particularly relevant to proactive intervention because it supports the consideration of possible developments before decisions and actions are taken. The feedback loop also shows that actions change the operational environment, requiring the situation to be reassessed as new information becomes available.

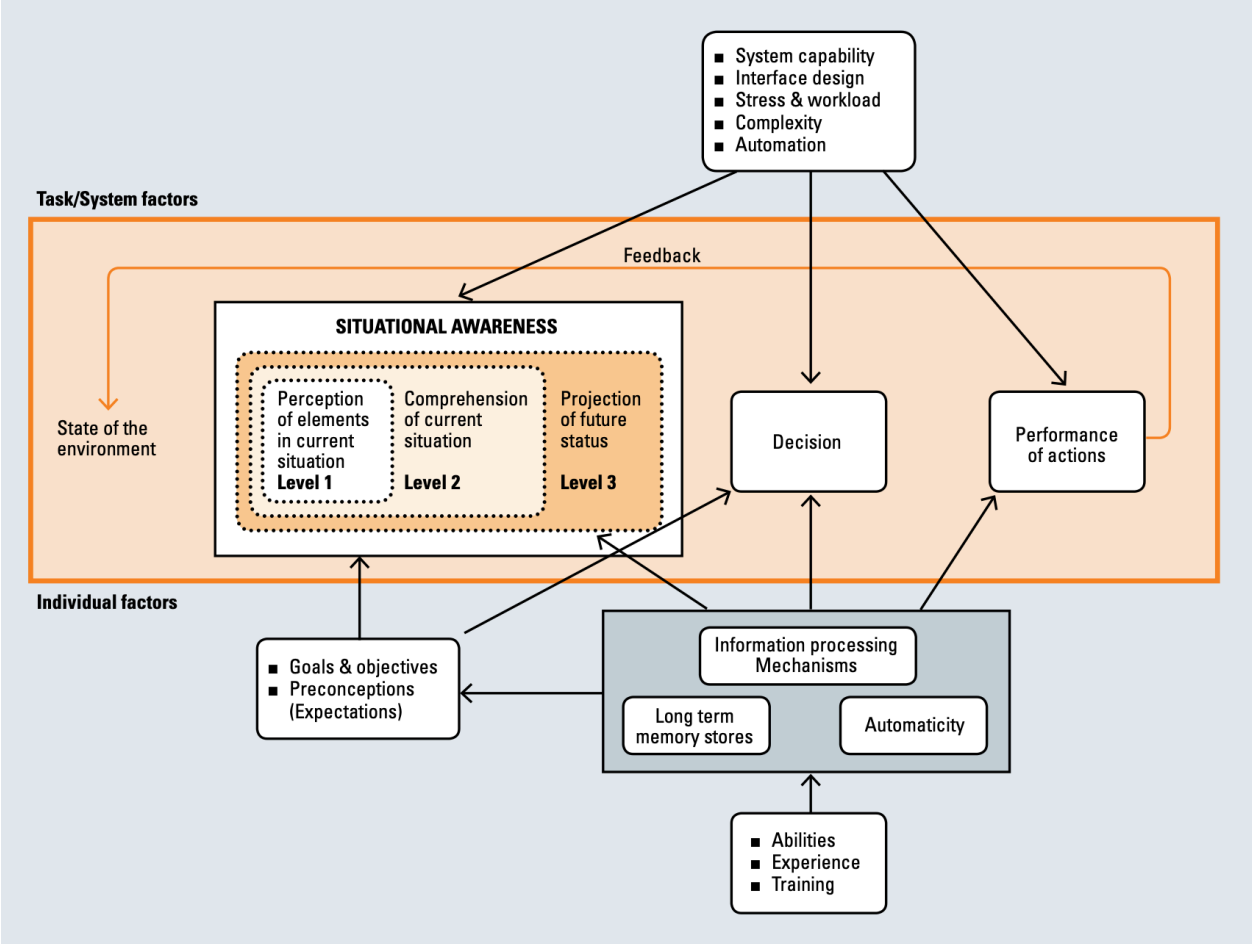



**Figure 1**: Endsley's model of situation awareness. Source: EUROCONTROL [19], based on Endsley [17].

Research related to proactive emergency intervention covers several complementary areas. Fire-behavior models estimate how a hazard may develop; situational-awareness systems organize information about the current incident; analytical models estimate demand, duration, or escalation; resource-allocation systems examine how available capabilities may be deployed; and generative systems support access to operational knowledge and the preparation of response plans. Together, these areas provide the broader context for the progressive anticipation process considered in this study. The following sections narrow the discussion to the French fire and rescue context and the operational environment of SDIS 42.

## 2.2 The French Fire and Rescue Context

In France, fire and rescue missions are mainly carried out by the *Services Départementaux d'Incendie et de Secours* (SDIS). Each SDIS organizes the staff, fire stations, vehicles, specialist teams, and operational facilities required to respond to emergencies within a department. Its missions extend beyond firefighting and include road traffic accidents, medical emergencies, natural disasters, technological incidents, pollution, and other events that may threaten people, property, or the environment [4], [20].

The present study was conducted in partnership with the *"Service Départemental d'Incendie et de Secours de la Loire"* (SDIS 42). Its territorial organization connects local fire and rescue centers with a departmental headquarters, or *"état-majeur"*, from which alerts and ongoing operations are monitored. This organization makes it possible to begin with a local response while progressively reinforcing command and coordination when an incident becomes larger, longer, or more complex [4], [6].

French emergency command also distinguishes legal direction from operational command. The *"Directeur des Opérations de Secours"* (DOS) exercises the legal direction of the response. This responsibility generally belongs to the mayor for a local incident and may pass to the prefect when the event exceeds municipal capabilities, concerns several municipalities, or requires departmental coordination. Under the authority of the DOS, the *"Commandant des Opérations de Secours"* (COS) directs the rescue operation and determines how the available emergency resources are employed [6], [20].

This distinction is important throughout this study. The COS is the operational decision-maker. Other command-post functions collect information, monitor actions, assess resources, and prepare alternatives, but they do not replace the COS's authority.

## 2.3 SDIS 42 Organization, Key Actors, and Operational Workflow

### 2.3.1 From the Emergency Call to Departmental Coordination

At departmental level, two entities support the continuity of an emergency operation: the *"Centre de Traitement de l'Alerte"* (CTA) and the *"Centre Opérationnel Départemental d'Incendie et de Secours"* (CODIS).

The CTA receives emergency calls, collects the first available information, identifies the type and location of the event, and dispatches the initial resources. At this stage, the operational picture is necessarily incomplete: the information mainly comes from the caller and must be confirmed once the first teams arrive.

The CODIS maintains a broader view. It follows the interventions taking place throughout

the department, monitors the availability of resources, coordinates reinforcements, informs the relevant authorities, and supports operations that increase in scale or duration. The CTA therefore initiates the response, whereas the "Centre Opérationnel Départemental d'Incendie et de Secours" (CODIS) maintains its departmental continuity [6], [21]. Figure 2 illustrates this division of responsibilities and the information exchanged between the alert center, operational coordination, field resources, and external partners. Although the figure was produced by another departmental fire and rescue service, it illustrates the general CTA–CODIS organization shared by French SDIS. The description specific to SDIS 42 is based on its operational regulations [6].

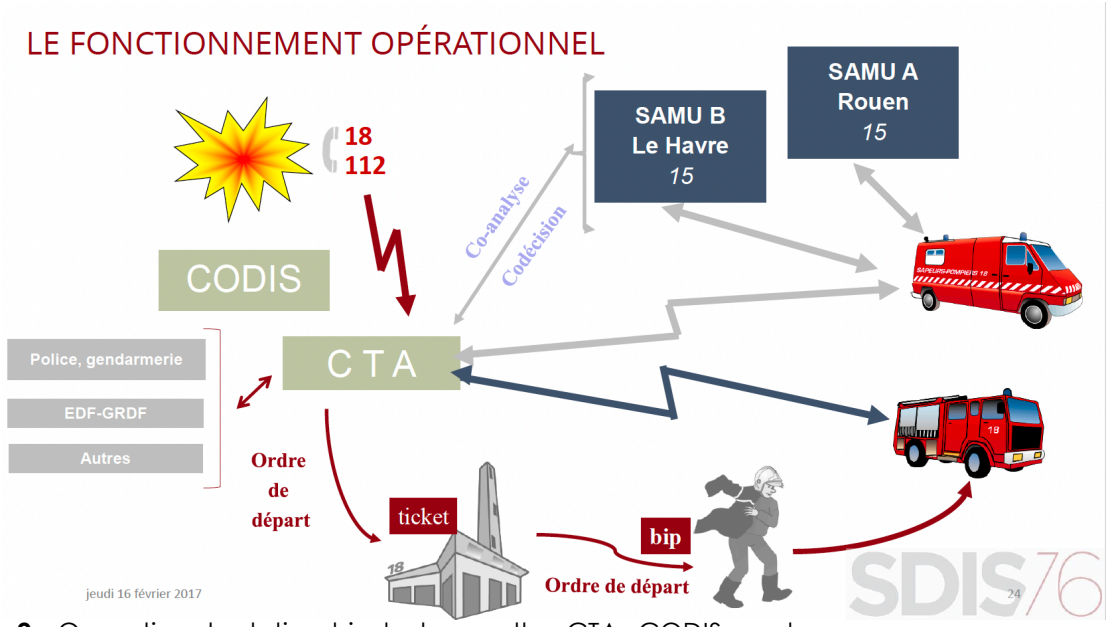



**Figure 2**: Operational relationship between the CTA, CODIS, and emergency response actors. Source: SDIS de la Seine-Maritime [22].

The CODIS does not receive information from a single channel. During an operation, it exchanges information with the COS, the command post, public authorities, partner emergency services, infrastructure operators, technical experts, and other organizations involved in the response. It must also preserve a department-wide view so that the reinforcement of one incident does not leave another area without sufficient operational coverage.

Figure 3 represents this position within the operational information network.

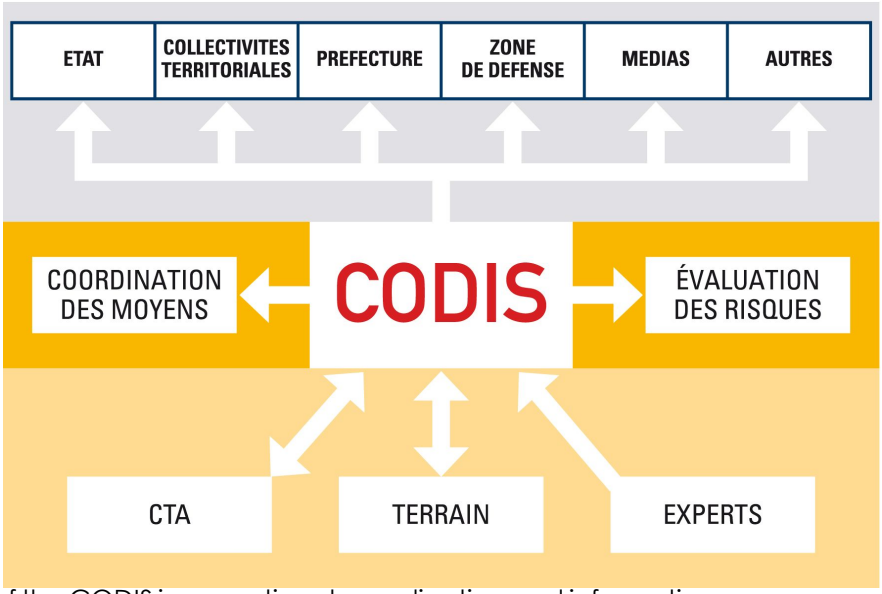



**Figure 3**: Role of the CODIS in operational coordination and information management during emergency response. Source: SDIS de l'Aveyron [23].

### 2.3.2 From Departmental Supervision to On-Site Command

The CODIS follows the operation from the departmental level, but it does not replace command on the incident ground. When the first resources arrive, their commander conducts reconnaissance, evaluates the immediate hazards, reports to the CODIS, and establishes the first operational priorities. As the operation grows, the command level may successively involve a *"chef d'agrès"*, *"chef de groupe"*, *"chef de colonne"*, and ultimately a *"chef de site"* [6].

A large, long-lasting, or multi-sector incident may require the activation of a *"poste de commandement"* (PC). The PC provides a working structure around the COS. Its purpose is not to add another decision-making authority, but to prevent the COS from having to process every radio message, resource request, field observation, and technical document alone.

The PC receives fragmented information and progressively turns it into a shared operational picture. It also maintains continuity between what has already happened, what is currently being done, and what may become necessary later. This work is distributed among several specialized functions: *"renseignements"*, *"action"*, *"moyens"* and *anticipation* under the coordination of the *"chef PC"*.

This command-post configuration defines the intended operational context of Anticip-42. The system is not designed for routine interventions managed by a small command structure. It is intended for complex operations that require the activation of a site command post and a dedicated *"anticipation"* function. At this level, the *"officier anticipation"* must continuously examine possible developments of the incident and prepare operational alternatives for the COS while information is arriving from several functions [6], [24], [25]. Anticip-42 is intended to support this officer by

organizing the available information, preserving its evolution, retrieving relevant operational knowledge, and assisting in the preparation of structured proposals. Its outputs remain working material for officer review and COS decision-making rather than operational orders.

### 2.3.3 Inside the Site Command Post

The site command post is not organized as a simple sequence in which one officer finishes a task before the next officer begins. Its functions exchange information continuously. Figure 4 distinguishes functional exchanges from authority relationships and shows the central coordinating position of the *"chef PC"*.

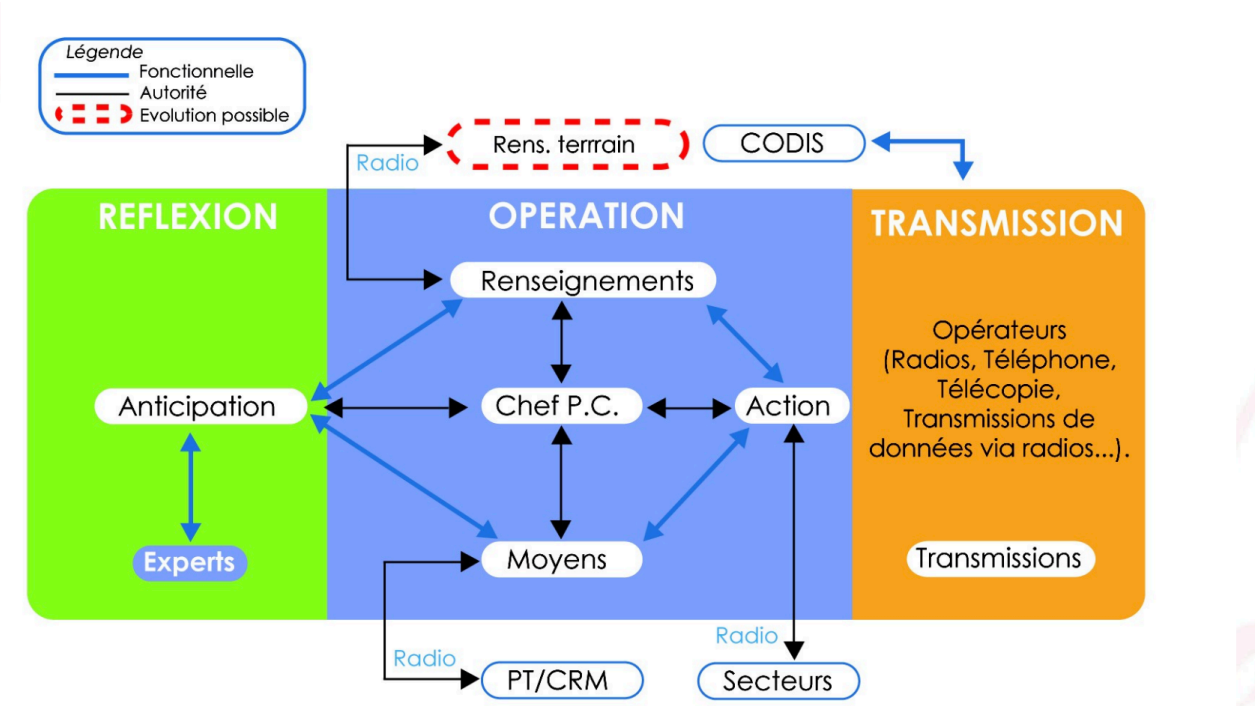



**Figure 4**: Functional organization of a site command post. Source: adapted from Freyermuth [26].

**The COS.** The COS commands the rescue operation. The officer analyzes the incident, defines the operational objectives, selects a general course of action, organizes the engaged resources, requests reinforcements, and ensures the safety of responders and exposed populations. The COS also reports to the CODIS and advises the DOS regarding any administrative or population-protection measures that may be required [6], [24].

The COS does not have to accept a proposal prepared by the command post. Operational objectives and maneuvers may be accepted, modified, combined, or rejected. This decision-

making responsibility remains with the COS throughout the workflow examined in this study.

**The "chef PC".** The *"chef PC"* coordinates the command-post functions and acts as the main interface between the PC and the COS. In practical terms, this officer may be described as the COS's "right-hand officer" (*"bras droit"*): the "chef PC" maintains the internal coherence of the command post while the COS concentrates on operational decisions and exchanges with the DOS, the prefect, the CODIS, or other senior authorities [24], [27].

When the COS is temporarily away from the PC, for example to attend a meeting, conduct a field assessment, or communicate with the prefectural authority, the "chef PC" ensures continuity within the command post. The officer coordinates the *"renseignements"*, *"action"*, *"moyens"*, and *"anticipation"* functions, organizes situation reviews, establishes working priorities, and ensures that operational documents remain up to date. This continuity prevents the different functions from working independently or from transmitting fragmented information directly to the COS.

The "chef PC" also performs an important filtering role. The COS does not need to receive every unprocessed radio message, field report, or resource request. The "chef PC" ensures that the appropriate functions verify, organize, and consolidate the information before presenting significant developments, identified risks, and operational proposals to the COS.

However, coordinating the PC during the COS's temporary absence does not automatically transfer operational command to the "chef PC". The COS remains responsible for selecting the objectives and directing the rescue operation unless a formal transfer of command has been established and communicated through the appropriate operational chain.

**The "renseignements" function.** The *"officier renseignements"* constructs and maintains the current operational picture. Information may arrive from sector commanders, recognition teams, the CODIS, witnesses, maps, weather services, sensors, partner organizations, or technical experts.

Collecting information is only the first part of the task. Each element should, when possible, be dated, located, and associated with its source. Confirmed facts must remain distinguishable from uncertain reports and working assumptions. The function maintains chronological records, prepares situation summaries, and updates the *"situation tactique"* (SITAC) [24], [28].

The renseignements function is closely connected to anticipation. The latter cannot explore a credible future if the description of the present is outdated. Conversely, anticipation may reveal missing information and ask the renseignements function to verify a particular hazard, exposed asset, access route, or change in the incident.

**The "action" function.** The *"officier action"* connects the COS's decisions to their execution in the operational sectors. The officer prepares or transmits validated orders, monitors the missions assigned to each sector, and records what has been completed, what remains in progress, and

what difficulties have been reported.

The action function is therefore mainly concerned with the current maneuver. It tells the PC whether the selected actions are producing the expected effects. If the situation changes or a mission becomes impossible, the resulting information returns to the command post and may require the COS to modify the operation [24], [25].

**The "moyens" function.** The *"officier moyens"* follows the personnel, vehicles, equipment, specialist teams, and logistical capabilities involved in the operation. The officer records which resources are engaged, in transit, waiting, available, unavailable, or likely to require replacement.

This view is essential for both action and anticipation. Action needs to know what can be deployed immediately. Anticipation needs to know whether a future option could be supported when it becomes necessary. The means function also considers endurance: crew relief, fuel, food, water, specialist equipment, and the duration for which the operation may have to be maintained [24], [25].

**The "anticipation" function.** The *"officier anticipation"* temporarily steps away from the immediate execution of the operation to study its possible future development. The officer considers how the hazardous phenomenon may evolve, which unwanted events may occur, what consequences they may produce, and which resources or actions may have to be prepared before the situation deteriorates.

This function remains connected to every other part of the PC. It receives the current picture from renseignements, the effects of ongoing actions from officier action, and the availability of resources from officier moyens. It may also consult experts when chemical, structural, meteorological, industrial, medical, or environmental knowledge is required [25], [26].

The anticipation officer prepares alternatives but does not independently select or order them. Proposals are coordinated through the "chef PC" and submitted to the COS. This human chain is particularly important for Anticip-42: the system is intended to assist the anticipation function, not to become an autonomous commander.

**Experts and technical advisers.** Experts contribute knowledge that is not always available within the standard command-post functions. Depending on the incident, they may include chemical risk advisers, industrial operators, meteorologists, hydrologists, structural specialists, veterinarians, medical personnel, environmental agencies, or network operators.

An expert may explain a phenomenon, provide calculation results, identify a threshold, or assess the feasibility of an action. This advice informs the analysis, but the expert does not replace the COS or independently direct the operation [6], [25].

## 2.4 Methods and Tools for Operational Anticipation

Operational anticipation does not begin by predicting an isolated event. It begins with the current operational picture and progressively transforms that picture into possible future situations, operational objectives, alternative courses of action, and executable tasks. Each stage depends on the previous one and must be reconsidered whenever the situation changes.

The tools used inside the command post make this reasoning visible. The SITAC represents the intervention and its current operational organization. Other supports allow officers to project the situation into the future, compare alternative responses, and prepare the elements required for execution. This section introduces this progression before examining each stage in detail.

### 2.4.1 The SITAC as the Starting Point

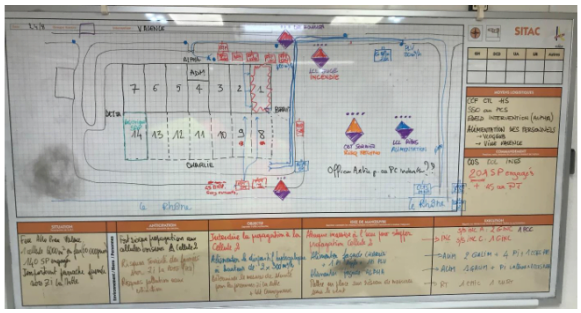
The *"situation tactique"* (SITAC) is one of the main supports used to share operational information inside the command post. The national command doctrine defines it as a graphical representation of the intervention area and its characteristics, the resources engaged or planned, the actions completed or planned, and the organization of command [28].

Depending on the incident, the SITAC may show the hazardous phenomenon, its location and extent, exposed people and infrastructure, access routes, operational sectors, water supplies, command locations, engaged resources, and actions in progress. These elements are represented through a common graphical chart based on standard colors, symbols, lines, and shapes. This shared language enables the COS, command-post officers, sector commanders, authorities, and partner services to understand the main operational information rapidly.

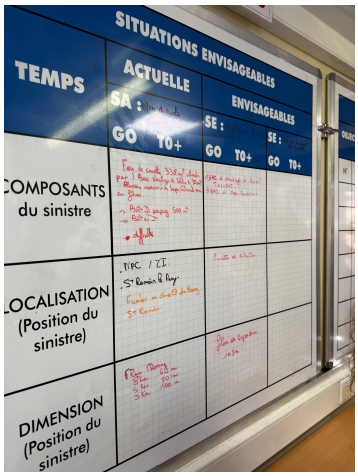
The SITAC must remain dynamic. The COS updates it progressively as the operation develops, using information received from the renseignements, action, moyens, and anticipation functions, as well as from sector commanders, the CODIS, partner services, and technical experts.Information that is no longer valid must be corrected so that the representation continues to reflect the most recent understanding of the incident.

In the command-post configuration observed during this study, the SITAC is maintained on a dedicated erasable whiteboard. Digital mapping and visualization tools, such as CRIMSON, may provide complementary information, but they do not replace the physical working board.

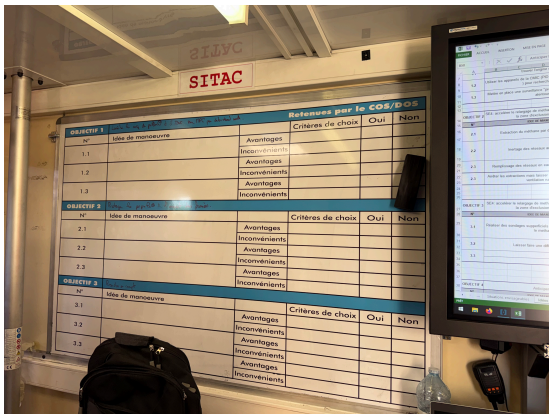
Figure 5 illustrates this progression.




(a) Graphical SITAC representing the intervention area, hazards, resources, and actions. Source: SDIS 26, reproduced from Rescue18 [29].




(b) Comparison of the current situation with possible future situations. Photograph by the author.




(c) Comparison of operational objectives and *"idées de manœuvre"*. Photograph by the author.

**Figure 5**: Command-post supports illustrating the progression from the current operational picture to possible future situations and operational choices. Panels b and c were photographed by the author during a simulation observed as part of this study.

Figure 5a represents the intervention spatially. It brings together the location of the incident, the building configuration, hazards, access points, water supplies, engaged resources, and command information. It consequently gives the command team a shared answer to the questions: what is happening, where is it happening, and what is currently being done?

The next two supports extend this picture. The board in Figure 5b introduces time by comparing the *"situation actuelle"* with several *"situations envisageables"*. The board in Figure 5c continues the reasoning by allowing objectives and alternative *"idées de manœuvre"* to be recorded and compared.

### 2.4.2 From the SITAC to the Elements of Anticipation

The three supports shown in Figure 5 illustrate the complete progression followed inside the command post. The SITAC first describes the intervention as it is currently understood. This picture is then projected into several possible futures, from which operational objectives, maneuvers, and

tasks can be prepared.

The terminology used throughout this study follows the ENSOSP anticipation method and the internal definitions and reference documents supplied by Capitaine Yoan, operational supervisor at SDIS 42 [24], [25], [30]. The different elements are defined as follows:

- **"Sinistre":** the hazardous phenomenon that is developing and requires an emergency response, such as a fire, explosion, chemical leak, collapse, or natural disaster.
- **"Situation actuelle" (SA):** the known state of the incident at a given moment. It describes the phenomenon, its location and dimensions, the exposed elements, the resources engaged, and the actions in progress.
- **"Évolution prévisible du sinistre" (EPS):** a credible unfavorable development of the incident without considering the effects of firefighter actions. It may result from the characteristics of the phenomenon, the site, environmental conditions, operational knowledge, and previous experience.
- **"Événement non souhaité" (ENS):** an event that could worsen the situation or cause harm to people, property, infrastructure, or the environment. Examples include an explosion, collapse, sudden weather change, loss of containment, or interruption of the water supply.
- **"Situation envisageable" (SE):** a coherent possible future situation resulting from the combination of an EPS and one or more ENS. Several SE may be considered because operational anticipation does not seek to predict one certain future.
- **Objective of the COS:** the operational effect that the COS wants to obtain in response to a validated SE, such as protecting exposed people, preventing propagation, stabilizing the incident, or limiting its consequences.
- **"Idée de manœuvre" (IM):** a general way of acting proposed to achieve an objective while considering the incident, risks, available time, constraints, and resources. Several IM may be associated to the same objective.
- **"Différentes tâches à accomplir" (DTA):** the concrete actions required to implement the selected IM. They translate the general maneuver into operational tasks, responsibilities, resources, locations, and priorities.

Together, these elements form a continuous reasoning chain. The SA provides the starting point. The EPS and ENS are used to construct several SE. Once a SE has been retained, an objective is defined, alternative IM are compared, and the selected maneuvers is finally translated into DTA.

This is the part of the operational process that Anticip-42 is intended to support. The system takes the current information communicated by the anticipation officer from the first part of the

SITAC and uses it to suggest the remaining elements of the chain: possible SE, corresponding COS objectives, alternative IM, and associated DTA.

The system does not generate the complete chain as one automatic decision. Each stage is reviewed by the anticipation officer and submitted through the "chef PC" to the COS. The COS validates, modifies, or rejects the proposed element before the process continues to the next stage. A validated SE is therefore required before objectives are proposed; an objective must be validated before IM are developed; and an IM must be selected before the corresponding DTA are prepared.

In this way, Anticip-42 assists the anticipation officer without changing the existing distribution of responsibilities. The system prepares structured proposals, the anticipation officer verifies their operational relevance, and the COS retains responsibility for every decision.

## 2.5 Technical Foundations of AI-Assisted Decision Support

The previous sections described how operational anticipation is performed within the command post. This section introduces the technical concepts needed to understand how Anticip-42 supports that work. It focuses on language models, document retrieval, structured generation, human-in-the-loop workflows, evaluation, and communication between system components.

### 2.5.1 AI and Large Language Models in Decision Support

Artificial intelligence can support decision-making without making the final decision. Depending on its purpose, an AI system may organize information, identify patterns, estimate possible developments, compare alternatives, or prepare recommendations for a human operator. In emergency response, this support must remain understandable and reviewable because the available information may be incomplete and the situation may change rapidly [18].

Large Language Models (LLMs) are AI models designed to process and generate language. Most recent LLMs are based on the Transformer architecture [31] and can perform tasks such as summarization, information extraction, question answering, and text generation [32].

An LLM generates its response from patterns learned during training and from the context supplied with the request. A fluent response is therefore not necessarily correct. The model may omit relevant information or produce unsupported statements, a problem commonly described as hallucination [33]. For this reason, an LLM should not be treated as an authoritative source of operational knowledge.

LLMs may be accessed through remote services or deployed locally. Local deployment offers greater control over sensitive information and system availability, but it also introduces hardware and response-time constraints. The reasons for choosing local deployment in this project are discussed in the architecture and implementation chapters.

### 2.5.2 Retrieval-Augmented and Structured Generation

Retrieval-Augmented Generation (RAG) connects a language model to an external collection of documents. Before generating a response, the system retrieves sections related to the current request and provides them to the model as context [34]. This allows the response to draw on selected operational documents instead of relying only on knowledge learned during model training.

Retrieval commonly uses embeddings, which represent the meaning of a text section as a numerical vector. Sections with similar meanings are located close to one another in the vector space, allowing a query to retrieve semantically related material [35]. Multilingual embeddings can also support retrieval across documents and queries written in different languages [36].RAG can improve documentary grounding, but it does not guarantee correctness. A relevant section may not be retrieved, and the model may misinterpret the material it receives. The source of retrieved information should therefore remain visible so that the user can examine the evidence supporting a generated proposal.

The model also requires clear instructions concerning the task and the expected output. This is commonly referred to as prompt engineering [37]. In a software workflow, structured output is often more useful than unrestricted text. A format such as JSON allows generated information to be validated, stored, displayed, and passed to another stage. However, a structurally valid response may still contain incorrect or unsupported information. Format validation does not replace content review.

### 2.5.3 Dynamic Human-in-the-Loop Workflows

A complex decision process cannot always be handled through one request to an AI model. It may contain several dependent stages in which the result of one stage becomes part of the input to the next. New operational information may also require an earlier proposal to be reconsidered.

A dynamic workflow maintains the state of this process and records the relationships between its stages. When information changes, the affected outputs can be reviewed or regenerated. This does not require the AI model to learn continuously; it can be achieved through state management, versioning, and controlled regeneration.

Human-in-the-Loop (HITL) interaction means that a human remains involved in producing, reviewing, or using an AI-generated result. The user may correct an intermediate output, reject a proposal, choose among alternatives, or make the final decision. The appropriate level of involvement depends on the consequences of error and the expertise required to assess the output [38].

Effective human–AI interaction should allow users to understand the system's limitations and

correct its outputs [39]. Human involvement must be meaningful rather than limited to confirming an answer that cannot be examined. This is particularly important for Anticip-42, where officers must remain responsible for interpreting the situation and validating proposals before they affect later stages of the workflow.

### 2.5.4 Reliability and Evaluation

The quality of an AI-generated proposal cannot be assessed from its writing style alone. A clearly written response may still be incomplete, operationally unsuitable, or unsupported by the available documents. Evaluation may therefore consider technical performance, retrieval quality, generated content, and interaction with the intended user. Technical evaluation examines factors such as response time and failures. Retrieval evaluation examines whether relevant documents were selected. Content evaluation considers completeness, coherence, and correspondence with the retrieved evidence. User-oriented evaluation considers whether the result is understandable, editable, and compatible with the operational workflow.

RAG evaluation similarly distinguishes the quality of retrieval from the relevance and faithfulness of the generated response [40]. Because several operational proposals may be defensible, automated measures may need to be combined with scenario-based testing and review by domain specialists.

### 2.5.5 Services, APIs, and External Data Sources

A modular system divides its functions among components that communicate through defined interfaces. An Application Programming Interface (API) specifies how one component requests information or an action from another and how the response is structured. Web APIs commonly exchange structured data using formats such as JSON.

Some APIs connect internal components, while others provide access to external services. In emergency response, external services may supply weather, geographic, or mapping information that contributes to the operational picture. These services avoid the need to reproduce existing capabilities, but their availability, response time, and data quality remain outside the system's direct control.

The system must therefore validate external data, retain its source and retrieval time, and handle unavailable services without stopping the complete workflow. These concerns influence the services and interfaces examined through SQ4.

The concepts presented in this section provide the technical foundation for the proposed system. The following section examines how previous studies have identified emergency-response requirements, translated them into system architectures, connected heterogeneous components,

and preserved human control.

## 2.6 Requirements and Architectures for Proactive Emergency Decision Support

The previous section introduced the technologies that can support AI-assisted decision-making. Their inclusion in an emergency-response system, however, must be guided by the work performed by its intended users. This requires an understanding of what proactive support means, which requirements the system must satisfy, how those requirements can be represented through system models, and how the resulting components can exchange information.

### 2.6.1 Requirements for Emergency-Response Systems

Emergency-response systems should be designed from the work and information needs of their intended users. The WORKPAD project used interviews, scenarios, and task analysis to identify requirements before developing an architecture connecting control rooms and field teams [41]. Prasanna et al. similarly examined the information and interaction needs of different firefighter roles and translated them into a layered information-systems architecture [42].

This research shows that emergency personnel do not all require the same information or interface. Information must be selected according to the user's role, current task, and operational environment [43], [44]. Systems must also distinguish confirmed information from uncertainty, preserve the source of information, and support work under changing conditions.

These requirements cannot be applied directly to Anticip-42 without considering its operational context. SQ1 therefore examines how existing requirements must be adapted to the responsibilities, information flows, and human-validation points of the anticipation function at SDIS 42.

### 2.6.2 System Modeling and Architectural Specification

Requirements describe what a system must support, but system models are needed to show how its users, components, and information exchanges work together. WORKPAD demonstrates how emergency scenarios and user requirements can guide the development of a distributed architecture [41]. The firefighter architecture proposed by Prasanna et al. similarly connects validated information needs to system functions and interfaces [42].

More recent work applies systems engineering to human-centered firefighting technologies. Innocente et al. identify stakeholders and requirements before describing the human, software, and hardware components of a UAV-supported wildfire-response system [45]. Although its purpose differs from Anticip-42, the study illustrates how operational needs can be translated into a system-

level design.

System modeling in emergency response is therefore not new. The purpose of SQ2 is to apply it to the specific progressive anticipation process examined in this study and to connect that process to the components and information exchanges required by Anticip-42.

### 2.6.3 Architecture, Interoperability, and Human Control

Emergency-response platforms often combine independently developed data sources and technical services. Interoperability allows these components to exchange information and use it correctly. The SAFERS wildfire platform, for example, connects a dashboard with weather, detection, forecasting, geospatial, and decision-support services through a modular service-oriented architecture [46].

IRIS-AI also separates incident-state processing, resource prediction, and resource allocation into distinct components. This modular structure allows parts of the system to be adapted while leaving the final allocation decision to the incident commander [47].

Human control must be considered as part of the architecture rather than added only at the interface. A hybrid human–AI framework for fire-related risk reduction emphasizes explanation, uncertainty communication, human validation, and the recording of operator overrides [48].

For Anticip-42, interoperability concerns the exchanges among the dashboard, backend, language model, document-retrieval mechanism, data storage, and contextual services. Human control concerns the points at which the anticipation officer reviews a proposal and the COS retains decision-making authority. These architectural concerns provide the basis for SQ3 and SQ4.

Table 1 summarizes two groups of related system-design research. The first includes WORK-PAD, the firefighter information architecture proposed by Prasanna et al., and the human-centered wildfire-response design of Innocente et al. The second includes the SAFERS platform, IRIS-AI, and a hybrid human–AI decision framework.

Table 1: Comparison of system-design approaches for emergency response.

| Approach and examples | Principal input | Principal output | Difference from the present study |
|---|---|---|---|
| Requirements and system architecture: WORKPAD [41]; firefighter information architecture [42]; and the UAV-supported wildfire-response design [45] | User activities, operational scenarios, information needs, and system constraints | Requirements specifications, conceptual models, and system architectures | The reviewed studies translate operational needs into system designs, but they do not model the complete SDIS anticipation process examined in this study, from SITAC to SE, COS objectives, IM, and DTA. |
| Modular platforms and human–AI frameworks: SAFERS [46]; IRIS-AI [47]; and the hybrid human–AI framework [48] | Incident information, contextual data, technical services, and human input | Integrated services, AI recommendations, and human–AI decision frameworks | The reviewed approaches support modular integration and human control, but they do not implement the progressive anticipation and validation workflow examined in this study. |

## 2.7 Existing Related Studies

Research on emergency decision support approaches anticipation from several perspectives. The following review distinguishes these perspectives according to the information they use, the outputs they produce, and the operational decisions they are intended to support.

### 2.7.1 Fire-Behavior Forecasting

Fire-behavior research uses physical and data-driven models to estimate how a fire may develop. Traditional methods include empirical correlations, zone models, and computational fluid dynamics, while recent systems increasingly use sensor data and machine learning to provide faster predictions [49], [50].AID-Fire combines sensor data, predictive models, cloud processing, and a digital-twin interface to estimate fire conditions in near real time [51]. Related models have been developed to forecast temperature and provide warnings of events such as flashover [52].

These systems are complementary to Anticip-42 rather than direct alternatives. Their fire forecasts could provide useful contextual information when possible situations are prepared. However, Anticip-42 addresses a broader command problem by combining incident development, operational resources, doctrine, and human validation. It does not replace specialized fire-behavior models, and such models were not implemented in the current proof of concept.

### 2.7.2 Situational Awareness and Operational Information

Several studies examine the information required by firefighters and incident commanders. Research on building fires and other emergency situations shows that information should be selected according to the user's role, current task, and operational goal rather than presented as an unfiltered collection of data [43], [44].Other systems organize information from multiple sources into a common operational picture. These approaches include firefighter ontologies, event-driven crisis models, and mobile decision-support tools [53], [54], [55]. Research has also examined the usability of digital tools for creating the French SITAC [56].

This research provides an important foundation for the present study because a reliable operational picture is required before anticipation can begin. The outputs of a SITAC or common operational picture could therefore be used as inputs to Anticip-42. The difference is that Anticip-42 continues beyond the representation of the current situation by supporting the successive preparation of SE, COS objectives, IM, and DTA. It complements situational-awareness systems rather than replacing them.

### 2.7.3 Intervention Analytics and Resource Allocation

Historical incident data have been used to forecast emergency demand, operation duration, and the likelihood that an incident will require escalation [57], [58], [59]. These models can support planning or monitoring, but their outputs are generally numerical estimates or classifications rather than operational proposals.

Resource-allocation research examines how limited emergency units can be assigned to incidents [60]. IRIS-AI extends this approach by using the evolving state of an incident to predict the characteristics of the resources that may be needed. A separate optimization component then matches those characteristics to available resources, while the final decision remains with the incident commander [47].

These approaches could support specific parts of Anticip-42. Predictions of incident development could help prepare possible situations, while resource-allocation models could help assess whether a maneuver is feasible with the available resources. However, they answer narrower questions and do not represent the complete anticipation workflow. Anticip-42 could use their outputs as additional inputs, but it does not replace their specialized prediction or optimization functions.

### 2.7.4 Generative AI and Operational Planning

Recent research has explored the use of generative AI for emergency planning and access to operational knowledge. DisasterResponseGPT generates candidate response plans from

a scenario and allows them to be revised through dialogue [61]. Ko et al. developed a generative decision-support system that provides field commanders with condensed information from operational manuals [62].

These studies demonstrate the potential of language models for plan drafting and procedural support. However, their evaluations are based mainly on fictional or simulated scenarios, and they do not implement the progressive French anticipation sequence examined in this study. They also do not combine local deployment, retrieval from an authorized SDIS corpus, contextual services, and human validation at each stage.

### 2.7.5 French SDIS Initiatives

French fire and rescue services are also developing digital and AI-based tools. NexSIS 18–112 provides national infrastructure for alert processing and operational management [63]. NexVA supports the preparation of intervention reports, while AIDCOS provides mobile access to operational doctrine [64].

MultiFireScan uses AI-assisted analysis of aerial imagery to update wildfire maps [65]. Other practitioner work proposes dashboards connecting operational systems, weather information, geographic data, and sensors [66]These initiatives form the practical digital environment in which a system such as Anticip-42 could operate. NexSIS could provide operational data, AIDCOS could support access to doctrine, and tools such as MultiFireScan could contribute updated geographic information. Their outputs could therefore be connected to the proposed system in future work.

Anticip-42 is not intended to replace these services. It examines how information from such systems could be brought together and used to prepare successive anticipation proposals under human control. Among the publicly documented French initiatives reviewed, no evaluated tool was identified that supports the complete progression from SITAC to SE, COS objectives, IM, and DTA.

Table 2 compares the main operational applications identified in the literature. It shows what information each research stream uses, what it produces, and how its purpose differs from the progressive anticipation process examined in this study.

Table 2: Comparison of operational research streams related to Anticip-42.

| Research stream | Principal input | Principal output | Difference from the present study |
|---|---|---|---|
| Fire-behavior forecasting [49], [51], [52] | Sensors, images, and simulated fire scenarios | Temperature estimates, fire-state predictions, or warnings | Concentrates on the physical development of a fire rather than the complete command situation. |
| Situational awareness [43], [53], [54] | Field reports and heterogeneous operational data | Selected information, a crisis model, or a common operational picture | Organizes the current situation but does not prepare the complete French anticipation chain. |
| Intervention analytics [57], [58], [59] | Historical incidents and contextual variables | Intervention demand, operation duration, or escalation class | Produces an estimate or classification rather than an operational proposal. |
| Resource allocation [47], [60] | Incident state, resource characteristics, and operational constraints | A resource schedule or recommendation | Focuses on the resources to assign rather than the comparison of possible future situations and maneuvers. |
| Generative planning [61], [62] | Scenario descriptions, objectives, manuals, and prompt instructions | A candidate response plan or procedural guidance | Does not implement the progressive French SDIS anticipation process or its specific validation stages. |
| French operational projects [63], [64], [65], [66] | Voice reports, doctrine, aerial imagery, sensors, and operational systems | Reports, document access, maps, or prospective dashboards | Among the publicly documented French initiatives reviewed, no evaluated tool was identified for supporting the anticipation officer from SITAC to SE, COS objectives, IM, and DTA. |

## 2.8 Research Gap: Supporting the Anticipation Officer

Existing research provides methods for identifying emergency-response requirements, architectures for firefighter information systems, modular platforms, AI-based recommendations, and human-controlled decision frameworks [41], [42], [45], [46], [47], [48]. Other studies address fire prediction, situational awareness, intervention analytics, resource allocation, and generative planning.

However, within the scientific and institutional sources reviewed, no evaluated system was identified for the complete progressive anticipation process examined at SDIS 42. Existing systems do not combine the current operational picture, command-post observations, local documentation, and external contextual information to successively prepare SE, COS objectives, IM, and DTA while preserving human validation and source traceability.

The gap is therefore not the absence of AI or system architecture in firefighting. It concerns the integration of existing technical approaches with a specific command-post role and operational reasoning process. Anticip-42 addresses this gap through a locally deployable, human-in-the-loop proof of concept that produces editable and traceable proposals without replacing the established chain of command.

# 3 METHODOLOGY

This chapter presents the methodological framework adopted in this study. The research did not begin with the implementation of an AI application. It began with the need to understand and formalize an operational process that is currently performed by experienced officers under demanding conditions. Only after this process had been studied could an appropriate technical artifact be designed.

The study therefore combines the investigation of an operational problem with the design and evaluation of a computational artifact. Design Science Research (DSR) was selected as the principal methodology because it supports the development of artifacts intended to address real-world problems while also producing knowledge about their requirements, design, and limitations [67], [68].

The resulting artifact, Anticip-42, is a proof of concept rather than a deployable operational system. Its purpose is to investigate how the anticipation process can be formalized, how recommendations can be generated progressively while preserving officer control, and how a generic technical architecture can use different sources of operational knowledge. Consequently, the evaluation focuses on requirements coverage, technical feasibility, workflow coherence, adaptability, and professional feedback. It does not attempt to demonstrate improved operational outcomes during real emergencies.

## 3.1 Design Science Research

Design Science Research connects three complementary forms of work. The *relevance cycle* connects the study to its operational environment and stakeholders. The *rigor cycle* grounds the design in scientific literature, doctrine, professional knowledge, and existing technical methods. Between them, the *design cycle* supports the repeated construction, evaluation, and refinement of the artifact [69].

In this study, the operational environment includes SDIS 42 officers, command-post practices, exercises, information constraints, and the requirements associated with operational anticipation. The knowledge base includes scientific literature, national doctrine, operational documentation, technical standards, and the professional definitions supplied for the project. Anticip-42 is developed between these two spaces, as illustrated in Figure 6.

The study does not treat these cycles as a rigid sequence. New information obtained during an exercise, a discussion with an officer, a scenario walkthrough, or a technical test may reveal that an earlier requirement or design decision must be reconsidered. The methodology therefore allows the study to move backwards as well as forwards.

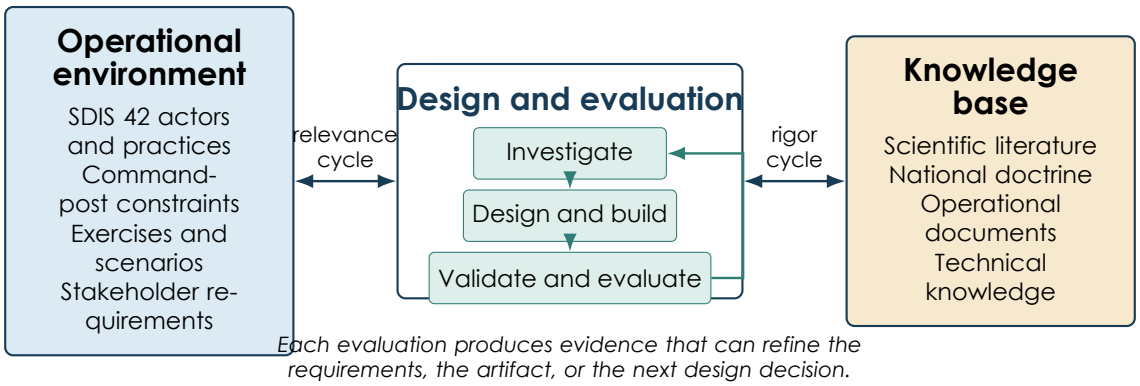



**Figure 6**: Application of the Design Science Research cycles to this study. Adapted from the three-cycle view of Hevner and the regulative cycle of Wieringa; the project-specific content is an original synthesis [67], [69].

## 3.2 Application of the DSR Cycle

The Design Science Research cycle was applied through five recurring activities. These activities provide the common structure used for each of the four iterations presented in Chapter 4.

1. **Problem Investigation:** understand the operational context, actors, constraints, information exchanges, and decision boundaries.
2. **Treatment Design:** translate the investigated problem into requirements, process models, interface designs, workflow rules, and architectural decisions.
3. **Treatment Validation:** examine whether the proposed design is consistent with operational practice, stakeholder expectations, and the defined system boundary before full implementation.
4. **Treatment Implementation:** transform the validated design into a working proof of concept.
5. **Implementation Evaluation:** examine the implemented artifact through scenario walkthroughs, functional and integration tests, structured-output controls, technical measurements, and professional feedback.

The cycle was not treated as strictly linear. Evaluation could reveal an incomplete requirement, an unsuitable interaction, or a technical limitation. In such cases, the study returned to problem investigation or treatment design. This iterative structure is appropriate because operational understanding and technical feasibility developed together [70].

Chapter 4 applies this structure to each iteration. It explains what was investigated, designed, validated, implemented, and evaluated. Detailed measurements, model comparisons, and findings are presented in the Results and Findings chapter.

Because Anticip-42 remains a proof of concept, evaluation concerns feasibility, operational coherence, workflow behavior, interoperability, and technical performance. It does not attempt to demonstrate improved emergency outcomes or readiness for deployment during a real incident.

## 3.3 Research and Iteration Plan

The study was organized into four connected iterations. Each iteration was primarily associated with one supporting research question and produced an artifact required by the following iteration. Together, the four iterations provide the evidence used to answer the main research question:

> *How can a proactive intervention system for firefighters be designed and implemented using human-in-the-loop AI to support operational decision-making during emergency response?*

Table 3: Mapping between supporting research questions, iterations, methods, and artifacts.

| SQ | Iteration | Principal method | Principal artifact |
|---|---|---|---|
| SQ1 | Iteration 1 | Documentary analysis, stakeholder exchanges, exercise observation, and requirements elicitation | Requirements baseline, prioritized user stories, acceptance criteria, and defined system boundary |
| SQ2 | Iteration 2 | UML, BPMN, data and architectural modelling, structured walkthroughs, and requirements traceability | Comprehensive operational, interaction, data, and architectural specification |
| SQ3 | Iteration 3 | Human-in-the-loop interaction design, high-fidelity prototyping, Streamlit feasibility prototyping, and scenario walkthroughs | Dynamic workflow for progressive generation, officer review, correction, validation, and state propagation |
| SQ4 | Iteration 4 | Service and interface implementation, API and integration testing, local-model execution, RAG evaluation, and exploratory technical evaluation | Integrated React–FastAPI proof of concept, interoperable services, evaluation dataset, summaries, charts, logs, and reports |

The iterations are cumulative. Iteration 1 determines what the system must support. Iteration 2 translates these requirements into formal models and an architectural specification. Iteration 3 investigates how human operators and AI components can collaborate through a progressive workflow. Iteration 4 implements the required services and interfaces, then evaluates their technical behavior across representative scenarios.

The main research question is therefore answered through the combined evidence from all four iterations rather than by one isolated implementation test.

## 3.4 Study Material and Data Collection

Data collection was organized around four complementary sources. This was intended to reduce the risk of defining the artifact from a single document, individual opinion, or technical assumption.

### 3.4.1 Documentary Analysis

The first source consisted of operational and technical documents. Public material included national *"Guides de doctrine opérationnelle"* (GDO), *"Guides de techniques opérationnelles"* (GTO), ENSOSP training material, experience reports, and technical publications related to industrial and environmental risks [12], [13], [14], [15], [28].

A limited collection of operational working material was also supplied by Capitaine Yoan, the operational supervisor of the project at SDIS 42. This material contained definitions, examples, ratios, vehicle information, and working documents related to the anticipation process [30]. These documents were used to understand the terminology and expected structure of the operational outputs. They were not treated as a complete representation of all SDIS 42 documentation.

Document analysis served two purposes. First, it supported the definition of the operational process and system requirements. Second, a selected subset of the documents formed the documentary corpus used to investigate Retrieval-Augmented Generation.

### 3.4.2 Operational Observation and Requirement Elicitation

The second source consisted of interactions with the project stakeholders. Working sessions with the SDIS supervisor were used to clarify the roles of the command-post actors, the sequence of the anticipation process, the information required at each stage, and the limits of acceptable AI support.

Observations made during operational training exercises provided additional context. They made it possible to examine how information is exchanged, rewritten, corrected, and displayed when an incident evolves. Particular attention was given to difficulties related to information traceability, the speed of updates, and the effort required to memorise and verbally reproduce multiple observations.

The observations were not treated as a statistical sample of SDIS operations. They were used as qualitative evidence for requirement elicitation and scenario construction. Interpretations derived from them were compared with doctrine and reviewed with the operational supervisor. This form of stakeholder verification is consistent with case-study guidance recommending the use of several evidence sources and the validation of interpretations with participants [71].

### 3.4.3 Scientific Literature

The third source was the scientific and professional literature reviewed in Chapter 2. It was used to identify existing approaches to fire prediction, resource recommendation, situational awareness, emergency planning, information retrieval, and Human-AI collaboration. The literature supported the theoretical foundation of the artifact and helped position the research gap addressed by the study.

### 3.4.4 Technical Evidence

The fourth source consisted of evidence produced during development and testing. This included API responses, retrieved document passages, similarity scores, generated structured outputs, execution logs, response times, error messages, and the corrections made during scenario walkthroughs.

Technical evidence was used to examine feasibility and reliability within the proof-of-concept environment. It was not interpreted as evidence that the system was ready for operational deployment.

## 3.5 Research Material

The research material consisted of documentary, operational, and technical elements used to design and assess the artifact.

### 3.5.1 Operational Knowledge Corpus

The documentary corpus assembled for this study included several types of material obtained from operational, institutional, and technical sources.

- national doctrine and operational guidance
- ENSOSP training and anticipation material
- public experience and incident reports
- industrial-risk and dangerous phenomena documentation
- working definitions and examples supplied for the project
- resource, vehicle, and ratio documents
- documents associated with different crisis families, including fires, chemical or industrial events, and natural hazards.

The available files were heterogeneous. They included PDF documents, word processing files, spreadsheets, and scanned documents. This heterogeneity was relevant to the proof of concept because the documentary retrieval service needed to extract, normalise, index, and preserve metadata from different forms of operational knowledge. It informed the architectural specification developed in SQ2 and the implementation of interoperable document-processing and retrieval services in SQ4.

The full internal documentary environment of SDIS 42 was not available. Access to some operational documents required an institutional agreement that was still being negotiated during the study. The corpus should therefore be understood as a feasibility corpus rather than a complete operational knowledge base.

### 3.5.2 Representative Crisis Scenarios

The proof of concept was examined through representative scenarios rather than live emergency data. The principal technical comparison concerned a fire in an industrial warehouse. A second scenario concerned a methane release from a pressurised pipeline, propagation through underground networks, and the risk of ignition or explosion.

These scenarios were used to test the progressive human-in-the-loop workflow defined in SQ3. In each case, the officer could provide or update operational information, inspect contextual and documentary inputs, review generated proposals, correct them, and decide which information could be retained for the following stage.

The scenarios also supported SQ4 by testing whether the same interface, workflow services, local language-model integration, retrieval pipeline, and data structures could be reused for different incident descriptions without rebuilding the complete proof of concept.

The scenarios were not intended to demonstrate that Anticip-42 can manage all types of emergencies. They provided bounded technical and interaction evidence under the selected test conditions only.

### 3.5.3 Operational and Contextual Inputs

The inputs examined during the study covered the main information needed to describe the incident and its operational context. These included:

- information entered by an officer about the current incident
- observations and updates received from the operational functions
- the location of the incident

- weather conditions
- nearby exposed people, infrastructure, services, and access routes
- available or engaged resources
- retrieved passages from the documentary corpus.

No live command decision was delegated to the proof of concept. The information used during development and testing was limited to simulated, publicly available, anonymised, or specifically authorised material.

## 3.6 Tools and Technologies

The technologies were selected according to four criteria: compatibility with the proof-of-concept objective, support for local processing, modularity, and the possibility of replacing individual components without redesigning the complete architecture. The project evolved during its development. Streamlit was first used as a deliberately temporary feasibility environment to determine whether local language models could support the progressive generation workflow. Once the pipeline had become sufficiently consistent under the tested conditions to justify further implementation, the actual dashboard was developed with React and TypeScript. The exact configurations and design decisions made during this evolution are presented in Chapter 4.

- **Python:** Python was the principal language used for document processing, backend services, language-model interaction, data transformation, contextual-data retrieval, and the first interface prototype.
- **Streamlit and PyDeck:** Streamlit was used during the early feasibility phase to connect the local LLM pipeline rapidly to input forms, session state, retrieval, and the SE–objectives–IM–DTA sequence. PyDeck supported the initial geographical representation of the incident and nearby contextual elements. This application was an experimental vehicle for testing whether the AI workflow could function; it was not the intended operational dashboard.
- **React, TypeScript, and Vite:** Following the feasibility phase, the actual dashboard was implemented with React. TypeScript supported explicit data structures and safer communication with the backend, while Vite provided the frontend development and build environment. The current interface separates the different stages of the operational workflow into dedicated pages and uses an embedded OpenStreetMap view to present the incident location and nearby stakes.

- **FastAPI, Uvicorn, Pydantic, and HTTPX:** FastAPI was used to separate the user interface from the application, retrieval, and generation logic. Uvicorn served the backend application, while Pydantic defined and validated the expected structure of incoming requests. HTTPX supported communication with Ollama and the external contextual services.
- **Ollama and local Large Language Models:** Ollama provided a local inference interface for evaluating different instruction-tuned language models. Local execution allowed the study to investigate the processing of operational information without transmitting the complete prompts and documentary context to a commercial cloud-based LLM service.
- **Sentence Transformers:** A multilingual sentence-embedding model transformed document passages and user queries into numerical vectors. These representations supported semantic comparison between an operational request and the documents stored in the knowledge base.
- **PostgreSQL and pgvector:** PostgreSQL stored extracted document passages, document types, filenames, and vector representations. The pgvector extension supported vector-distance calculations for semantic retrieval. The retrieval process was complemented by keyword matching and the prioritization of documents associated with the current stage of the workflow.
- **Document-processing libraries:** The document-indexing pipeline used `pdfplumber`, `python-docx`, `pandas`, `openpyxl`, and `xlrd` to extract content from PDF, Word, and spreadsheet files. Additional extraction functions supported ODT, ODS, TXT, and Markdown documents. An optional OCR fallback based on `pdf2image` and `pytesseract` was implemented for scanned PDF documents from which text could not be extracted directly.
- **External contextual services:** Open-Meteo provided weather information. OpenStreetMap and the Overpass API supplied geographical elements such as water points, sensitive facilities, infrastructure, and access routes. Nominatim and the Base Adresse Nationale supported reverse geocoding, while Géorisques provided information concerning nearby classified industrial installations.
- **Docker and Git:** Docker Compose provided a reproducible configuration for PostgreSQL, pgvector, Redis, and persistent database storage. In the current proof of concept, Docker was used for the data infrastructure; the React frontend, FastAPI backend, document indexer, and Ollama service were started separately. Git was used for source control and for maintaining the history of the successive prototype and implementation changes.

The selection of these technologies does not imply that they constitute the only possible implementation of the proposed system. They were selected within the constraints of the proof

of concept to investigate the technical feasibility of the architecture and workflow. The transition from Streamlit to React also illustrates this modularity: the interface could be replaced while the FastAPI backend, local language-model services, retrieval pipeline, and external contextual services remained separated from the presentation layer.

## 3.7 Evaluation Strategy

Evaluation is central to Design Science Research because it examines whether an artifact addresses the investigated problem as well as whether it runs technically. The evaluation strategy combined formative evaluation during design and implementation with exploratory technical evaluation of the integrated proof of concept [72].

### 3.7.1 Formative Evaluation

Formative evaluation took place throughout the four iterations. Requirements, process models, interface proposals, workflow rules, and architectural decisions were reviewed as they were produced. Feedback was used to identify misunderstandings, missing information, unsuitable terminology, unclear allocation of responsibility, and interaction choices that did not correspond to the intended operational process.

This formative evaluation was particularly important for SQ1, SQ2, and SQ3. The requirements had to preserve the operational boundary of the system; the models had to distinguish human authority from software support; and the interactive workflow had to maintain meaningful officer participation.

### 3.7.2 Integrated Proof-of-Concept Evaluation

The integrated artifact was evaluated using representative scenarios and predefined criteria. The evaluation covered the following dimensions:

1. **Operational alignment:** whether the workflow, terminology, information dependencies, and validation points corresponded to the anticipation process identified during requirements elicitation.
2. **Progressive coherence:** whether information validated or modified at one stage was correctly reflected in later inputs and proposals.
3. **Human control:** whether generated proposals remained editable and whether an officer could accept, reject, modify, or request a new proposal before the workflow continued.

4. **Knowledge grounding and retrieval:** whether the expected documentary sources were retrieved, whether source metadata was visible, and whether the retrieval process remained available for each stage.
5. **Technical interoperability:** whether the React interface, FastAPI services, local models, retrieval service, persistence layer, and contextual services could exchange information through the defined interfaces.
6. **Cross-scenario adaptability:** whether the same technical workflow could be applied to the industrial-fire and methane-release scenarios without rebuilding the architecture.

### 3.7.3 Local-Model Technical Comparison

Several locally deployed language models were tested using the same progressive generation pipeline. The automatic evaluation checked whether each model produced the expected number of possible situations, COS objectives, maneuver ideas, and detailed tasks. It also checked whether the required fields were completed, whether the outputs remained consistent across the successive stages, and whether quality issues or malformed responses were detected.

In addition, the evaluation recorded automatic repair attempts, documentary retrieval coverage, the executability of detailed tasks, and the total execution time of the complete workflow.

The weighted automatic score assigned 30% to possible situations, 20% to COS objectives, 20% to maneuver ideas, and 30% to detailed tasks. This weighting reflects the importance of producing complete and usable outputs throughout the progressive workflow. However, the score mainly measures structure, completeness, and predefined quality-control conditions. It was therefore not interpreted as a direct measure of tactical correctness or operational safety.

### 3.7.4 LLM-as-Judge Assessment

A fixed local language model was used as an LLM judge to assess the quality of the generated outputs. The judge received the incident context and the generated situations, objectives, maneuver ideas, and detailed tasks. It did not know the name of the model that produced the output, its automatic score, or its previous ranking.

The assessment considered operational relevance, tactical coherence, realism, specificity, risk anticipation, task executability, grounding, and hallucination control. This semantic score was combined with the automatic score, task executability, reliability, and execution time.

This evaluation remains exploratory. Qwen3 30B was used as the judge but was also one of the tested models. Therefore, the LLM-as-judge results were used as additional evidence only and

do not replace evaluation by SDIS professionals.

### 3.7.5 Evaluation by Supporting Research Question

Table 4: Evaluation strategy for the supporting research questions.

| SQ | Evaluation approach | Main criteria |
|---|---|---|
| SQ1 | Review of requirements and user stories against doctrine, exercise observations, stakeholder explanations, and the system boundary | Process coverage, correctness of roles, preservation of authority, traceability, prioritization, and acceptance criteria |
| SQ2 | Structured walkthroughs and cross-model consistency checks across UML, BPMN, data, interaction, and architectural models | Requirements coverage, consistency of actors and information objects, validation loops, responsibility separation, and traceability |
| SQ3 | Scenario walkthroughs of the human-in-the-loop prototype involving updates, generated proposals, corrections, selections, regeneration, and stage progression | Editability, preservation of officer input, propagation of validated information, visibility of failures, and validation clarity |
| SQ4 | API and integration tests followed by repeatable execution with different local models and representative scenarios | Interface compatibility, structured-output validity, retrieval coverage, task executability, failure handling, repair behavior, runtime, semantic quality, and component replaceability |

Professional feedback was treated as qualitative evidence rather than proof that the artifact was operationally safe. Similarly, successful completion of a scenario demonstrated technical feasibility only under the tested conditions.

## 3.8 Validity Measures and Ethical Considerations

Validity was considered throughout the design and evaluation of the study. Operational concepts were based on national doctrine, ENSOSP material, project documents, command-post observations, and stakeholder explanations. Traceability was maintained between the requirements, system models, design decisions, implementation, and evaluation criteria. The proof of concept was also tested using an industrial warehouse-fire scenario and a methane-release scenario, while documentary evidence, technical observations, automated measures, and formative stakeholder feedback were combined during the evaluation. The remaining threats to construct, internal, external, and conclusion validity are discussed in Section 6.3.

The study also considered the sensitive nature of emergency-response information. Language models and document storage were deployed locally to maintain greater control over operational material. External services were used only to retrieve contextual information, such as weather and geographic data. No personal or victim information was intentionally processed.

Anticip-42 was not deployed in a real command post and was not used to direct an actual intervention. All generated outputs remained proposals requiring human review.

# 4 ITERATIONS

This chapter explains how the methodological plan was applied through four successive design iterations. Each iteration follows the same structure: problem investigation, treatment design, treatment validation, treatment implementation, and implementation evaluation. The evaluation subsection states what was examined, how it was examined, and the immediate outcome that motivated the following iteration. Detailed measurements, comparisons, and research findings are presented in the Results and Findings chapter, while their broader interpretation is reserved for Discussion.

The iterations are cumulative. The requirements identified in Iteration 1 constrain the models produced in Iteration 2. Those models guide the human-in-the-loop workflow created in Iteration 3. Iteration 4 implements the services and interfaces needed to connect the workflow and conducts an exploratory technical evaluation of the integrated proof of concept.

Table 5: Progression of the four design iterations.

| Iteration | Starting question | Main output | RQ |
|---|---|---|---|
| 1 | What must the system support in its operational context? | Requirements and user stories | SQ1 |
| 2 | How can those requirements become a coherent system specification? | Operational, interaction, data, and architectural models | SQ2 |
| 3 | How should officers and AI components collaborate throughout the anticipation process? | Dynamic human-in-the-loop workflow and prototype | SQ3 |
| 4 | Which services and interfaces are needed to implement and connect the system? | Integrated proof of concept and exploratory technical evaluation | SQ4 |

## 4.1 Iteration 1: Operational Investigation and Requirements

The first iteration focused on understanding the operational need before defining the structure or technologies of Anticip-42. It established the requirements addressed by the following iterations and therefore contributed mainly to SQ1.

### 4.1.1 Problem Investigation

At the beginning of the project, the general objective was to support anticipation during emergency response. However, this objective was not yet detailed enough to guide the design of a software system. A clearer understanding of the operational work was required.

The anticipation process involves information provided by several operational functions. This information may change during an intervention and may be communicated through documents,

digital systems, meetings, or oral reports. The anticipation officer must combine these inputs to prepare possible situations, objectives, maneuver ideas, and detailed tasks. These proposals must then be reviewed by the appropriate officer or validated by the COS.

National doctrine, ENSOSP material, and the documents available for the project described the command structure and the main anticipation concepts [30]. However, they did not provide a direct software specification. They did not state precisely which information the system should receive, how that information should be updated, which outputs should be generated, or where human validation should take place.

The problem addressed in this iteration was therefore the absence of a clear and structured definition of what Anticip-42 should support in its operational context.

### 4.1.2 Treatment Design

The treatment planned for this iteration was a requirements study based on several complementary sources. The plan combined document analysis, meetings with the SDIS 42 operational supervisor, and observations made during operational exercises and simulations.

The documents were expected to provide the official terminology, command structure, operational concepts, and existing working practices. The meetings were planned to clarify information that could not be fully understood from the documents. The exercise observations were intended to show how information was exchanged and updated in a more realistic setting.

The collected information would then be organized as requirements and user stories. The requirements would describe the expected system behavior, while the user stories would express the main capabilities from the perspective of the operational user. A simple interface prototype was also planned to make these requirements easier to discuss and visualize.

This approach was intended to answer four main questions: what information the system should receive, what outputs it should prepare, who should interact with it, and which actions must remain under human control.

### 4.1.3 Treatment Validation

Before applying the plan, its scope and proposed activities were discussed with the operational supervisor. The combination of documents, meetings, and exercise observations was considered suitable because no single source provided a complete description of the anticipation process.

The planned focus on possible situations, objectives, maneuver ideas, and DTA was also checked against the expected work of the anticipation function. Particular attention was given to the role of the COS and to the requirement that generated proposals remain subject to human

review and validation.

The scope was limited to activities that could be represented and examined through a proof of concept. Functions requiring direct access to live SDIS systems, continuous operational data, or organizational decisions were identified as future requirements. The interface prototype was retained in the plan as a visual aid for evaluating whether the requirements could be represented as a clear workflow.

This validation established that the planned investigation was appropriate for identifying the requirements of Anticip-42 without making claims about its use during real emergency operations.

### 4.1.4 Treatment Implementation

The validated plan was then carried out through document analysis, meetings, exercise observations, and the preparation of an initial interface prototype. These activities produced the requirements baseline used throughout the remainder of the project.

**Operational investigation.** National doctrine, ENSOSP material, and the working documents available for the project were reviewed. These sources were compared with explanations provided during meetings with the operational supervisor and with observations made during exercises and simulations.

The investigation identified the main actors involved, the information they provide, and the successive outputs prepared during anticipation. It also showed that operational information may change during an intervention and that the system must allow this information to be corrected or completed.

The analysis confirmed that Anticip-42 should support a progressive workflow. Operational information is first collected and reviewed. The system may then assist with the preparation of possible situations, objectives, maneuver ideas, and DTA. Each result must remain visible and editable, and it must be validated before being used in the following stage.

**Requirements and user stories.** The identified needs were organized into a prioritized user-story backlog. The stories described the principal capabilities expected from Anticip-42 without linking them to a specific programming language, interface framework, or language model.

Table 6 summarizes the user stories developed during this iteration.

Table 6: User-story backlog developed during Iteration 1.

| ID | Expected system capability | Priority |
|---|---|---|
| US1 | Generate possible situations to support the anticipation officer in preparing for different developments of an incident. | High |
| US2 | Propose several maneuver ideas so that the COS can compare possible operational approaches. | High |
| US3 | Allow the anticipation officer to enter and modify the operational situation, orders, and constraints used by the system. | High |
| US4 | Record and analyze previous interventions so that available experience can support later operations. | High |
| US5 | Connect to existing SDIS data sources to reuse available information and reduce unnecessary manual entry. | High |
| US6 | Import images from drones, satellites, or the field to provide additional information about the intervention. | High |
| US7 | Generate proposed operational objectives and submit them to the COS for validation. | High |
| US8 | Generate a structured DTA for each selected maneuver idea to provide an initial basis for operational planning. | High |
| US9 | Print or export the generated tables for use in physical or digital form. | Medium |
| US10 | Provide an interactive map showing the intervention area, nearby stakes, and other relevant operational elements. | Medium |

The complete workbook also recorded conditions used to clarify the expected behavior of each user story. For example, entered information had to remain editable, generated objectives had to be submitted for validation, maneuver proposals had to be selectable or rejectable, and the resulting tables had to be exportable. These conditions guided the later design and implementation, but they did not imply that every function had already been developed.Figure 7 presents an extract from the user-story backlog, illustrating the identified requirements, their priorities, and the expected system behavior.

| Titre | USER STORY | | Critères d'acceptation | Priorité | ESTIMATION | DESCRIPTION |
|---|---|---|---|---|---|---|
| **Prédiction des situations envisageables** | En tant qu'officier anticipation, je veux que l'IA génère automatiquement les scénarios d'évolution du sinistre afin de préparer mes décisions sans perdre de temps. | 1 | L'IA génère le nombre minimum des scénarios ( à definir ) | High | | Cette fonctionnalité permet à l'IA d'analyser les données du sinistre en cours et de générer automatiquement les scénarios d'évolution possibles, du plus probable au plus dangereux. Elle s'appuie sur le référentiel de risques existant et les interventions passées similaires. |
| | | 2 | Chaque scénario inclut une probabilité en pourcentage | | | |
| | | 3 | Les scénarios s'affichent en moins de 10 secondes | | | |
| | | 4 | Le scénario rare est toujours signalé si le risque existe | | | |
| | | | | | | |

**Figure 7**: Extract from the user-story backlog developed during Iteration 1.

## 4.1.5 Implementation Evaluation

The first iteration produced a prioritized user-story backlog, a requirements baseline, and an initial representation of the interface. These outputs clarified what information Anticip-42 should receive, what assistance it should provide, and where human intervention was required.

The prototype helped evaluate whether the requirements could be represented as a coherent user workflow. It showed that generation, modification, selection, and validation needed to remain separate actions. It also made the progression from possible situations to objectives, maneuver ideas, and DTA easier to understand.

Figure 8 presents one of the screens used to examine how the identified requirements could be represented in the interface.

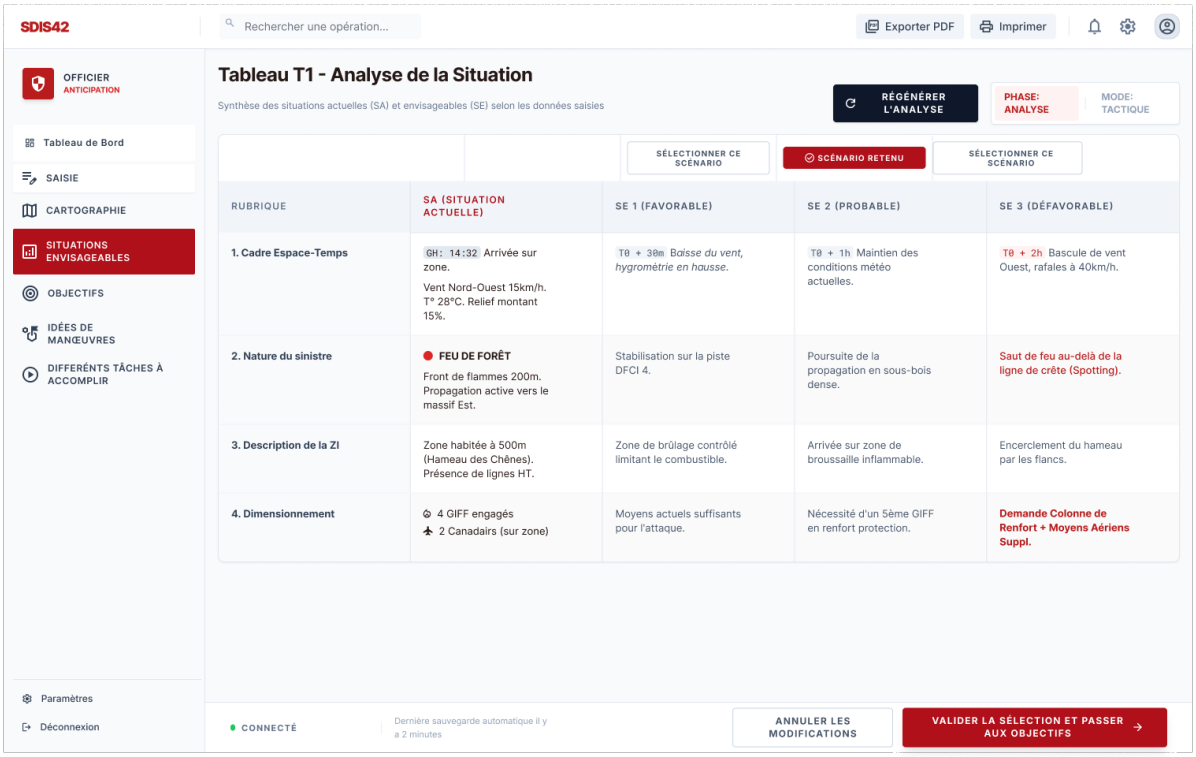



**Figure 8**: Initial interface prototype used to evaluate the progressive anticipation workflow.

The iteration addressed SQ1 by identifying requirements related to operational information, progressive generation, human validation, document use, cartography, interoperability, traceability, and output export. It also established an important boundary: Anticip-42 could prepare proposals, but operational authority and final decisions had to remain with the responsible officers.

In summary, the first iteration identified and organized the main requirements of Anticip-42 through document analysis, meetings, exercise observations, and user stories. The initial interface prototype helped show how these requirements could be represented through a progressive workflow with visible human validation. However, the requirements and prototype did not yet provide a complete description of the actors, activities, information exchanges, and technical functions of the system. This led to the next Knowledge Question: *Can the identified operational requirements be translated into a coherent system specification using complementary system and process models?* The next iteration addresses this question through the development of UML and BPMN models.

## 4.2 Iteration 2: System Modeling

The second iteration translated the requirements and user stories from Iteration 1 into models describing the operational process and the functions expected from Anticip-42. It examined whether complementary modeling views could provide a sufficiently clear specification for later development.

### 4.2.1 Problem Investigation

Iteration 1 established what Anticip-42 was expected to support, but the requirements did not explain precisely how the actors, activities, information, and validation decisions were connected.

The initial interface prototype presented the intended screens, but an interface alone could not describe the complete system. It did not fully represent the responsibilities of the different officers, the order of their exchanges, the correction loops, or the relationship between operational activities and technical functions.

A modeling stage was therefore required before developing the AI workflow. The models needed to describe the system boundary, the operational sequence, the information exchanged, the responsibilities of each actor, and the points at which the COS validates or rejects a proposal.

### 4.2.2 Treatment Design

The treatment planned for this iteration was the development of several complementary UML and BPMN models. Each model was selected to answer a different part of the modeling problem.

A use-case diagram was planned to identify the actors and define the system boundary. An activity diagram was selected to show how work is distributed between the participants. A sequence diagram was planned to represent the order of their interactions and the exchanges that occur when a proposal is accepted, corrected, or rejected.

These UML diagrams were to be created using diagrams.net, also known as draw.io. This tool was selected because it supported the required UML notations and made the diagrams easy to modify as the understanding of the process developed.

BPMN was selected to provide an end-to-end representation of the anticipation process. Two BPMN models were planned. The business-level model would present the overall operational process, while the technical-level model would provide a more detailed view of the functions required to support that process. Camunda Modeler was selected to create the BPMN diagrams because it provided the events, tasks, gateways, message flows, and swimlanes required to repre-

sent the workflow.

The planned models were not intended to repeat the same information. The UML diagrams would provide focused views of the actors, activities, and interactions, while the BPMN models would connect these elements within a complete process.

### 4.2.3 Treatment Validation

Before the models were developed, the proposed modeling plan was checked against the requirements and user stories from Iteration 1. This validation examined whether the selected diagrams would provide enough information to guide the following design and implementation work.The plan was first examined qualitatively. The selected models had to represent the operational actors, the distribution of responsibilities, the information exchanges, the four anticipation stages, and the validation authority of the COS. The distinction between the high-level operational process and its supporting technical functions also had to remain clear.

A simple quantitative coverage check was also used. The planned set of models was compared with the ten user stories identified during Iteration 1. Each user story could be connected to at least one planned modeling view. The plan also covered all four operational outputs: possible situations, objectives, maneuver ideas, and DTA.

The review confirmed that no single diagram could represent all the required information clearly. The combination of three UML diagrams and two BPMN models was therefore retained. The proposed modeling plan was considered suitable for translating the requirements into a more detailed system specification.

Figure 9 summarizes the validated approach. It shows how the requirements were to be translated into UML and BPMN models and then checked again against the operational needs.

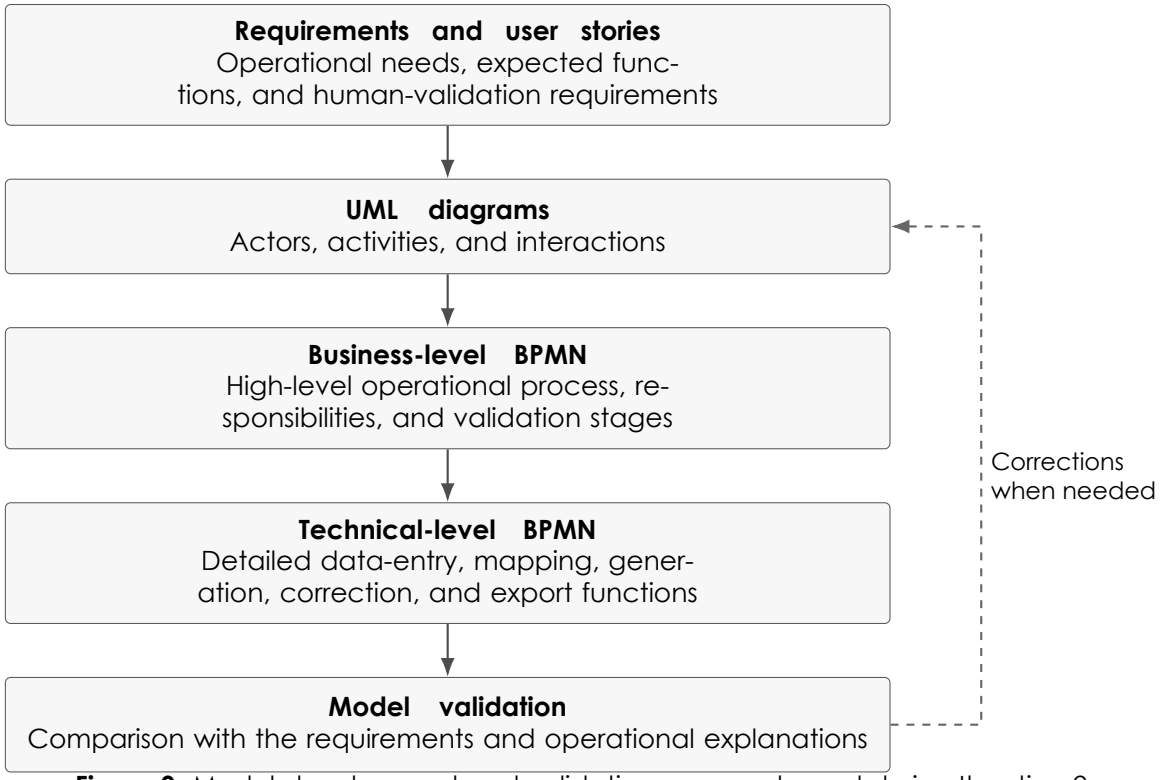



**Figure 9**: Model-development and validation approach used during Iteration 2.

### 4.2.4 Treatment Implementation

The validated modeling plan was then implemented using diagrams.net for the UML diagrams and Camunda Modeler for the BPMN models. The resulting artifacts provided different but connected views of the proposed system.

Table 7 summarizes the five modeling artifacts produced during the iteration, their purpose, and their contribution to the system specification.

Table 7: Process-modeling artifacts produced during Iteration 2.

| Artifact | Purpose | Contribution |
|---|---|---|
| Use-case diagram | Define the system boundary | Identified the operational actors and their main interactions with Anticip-42. |
| Activity diagram | Represent the distribution of work | Connected information collection, officer proposals, validation, correction, and execution. |
| Sequence diagram | Represent the order of exchanges | Showed the dependencies between the successive stages and their validation loops. |
| Business-level BPMN | Formalize the operational process | Described the operational workflow independently of the technical solution. |
| Technical-level BPMN | Identify potential system support | Connected operational tasks to data capture, storage, retrieval, generation, mapping, and export functions. |

**UML models.** The first models were produced with Unified Modeling Language (UML) [73]. The use-case diagram identified the operational actors and the main functions expected from Anticip-42. It also helped define what was inside the system boundary and what remained under the responsibility of the operational participants.

The activity diagram represented the distribution of work between the COS, the Chef du PC, the anticipation officer, and the officers responsible for action, resources, and intelligence. It connected the collection of operational information with proposal preparation, correction, validation, and execution.

The sequence diagram focused on the order of exchanges. It showed how information moves between the participants and how a proposal can be returned for correction before the process continues. It also made the dependency between the successive anticipation stages more visible.

Together, the three UML diagrams described who interacts with the system, what activities are performed, and when the main exchanges occur. However, separate UML views did not provide a single end-to-end representation of the complete anticipation process. BPMN was therefore used to connect these elements within one workflow.

**Use of BPMN for process modeling.** UML is a general-purpose modeling language used to describe different views of a system. In this study, the use-case diagram showed who interacts with Anticip-42, the activity diagram showed the main activities, and the sequence diagram showed the order of interactions. Each UML diagram therefore described one particular aspect of the proposed system.

BPMN has a more specific purpose. It is designed to represent a complete operational or business process from its starting event to its final outcome. It shows which participant performs each task, how information moves between participants, where decisions are made, and how the process continues when a proposal is accepted or rejected. Although a UML activity diagram can also represent a flow of activities, BPMN makes responsibilities, message exchanges, events, and decision paths more explicit.

BPMN was needed because anticipation is not performed by one actor or through one system interaction. It involves information received from several officers, proposals prepared in successive stages, exchanges through the command post, and repeated validation by the COS. Representing the complete process made it possible to understand how these activities are connected before deciding which of them could be supported by Anticip-42.

Two BPMN models were consequently developed. The business-level BPMN represents the operational process at a high level, without describing the internal technical solution. The technicallevel BPMN follows the same process but adds Anticip-42 and specifies the system functions

required to support each stage.

**Business-level process.** The business-level BPMN represents the anticipation process independently of a specific software architecture. Its swimlanes correspond to the COS, the Chef du PC, the anticipation officer, and the three officers:action, moyens and renseignements. The purpose of this model is to show who participates in the process, what information is exchanged, and where operational validation occurs.

The process starts when the COS issues an initial order or when a new situation is reported. The chef du PC communicates the order to the relevant officers. The renseignements officer analyzes the available data, the action officer reports the current field situation, and the moyens officer reports the availability and development of the deployed resources. These inputs are transmitted to the anticipation officer. The anticipation work is then represented through four successive operational outputs: possible situations (SE), objectives, maneuver ideas (IM), and detailed tasks (DTA). For each stage, a proposal is prepared and submitted to the COS. The COS may approve the proposal or return it for correction. An approved output becomes an input to the following stage. At the end of the process, the final decision is communicated through the chef du PC and transmitted to the officer action.

This model remains at a high level of abstraction. It identifies the operational responsibilities, information flows, successive outputs, and validation points without specifying software components, interfaces, or implementation technologies.Because the complete business BPMN contains several swimlanes, message flows, and validation loops, reducing it to a standard report page would make its labels unreadable, Figure 10 presents the beginning of the process, in which the COS issues an initial order and communicates it to the chef PC. The complete business-level model is provided in Appendix A.

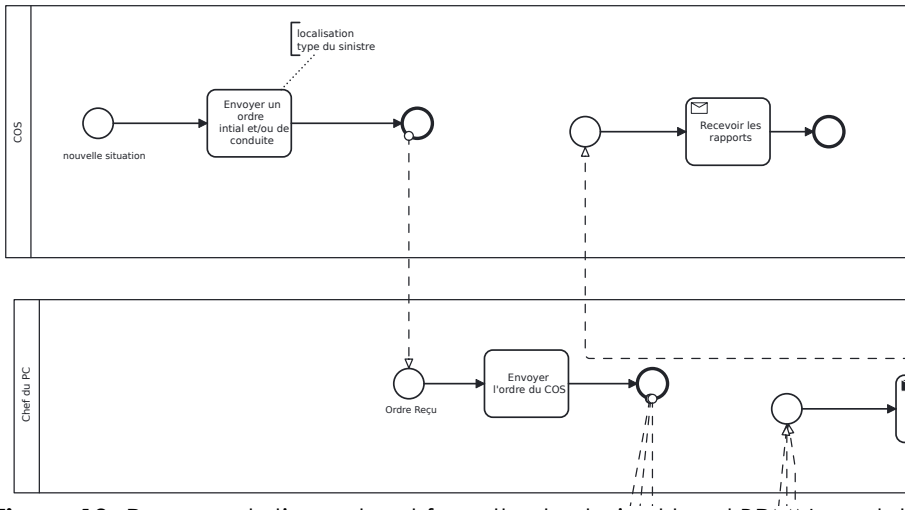



**Figure 10**: Representative extract from the technical-level BPMN model.

**Technical-level process.** The technical-level BPMN is a more detailed version of the business-level process. It retains the same operational actors, sequence of outputs, and COS validation decisions; still, its purpose is to show the functions required to support each activity represented at a high level in the business BPMN. The first detailed part concerns the preparation of the input data. The model shows the entry and storage of operational information, the addition of related documents, images, and geographical data, and the handling of missing or incorrect information. It also details access to an editable map containing annotations about the disaster area.

The model then expands the four main anticipation stages. For possible situations, objectives, maneuver ideas, and DTA, the process follows a similar pattern. The anticipation officer requests a proposal, the corresponding table is generated, and the result is transferred for evaluation. If the proposal is rejected, the input data can be corrected or completed before a new version is generated. Once approved by the COS, the output is used in the following stage.The technical BPMN also describes supporting functions that were represented only generally in the business model. These functions include saving and updating information, loading related documents, displaying operational maps, generating the different tables with AI support, reporting errors, and exporting or printing the produced tables and annotated maps. Speech-to-text is also shown as a pending function rather than as an implemented capability. The difference between the two BPMN models is therefore their level of detail. The businesslevel BPMN explains the overall operational workflow: the actors, the information exchanged, the four anticipation stages, and the COS validation points. The technical-level BPMN explains the functions needed within those same stages. It translates each high-level activity into more precise data entry, storage, mapping, generation, correction, transfer, and export functions.

The complete technical BPMN contains several detailed subprocesses for data preparation, mapping, generation, correction, and output management. Reducing the full model to a standard report page would make its labels unreadable. Figure 11 presents the technical process for entering operational data, completing missing information, and storing the validated inputs in the database. It also represents the error-handling path followed when the data cannot be saved. The complete technical-level model is provided in Appendix A.

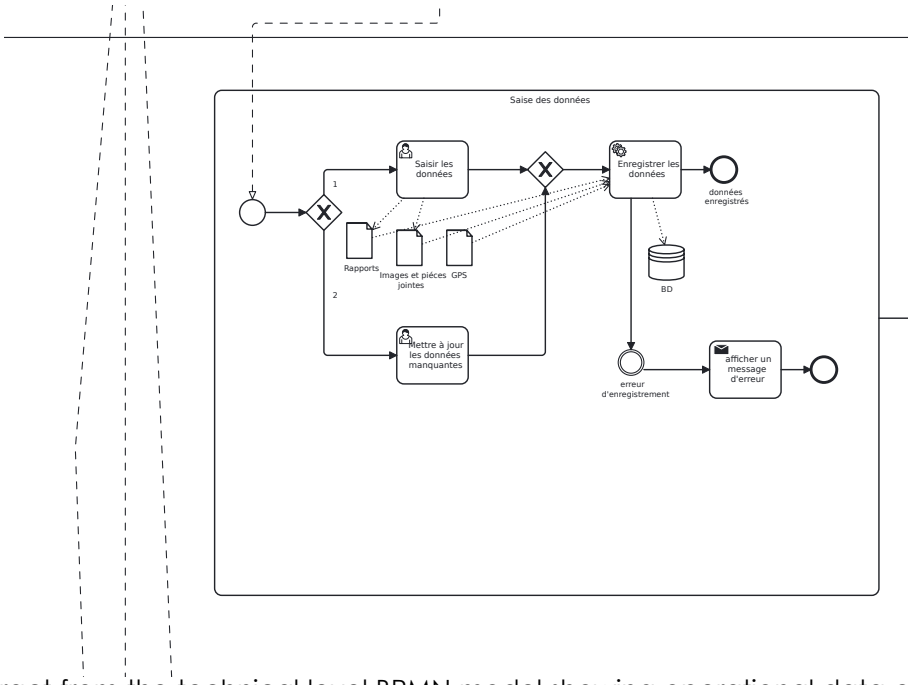



**Figure 11**: Extract from the technical-level BPMN model showing operational data entry, completion of missing information, database storage, and error handling.

Together, the UML and BPMN models connected the requirements from Iteration 1 to a more detailed system specification. They described the system boundary, the operational workflow, the information exchanges, the validation loops, and the functions required to support the process.

### 4.2.5 Implementation Evaluation

The completed models were evaluated both quantitatively and qualitatively. The quantitative evaluation was limited to completeness and traceability; it was not a statistical evaluation of model quality. The final set contained three UML diagrams and two BPMN models. Together, they covered the ten user stories identified during Iteration 1 and represented all four anticipation stages: possible situations, objectives, maneuver ideas, and DTA.

The qualitative evaluation examined the meaning and consistency of the models. The diagrams were compared with the requirements, operational documents, and explanations provided by the operational supervisor. Particular attention was given to the responsibilities of the different actors, the sequence of information exchanges, and the distinction between an AI-generated proposal and a decision validated by the COS.

The comparison between the diagrams also helped identify unclear or missing elements. The models were corrected when a responsibility, information exchange, or validation step was not represented consistently. Following these corrections, the UML and BPMN models provided compatible views of the same anticipation process.

The business BPMN provided the high-level operational view, while the technical BPMN detailed the functions needed to support it. The UML diagrams provided additional views of the system boundary, activities, and interactions. The combination was considered sufficient to guide the design of the human-in-the-loop workflow in the following iteration.

Iteration 2 therefore addressed SQ2 by showing how the requirements could be translated into a more complete system specification through complementary modeling views. However, the models did not demonstrate whether the proposed collaboration between the officers and the AI component could be implemented as a practical and understandable workflow. This led to the next Knowledge Question: *Can the modeled anticipation process be implemented as a progressive human-in-the-loop AI workflow while keeping the officer in control at every generation stage?* The next iteration addresses this question through the design and implementation of the human-in-the-loop workflow.

## 4.3 Iteration 3: Design of the Human-in-the-Loop AI Workflow

The third iteration used the models from Iteration 2 to design and implement the collaboration between the operational user and the AI component. It focused on maintaining human control throughout the successive generation stages and therefore contributed mainly to SQ3.

### 4.3.1 Problem Investigation

The models developed during Iteration 2 described how the anticipation process should progress from operational information to possible situations, objectives, maneuver ideas, and DTA. The next challenge was to determine how AI could support this process without replacing the judgment of the operational user.

Generating a complete operational plan through a single AI request would provide insufficient control. An incorrect proposal produced at an early stage could affect all the outputs generated afterward. Operational information may also change during an intervention, requiring previous proposals to be reviewed or generated again.

The system therefore needed to divide the anticipation process into separate stages. At every stage, the generated content had to remain a proposal that could be examined, modified, rejected, or validated by an officer. Only validated information could be transferred to the following stage.

This iteration investigated whether a locally deployed language model could be integrated into such a progressive workflow. The objective was to examine the feasibility of the human–AI interaction, not to demonstrate the tactical correctness of the generated proposals or readiness for operational deployment.

### 4.3.2 Treatment Design

The treatment planned for this iteration was a progressive human-in-the-loop workflow based on the four stages identified in the BPMN models. The existing high-fidelity interface prototype would provide the visual reference, while a working Streamlit prototype would be developed to test the interaction with a local language model.

Operational information would first be used to generate possible situations. The situations selected by the user would then support the generation of objectives. Validated objectives would be used to propose maneuver ideas, and the selected maneuvers would finally be developed into DTA.

The same interaction pattern was planned for every stage. The officer would initiate the generation, examine the proposed content, and decide whether it should be modified, rejected, regenerated, or validated. The workflow would continue only after explicit human validation.

Structured output formats were also included in the plan. Instead of returning unstructured paragraphs, the language model would be asked to produce structured JSON corresponding to the information required at each stage. This would allow the proposals to be displayed in tables and reused by the following stages.

Figure 12 presents the planned interaction cycle. A rejected proposal returns to correction or regeneration, while a validated result becomes an input to the following stage.

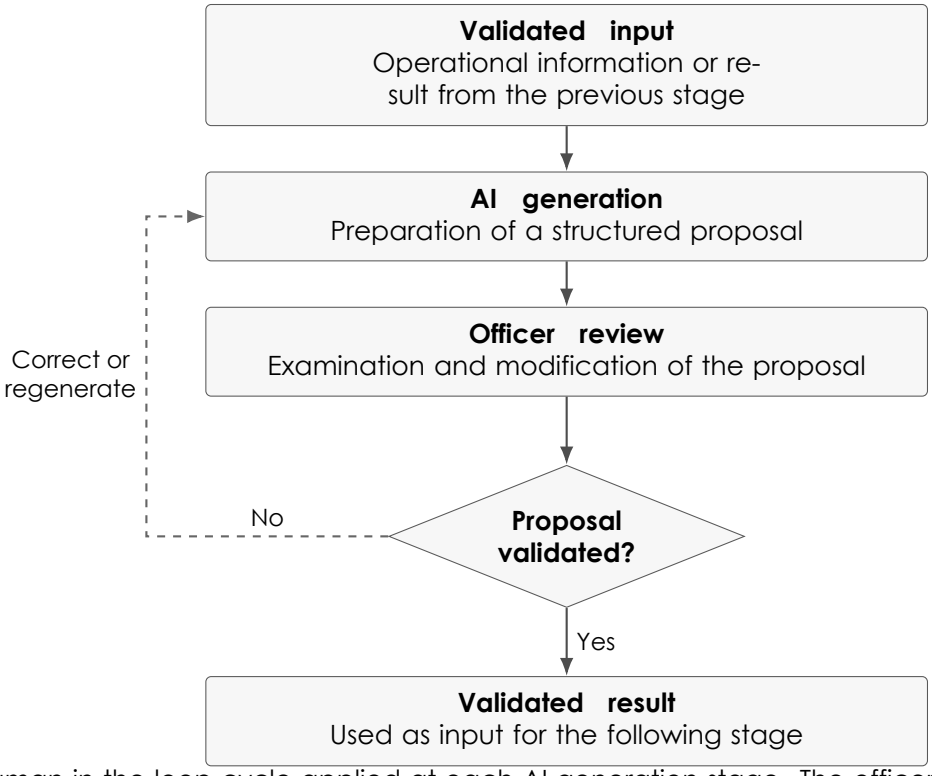



**Figure 12**: Human-in-the-loop cycle applied at each AI generation stage. The officer reviews the generated proposal and either validates it for use in the following stage or requests correction or regeneration.

The workflow was considered dynamic because the input information and generated proposals could be updated while the process was in progress. However, the system was not designed to change an operational plan automatically. An officer would still need to confirm the updated information and initiate a new generation.

### 4.3.3 Treatment Validation

Before implementation, the proposed workflow was checked against the requirements from Iteration 1 and the models developed during Iteration 2. The purpose was to verify that the planned interaction preserved the operational roles and validation decisions already identified.

The qualitative check examined the distinction between an AI proposal and an operational decision. Generated content had to be clearly presented as a proposal, while validation authority had to remain with the appropriate human actor. Generation, review, correction, and validation also had to appear as separate actions.

A completeness check was used to confirm that the plan covered all four anticipation outputs. Possible situations, objectives, maneuver ideas, and DTA were each assigned their own generation and validation stage. The plan also specified that only a validated result could be used as

input for the next stage.

The proposed workflow was considered consistent with the requirements and system models. It was therefore retained as the basis for the Streamlit feasibility prototype. The validation did not demonstrate that the language model would produce tactically correct results; it only confirmed that the planned interaction maintained the required human control.

### 4.3.4 Treatment Implementation

The validated workflow was implemented as a Streamlit prototype connected to locally deployed language models. The implementation made it possible to test the complete progression from operational data entry to possible situations, objectives, maneuver ideas, and DTA.

**Progressive generation pipeline.** A separate generation step was implemented for each operational output. The output selected or validated during one stage was retained and added to the input of the following stage.

The language models were instructed to return structured JSON instead of unstructured text. A structure was defined for each stage so that the generated information could be displayed in tables and transferred between the different parts of the workflow.

The models did not always return valid JSON or all the expected fields. Validation checks were therefore added before displaying a response. When a result could not be interpreted, the application reported an error instead of presenting incomplete content as a valid proposal.

Streamlit session state was used to preserve the information entered and selected by the user. This allowed validated situations, objectives, and maneuver ideas to remain available when moving between tabs. When earlier information was modified, the related proposal could be generated again.

**Streamlit feasibility interface.** Streamlit was selected because it allowed the workflow and the local AI pipeline to be implemented and modified quickly. The prototype brought together operational input forms, contextual information, cartography, document retrieval, and the four generation stages within one application.

The high-fidelity prototype introduced during Iteration 1 remained the visual reference for the intended navigation and validation controls. The Streamlit prototype had a different purpose: it provided a working implementation for testing the transfer of information between the user interface, local language models, document retrieval, and the successive generation stages.

Figure 13 shows the initial operational-input screen. It contains the information required to describe the incident, available resources, command situation, exposed people and assets, anticipated developments, preliminary objectives, maneuver intentions, and reports from the opera-

tional functions.

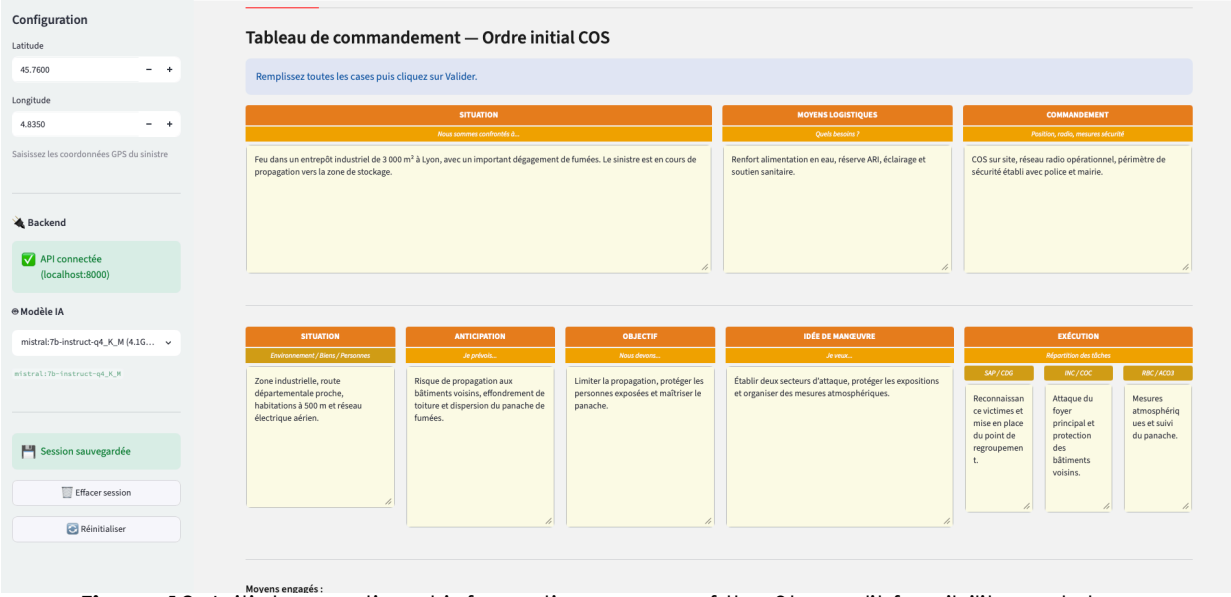



**Figure 13**: Initial operational-information screen of the Streamlit feasibility prototype.

After the initial information was validated, the prototype requested contextual information using the incident coordinates. As shown in Figure 14, the screen combined the location, weather conditions, fire-weather indicator, nearby operational stakes, and an interactive map.

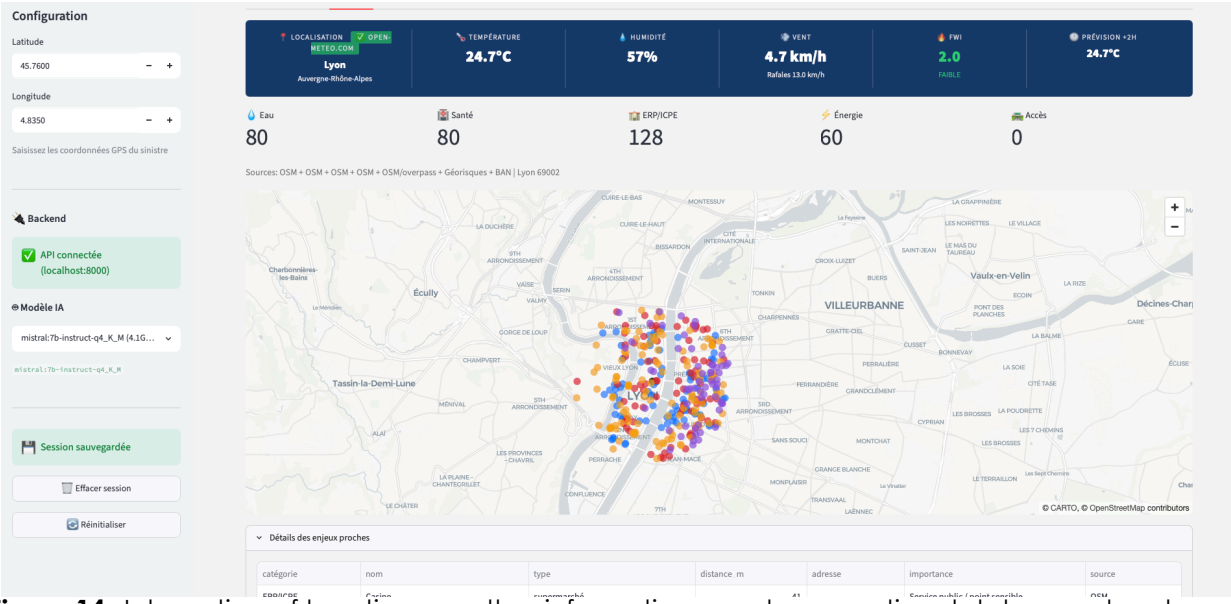



**Figure 14**: Integration of location, weather information, nearby operational stakes, and cartography in the Streamlit prototype.

The contextual information could also be examined in greater detail. Figure 15 shows how nearby stakes were presented with their category, name, type, distance, operational importance, and source. This information remained visible for review instead of being treated as automatically

validated operational information.

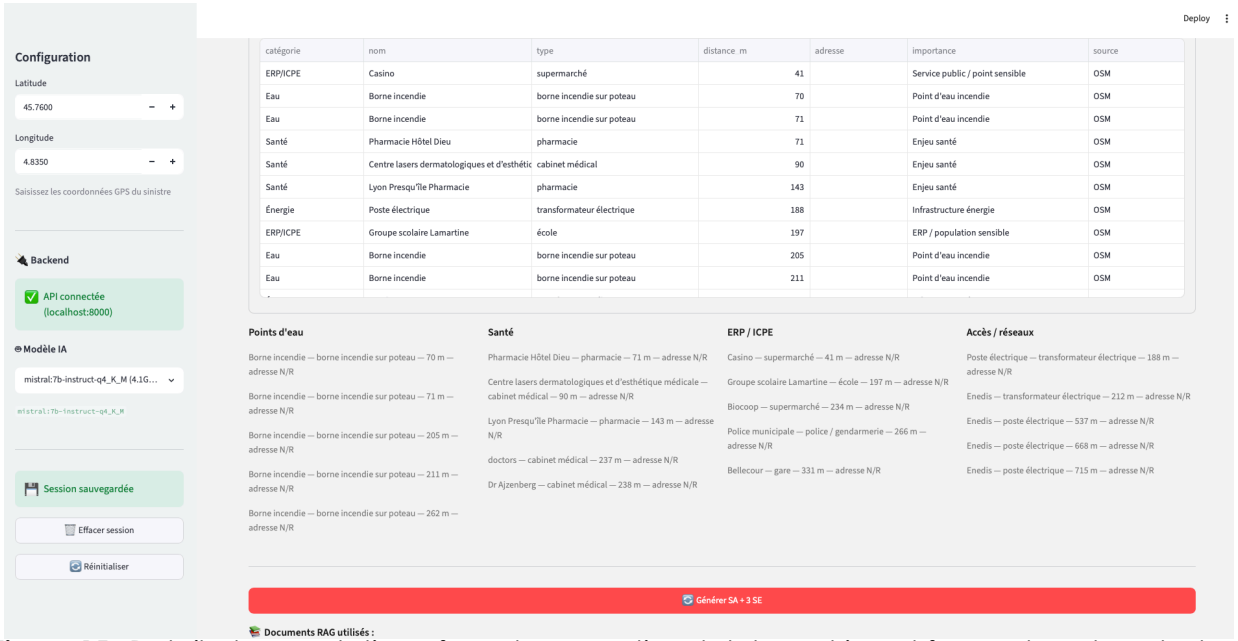



**Figure 15**: Detailed presentation of nearby operational stakes retrieved from external contextual services.

Figure 16 presents the first AI-supported stage. The retrieved documents and their relevance scores were displayed above the generated table. The interface then presented the current situation together with three possible situations organized according to their time horizon, incident characteristics, affected area, estimated scale, and monitoring indicators.

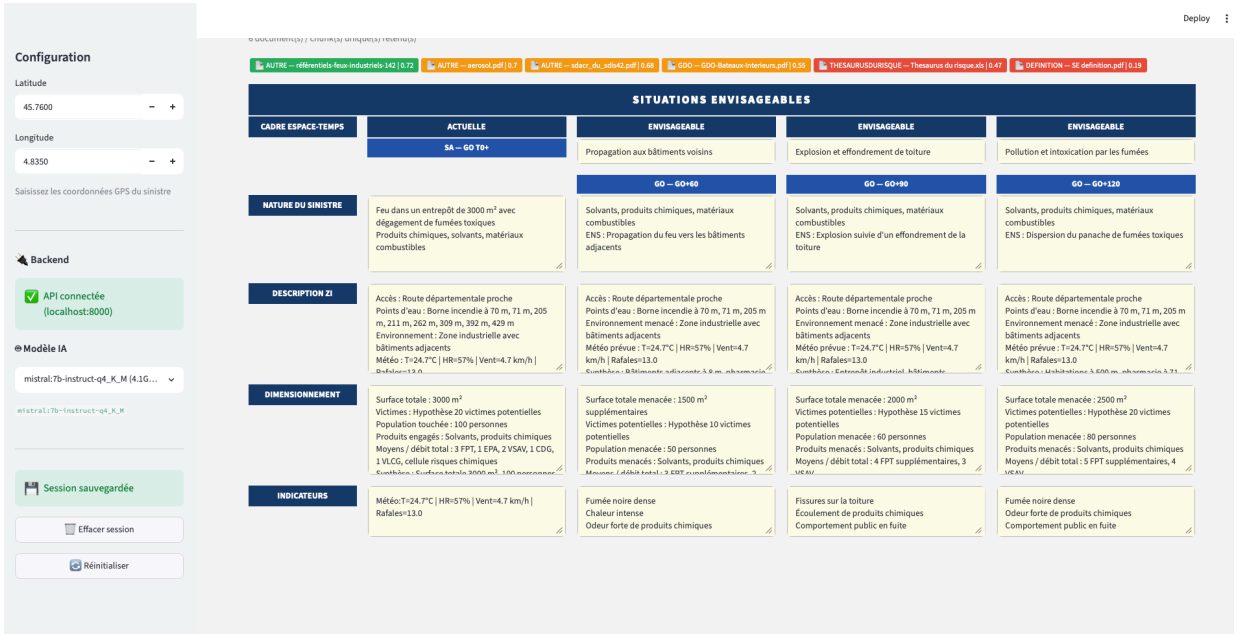



**Figure 16**: Presentation of the current situation, three possible situations, and retrieved documents in the Streamlit prototype.

Together, Figures 13–16 show how the working prototype connected operational input, contextual information, document retrieval, and structured AI generation. The officer remained involved by providing the initial information, examining the retrieved context, reviewing the generated proposals, and validating the information passed to the following stage.

### 4.3.5 Implementation Evaluation

The implemented workflow was evaluated through scenario walkthroughs. These walkthroughs examined whether the four generation stages could be completed in the intended order, whether validated information was transferred correctly, and whether corrections remained available when moving between stages.

The evaluation confirmed that the prototype could separate AI generation from human validation. The officer initiated each generation step, reviewed the result, and decided whether it should be modified, generated again, or accepted. A proposal could not become an input to the following stage until it had been validated.

The evaluation also identified several limitations. The language models did not always return the expected JSON structure, and missing or incorrect information from an early stage could influence later proposals. Generation could also be slow on the available local hardware. These observations showed the need for response checks, error reporting, and careful management of the information transferred between stages.

As the prototype grew, Streamlit also became less suitable for managing the complete interface and its different states. The interface, workflow logic, language-model calls, document retrieval, and contextual services were closely connected within the same application. This made further development and maintenance more difficult.

The evaluation was limited to technical feasibility and interaction design. It did not establish the tactical correctness of the generated content or the operational effectiveness of Anticip-42. The detailed model comparisons, automatic quality checks, runtime measurements, and LLM-as-judge results are therefore presented in the Results and Findings chapter.

Iteration 3 addressed SQ3 by demonstrating how a progressive human-in-the-loop workflow could support collaboration between the officer and the AI component. The AI prepared structured proposals, while the officer controlled their correction, selection, and validation. Streamlit was suitable for testing the feasibility of this workflow quickly, but it became difficult to maintain as the prototype expanded. The interface, session state, workflow logic, AI calls, document retrieval, and external data processing were too closely connected. This made it difficult to modify the interface independently, reuse backend functions, manage complex interactions between stages, replace individual components, and support future system growth.

A React interface was therefore introduced to provide a clearer and more flexible user interface for the complete operational workflow. React allowed the interface and its state-management logic to be separated from the FastAPI backend, where the AI, retrieval, storage, and contextual-data services could be exposed through defined APIs. This separation improved maintainability, scalability, component replaceability, interoperability, and failure isolation.

These limitations led to the following Knowledge Question: *How can the human-in-the-loop workflow be implemented through a separate React interface and service-based backend to support maintainability, scalability, interoperability, component replacement, and resilience to individual service failures?*

Iteration 4 addresses this question through the React interface, FastAPI backend, and supporting internal and external services.

## 4.4 Iteration 4: Service Integration and System Interoperability

The final iteration transformed the human-in-the-loop workflow from Iteration 3 into a modular application. It focused on the services and interfaces needed to connect the user interface, AI generation, document retrieval, storage, and external contextual information, and therefore contributed mainly to SQ4.

### 4.4.1 Problem Investigation

The Streamlit prototype developed during Iteration 3 showed that the human-in-the-loop workflow was technically feasible. However, it also combined many responsibilities in one application. The interface, workflow state, language-model calls, document retrieval, cartography, and external data services were closely connected.

As the prototype grew, this organization became more difficult to maintain. Changes to one part of the application could affect several other parts, and the interface could not easily evolve independently from the AI and data services.

External services introduced another problem. Weather, geocoding, mapping, and risk information depend on network access and third-party providers. These providers may be slow, temporarily unavailable, or return incomplete information. A failure in one external service should not stop the complete anticipation workflow or be presented as a valid empty result.

The problem addressed in this iteration was therefore how to separate the main components of Anticip-42 while allowing them to exchange the information required by the human-in-the-loop workflow.

### 4.4.2 Treatment Design

The treatment planned for this iteration was a service-based architecture with clear interfaces between the application components. The user interface would be separated from the workflow, AI, retrieval, storage, and contextual-data services.

React and TypeScript were selected for the new user interface. Separate pages were planned for operational data entry, cartography, possible situations, objectives, maneuver ideas, and DTA. The pages would share the same application state so that validated information could remain available throughout the workflow.

FastAPI was selected as the common backend interface. Instead of contacting language models, databases, or external providers directly, the React application would send HTTP requests to the backend and receive structured JSON responses. This separation would allow the interface and backend services to be modified independently.

The planned backend was divided into four main service groups:

- workflow services for the successive possible-situation, objective, maneuver-idea, and DTA stages;
- local AI services for generating the proposed operational content;
- document-retrieval and storage services for indexed documents and saved operations; and
- external contextual services for weather, geocoding, nearby stakes, infrastructure, and industrial risks.

A failure-handling strategy was also included in the plan. When two providers offered the same type of information, a secondary provider or server could be used. When a source only provided additional information, its failure would not prevent the other services from returning their results. Missing information would remain visible to the user.

Figure 17 presents the planned organization. The React interface supports interaction with the officer, while FastAPI connects the interface to the different services required by the workflow.

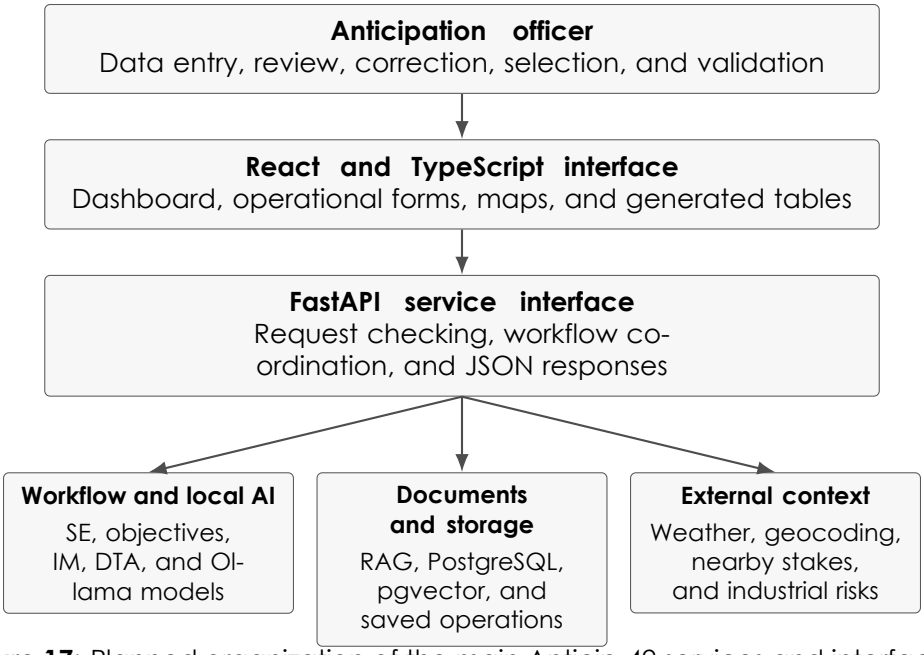



**Figure 17**: Planned organization of the main Anticip-42 services and interfaces.

### 4.4.3 Treatment Validation

Before implementation, the proposed service architecture was checked against the human-in-the-loop workflow developed during Iteration 3. The purpose was to verify that separating the components would not interrupt the progression between the four anticipation stages.

The qualitative check examined the responsibilities assigned to each component. React would manage the user interaction, FastAPI would coordinate the requests, Ollama would provide access to local language models, the RAG service would retrieve relevant documents, and the database would preserve indexed documents and operation records. External providers would supply supporting contextual information.

The plan was also checked for completeness. Each of the four generation stages had a planned request and response interface. The plan covered operational data entry, contextual-data retrieval, document retrieval, local AI generation, operation storage, and backend availability checks. It also preserved the rule that only information validated through the interface could be reused in a later stage.

The failure-handling plan was examined separately. Weather required an alternative provider, OpenStreetMap requests required more than one Overpass server, and independent sources such as Géorisques and the Base Adresse Nationale had to remain optional. The interface also had to distinguish missing information from a valid empty response.

This review confirmed that the planned architecture covered the main exchanges required by the workflow and provided a reasonable way to limit the effect of individual service failures. The

plan was therefore retained for implementation.

### 4.4.4 Treatment Implementation

The validated architecture was implemented by replacing Streamlit as the main interface with a React and TypeScript application connected to a FastAPI backend. The backend provided a common access point for the workflow, local AI, document retrieval, storage, and external contextual services.

**React and FastAPI integration.** The React application provided separate pages for the main stages of the workflow. These pages shared the same application state, which preserved the operational information, generated proposals, selected alternatives, document references, and officer corrections during navigation.

React sent HTTP requests to FastAPI and received structured JSON responses. The interface handled request timeouts and displayed error messages when the backend or another required service was unavailable. An unavailable service was therefore not presented as a valid empty result.

**AI-generation interfaces.** Separate API endpoints were implemented for possible situations, objectives, maneuver ideas, and DTA. Each endpoint received the information required for its stage and returned a structured response that could be displayed and reused by the interface.The local language models were accessed through Ollama. The selected model was included in the request, allowing different models to be used without changing the React pages or the general workflow.

The backend checked the structure of each generated response. When expected fields were missing or the required number of proposals was not returned, the backend attempted to normalize or correct the response. If a usable result could not be obtained, the interface displayed an error.

**Document retrieval.** The RAG service retrieved relevant passages from the local document collection before each generation. Retrieval was adapted to the current stage. For example, the possible-situation stage prioritized documents related to risks and situation analysis, while the maneuver and DTA stages prioritized material related to equipment, operational tasks, ratios, and feared events.

The retrieved passages and document names were returned with the generated result. This made the documentary context used during generation visible and allowed the application to preserve references to the retrieved sources.

**Containerized data infrastructure.** Docker Compose was used to provide the database infrastructure required by the proof of concept. The configuration started PostgreSQL with the pgvector extension and created a persistent volume for indexed documents, vector embeddings, and saved operation records.This configuration provided a reproducible data environment without requiring PostgreSQL and pgvector to be configured manually on each development machine. It also allowed the database service to be started again without losing its stored information.

Redis was included in the Docker configuration as a possible supporting service, but it was not used by the implemented workflow. It was retained for possible future caching or temporary state-management needs.Docker did not containerize the complete application. The React interface, FastAPI backend, document indexer, and Ollama service were started separately on the host machine. Docker was therefore used to standardize the data infrastructure rather than package the entire proof of concept.

**External context and fallback services.** External APIs supplied contextual information that was not available in the local document collection. Open-Meteo provided current weather information, while `wttr.in` was used as its secondary provider. Nominatim converted incident coordinates into a location.

OpenStreetMap data was requested through three Overpass servers to identify nearby water points, health facilities, public buildings, roads, energy infrastructure, and other operational stakes. Results returned by the available servers were combined, and duplicates were removed.

Géorisques supplied information about classified industrial installations, while the Base Adresse Nationale provided additional locality information. These services were independent sources of additional context rather than direct replacements for the Overpass services.

The fallback behavior was adapted to the type of service. Weather had a direct secondary provider, and Overpass used several servers providing the same type of data. If Géorisques or the Base Adresse Nationale failed, the results obtained from the remaining sources could still be returned. If reverse geocoding failed, the location was marked as unknown without stopping the rest of the workflow.All contextual information was presented as supporting information for officer review. It was not automatically treated as validated operational information.

**Operation storage.** The application state included the operational order, location, contextual information, generated proposals, selected alternatives, document references, and model selection.

FastAPI provided interfaces for saving, listing, loading, and deleting operation records. An operation could therefore be reopened without losing the information preserved from its earlier stages.

The proof of concept did not implement a complete audit history of every individual modification. The storage functions preserved the current operation state but did not provide all the auditing functions that would be required in an operational system.

Table 8 summarizes the principal interfaces implemented through FastAPI and the behavior used when a required service did not return an expected result.

Table 8: Main service interfaces and failure handling.

| Interface | Purpose | Behavior when unavailable |
|---|---|---|
| `/geocode` | Convert incident coordinates into a location using Nominatim. | Return an unknown location without stopping the remaining workflow. |
| `/meteo` | Obtain current weather information from Open-Meteo. | Try `wttr.in` if Open-Meteo fails. If both providers fail, return unavailable values marked as fallback data. |
| `/enjeux` | Find nearby water points, health facilities, public buildings, roads, energy infrastructure, and industrial risks. | Query several Overpass servers and retain the available results. Failure of Géorisques or BAN does not remove results obtained from the other providers. |
| `/v2/se`, `/v2/objectifs`, `/v2/im`, and `/v2/dta` | Run the four AI-supported generation stages and return structured JSON with the retrieved document references. | Check and normalize the response, request a correction when possible, and display an error when a usable result cannot be produced. |
| `/operations` and `/health` | Save, list, load, or delete operations and check backend availability. | Inform the interface when the backend or persistence service is unavailable. |

### 4.4.5 Implementation Evaluation

The integrated application was evaluated through technical walkthroughs of the complete workflow. These walkthroughs examined the exchanges between the React interface, FastAPI backend, local language models, document-retrieval service, database, and external contextual services.The evaluation confirmed that each of the four generation stages could be requested through a separate API interface. The structured responses could be returned to React, displayed to the user, and preserved for the following stage after validation. The interface could also use the backend services for contextual information, document retrieval, health checks, and operation storage.

The fallback mechanisms were examined by considering the unavailability of individual external providers. Weather retrieval could move from Open-Meteo to `wttr.in`, and nearby stakes could be requested through several Overpass servers. Independent contextual sources could fail without preventing the remaining services from returning their information.

These checks showed that the application could continue with partial context when an external provider was unavailable. Missing information remained visible instead of being silently re-

placed with invented values. However, fallback mechanisms could not guarantee that the available contextual information was complete or current.The migration also showed that the interface and backend could evolve separately. React replaced Streamlit as the main user interface, while the AI, retrieval, storage, and contextual functions remained accessible through the FastAPI interfaces. This separation reduced the direct dependencies between the user interface and the individual services.

The evaluation remained technical. Anticip-42 was not connected to live SDIS systems, and the external services could still provide incomplete or outdated information. The walkthroughs did not demonstrate operational effectiveness or the tactical correctness of the generated proposals. Detailed model comparisons, runtime measurements, automatic quality checks, LLM-as-judge results, and scenario findings are presented in the Results and Findings chapter.

In summary, the final iteration connected the React interface, FastAPI backend, local language models, document retrieval, database, and external context providers through defined HTTP and JSON interfaces. It also introduced fallback providers, partial responses, and visible error handling to reduce the effect of individual service failures. The Knowledge Question raised at the end of Iteration 3 can therefore be answered positively: the human-in-the-loop workflow can be implemented through separate and interoperable services while preserving the validated progression between its stages. This result answers SQ4 at the proof-of-concept level, while the quality and limitations of the generated outputs are examined in the Results and Findings chapter.

**Chapter synthesis.** The four iterations progressively developed Anticip-42 from an operational need into an integrated proof of concept. Iteration 1 identified and organized the requirements and user stories. Iteration 2 translated them into UML and BPMN models. Iteration 3 designed and tested the progressive human-in-the-loop AI workflow. Iteration 4 connected the interface, backend, local AI, document retrieval, storage, and external contextual services.

Together, these iterations describe how the proposed system was designed and implemented in relation to the four research subquestions. The detailed evaluation results are presented in the Results and Findings chapter, while their meaning, limitations, and implications for future operational use are examined in the Discussion chapter.

# 5 RESULTS AND FINDINGS

This chapter presents the results obtained from the implementation and preliminary evaluation of Anticip-42. It first demonstrates the integrated React proof of concept and its progressive human-in-the-loop workflow. It then compares the locally deployed language models through deterministic measures and a complementary LLM-as-judge assessment. Finally, the combined findings are used to answer the four supporting research questions and the main research question.

Anticip-42 must be understood as a proof of concept rather than a production system. It was not deployed on the final SDIS infrastructure, connected to live operational information systems, nor evaluated with the complete authorized SDIS documents and database. The model runtimes were measured on the hardware available during the project and therefore provide an initial baseline for future scaling rather than guaranteed production response times.

Under these boundaries, the results establish the technical feasibility of the proposed architecture and workflow. They do not establish that every generated proposal is tactically correct or ready for operational use.

## 5.1 Implemented Anticip-42 Proof of Concept

The design iterations resulted in an integrated proof of concept connecting a React and TypeScript interface, a FastAPI backend, locally deployed language models, a stage-specific Retrieval-Augmented Generation pipeline, PostgreSQL and pgvector storage, and external contextual-information services.

The interface implements a progressive workflow. The officer enters and verifies the operational information, after which the system collects geographic and environmental context. The different stages of AI then generate the current and possible situations (SE), COS objectives, maneuver ideas (IM), and detailed tasks (DTA). Information entered, generated, corrected, or retained during one stage remains available in the following stages through a shared operation state.

The model-selection drop-down in the application header allows the user to choose among the locally available language models. The selected model changes without requiring a different interface or workflow because the models use the same FastAPI endpoints and structured response contracts.

The interfaces were also based on the tables used at the command post, making the system more familiar to officers and consistent with their existing working practices.

### 5.1.1 Structured Operational-Information Input

The first page corresponds to the initial order given by the COS. The anticipation officer uses this page to enter and organize the information provided by the COS. As shown in Figure 18, the page separates the incident situation, logistical requirements, command arrangements, exposed people and assets, anticipated developments, preliminary objectives, maneuver intentions, and execution information.

The page also includes fields for geographic coordinates, attachments, additional observations, and an operation reference. This organization avoids placing the entire operational picture in one unstructured paragraph and creates structured information that can be reused during the following stages. The information remains editable, allowing the anticipation officer to review and correct it before continuing the workflow.

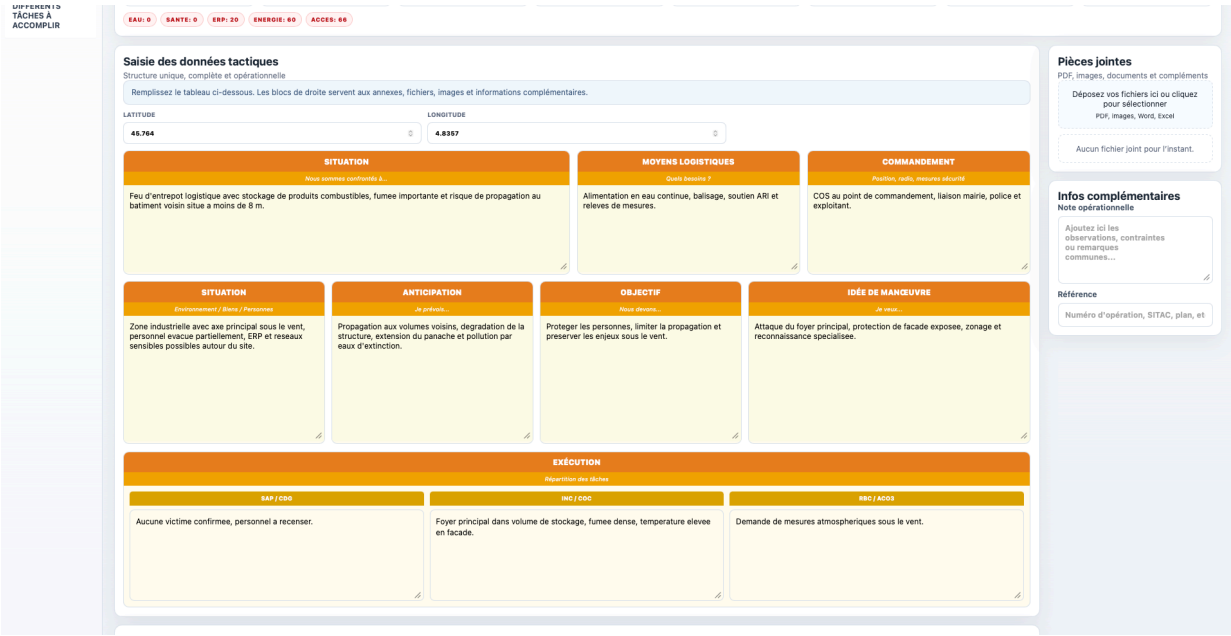



**Figure 18**: Structured input of the initial order provided by the COS in the React proof of concept.

### 5.1.2 Geographic and contextual-information support

Once the incident coordinates have been entered, the FastAPI backend collects reverse-geocoding, weather, and nearby operational information. The cartography page provides both a spatial view and a detailed tabular view.

Figure 19 combines the incident position, local weather conditions, reverse-geocoding results, and nearby operational features. The markers distinguish water points, health facilities, ERP/ICPE sites, energy infrastructure, and access routes. Selecting a marker displays its category, name, type, distance, operational importance, and source. The marker colors and symbols follow those used

in SDIS operational practice, making the map familiar to officers and consistent with the identified requirements and operational doctrine.

The map supports spatial interpretation of the intervention environment. It allows water supplies, sensitive establishments, infrastructure, and access routes to be examined in relation to the reported incident location.

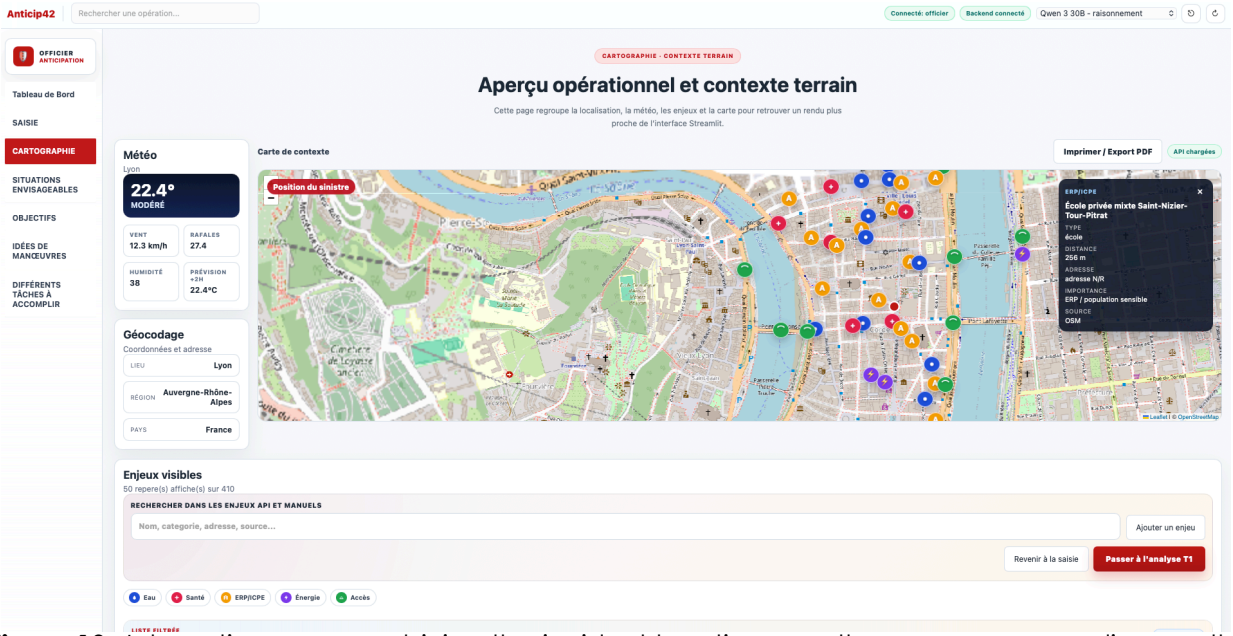
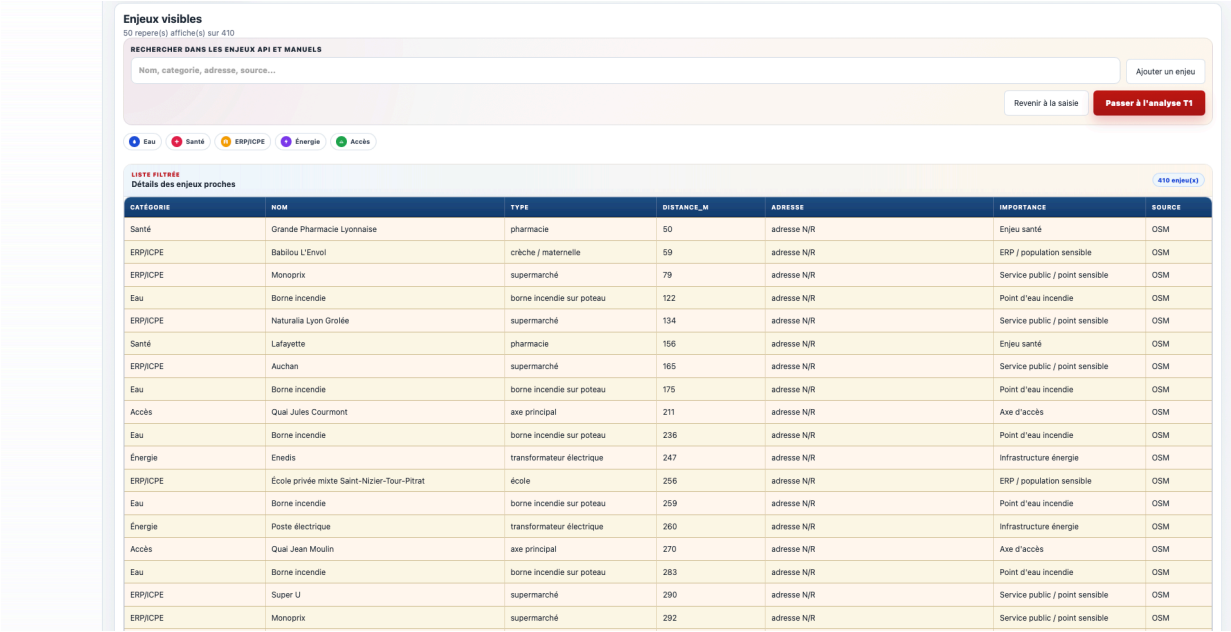



**Figure 19**: Interactive map combining the incident location, weather, reverse-geocoding results, and nearby operational features.

Figure 20 presents the same contextual information as a searchable and filterable table. Each row includes the category, name, type, distance, available address, importance, and source. In the illustrated test, 410 elements were collected, while the map initially displayed 50 markers to preserve readability.




| CATÉGORIE | NOM | TYPE | DISTANCE_M | ADRESSE | IMPORTANCE | SOURCE |
|---|---|---|---|---|---|---|
| Santé | Grande Pharmacie Lyonnaise | pharmacie | 50 | adresse N/R | Enjeu santé | OSM |
| ERP/ICPE | Babilou L'Envol | crèche / maternelle | 59 | adresse N/R | ERP / population sensible | OSM |
| ERP/ICPE | Monoprix | supermarché | 79 | adresse N/R | Service public / point sensible | OSM |
| Eau | Borne incendie | borne incendie sur poteau | 122 | adresse N/R | Point d'eau incendie | OSM |
| ERP/ICPE | Naturalia Lyon Grolée | supermarché | 134 | adresse N/R | Service public / point sensible | OSM |
| Santé | Lafayette | pharmacie | 156 | adresse N/R | Enjeu santé | OSM |
| ERP/ICPE | Auchan | supermarché | 165 | adresse N/R | Service public / point sensible | OSM |
| Eau | Borne incendie | borne incendie sur poteau | 175 | adresse N/R | Point d'eau incendie | OSM |
| Accès | Quai Jules Courmont | axe principal | 211 | adresse N/R | Axe d'accès | OSM |
| Eau | Borne incendie | borne incendie sur poteau | 236 | adresse N/R | Point d'eau incendie | OSM |
| Énergie | Enedis | transformateur électrique | 247 | adresse N/R | Infrastructure énergie | OSM |
| ERP/ICPE | École privée mixte Saint-Nizier-Tour-Pitrat | école | 256 | adresse N/R | ERP / population sensible | OSM |
| Eau | Borne incendie | borne incendie sur poteau | 259 | adresse N/R | Point d'eau incendie | OSM |
| Énergie | Poste électrique | transformateur électrique | 260 | adresse N/R | Infrastructure énergie | OSM |
| Accès | Quai Jean Moulin | axe principal | 270 | adresse N/R | Axe d'accès | OSM |
| Eau | Borne incendie | borne incendie sur poteau | 283 | adresse N/R | Point d'eau incendie | OSM |
| ERP/ICPE | Super U | supermarché | 290 | adresse N/R | Service public / point sensible | OSM |
| ERP/ICPE | Monoprix | supermarché | 292 | adresse N/R | Service public / point sensible | OSM |

**Figure 20**: Searchable table of nearby operational features, including their categories, distances, importance, and data sources.

The table complements the map by supporting the inspection of exact distances and descriptive information. The user can search the results, filter them by category, or manually add relevant information that is absent from the external sources.

Testing showed that the public contextual APIs could respond slowly or become temporarily unavailable. The backend therefore applies timeouts and fallback behavior, while the interface displays warnings or partial results instead of inventing missing information. This limitation must be considered when the system is scaled to a production environment.

### 5.1.3 Comparison of current and possible situations

The first AI-supported stage transforms the validated input and contextual information into a structured current situation and three possible future situations. Figure 21 shows the results of this phase.

This table reproduce the structure and visual organization of the operational tables observed at the command post. They include the time horizon, nature of the incident, threatened area, estimated scale, available means, and monitoring indicators. Maintaining this familiar format reduces the effort required by users to interpret the interface and satisfies the requirement that Anticip-42 remain consistent with existing operational practices.

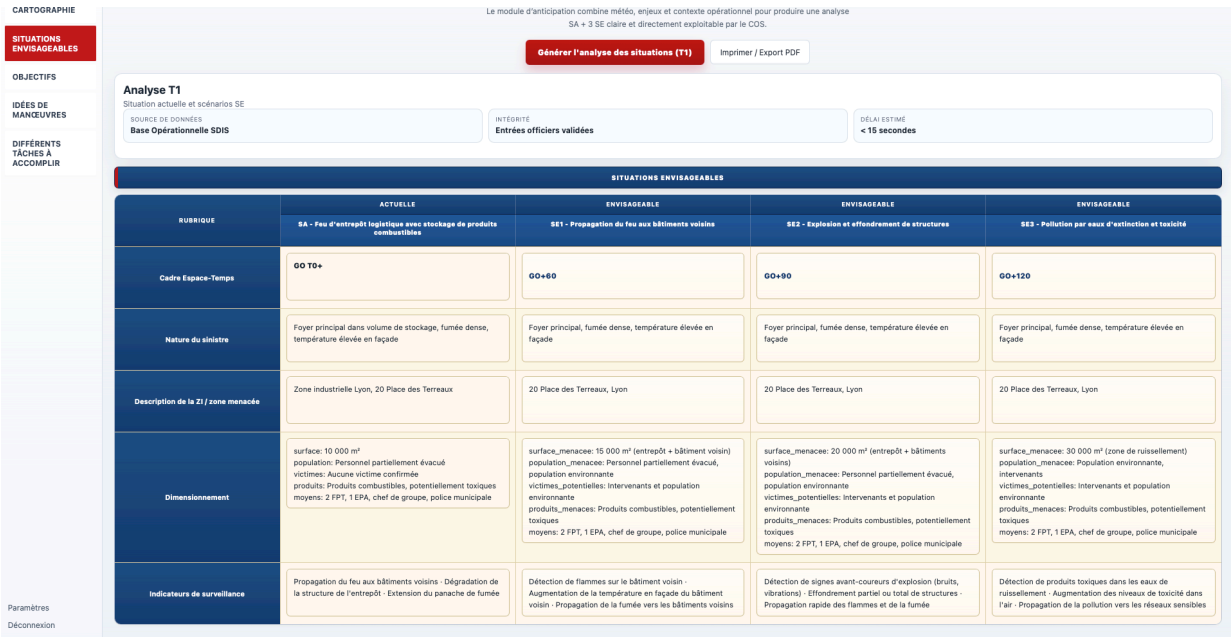



**Figure 21**: Structured comparison of the current situation and three possible future situations.

This result demonstrates that the model output can be transformed into a comparative operational representation rather than displayed as an unstructured response. The application also retains the references retrieved by the stage-specific RAG pipeline, providing an initial level of documentary traceability.

### 5.1.4 Generation and validation of COS objectives

The second AI-supported stage translates the possible situations into proposed COS objectives. As shown in Figure 22, each objective includes its description, expected effect, justification in relation to a possible situation, required means and constraints, and measurable success criteria.

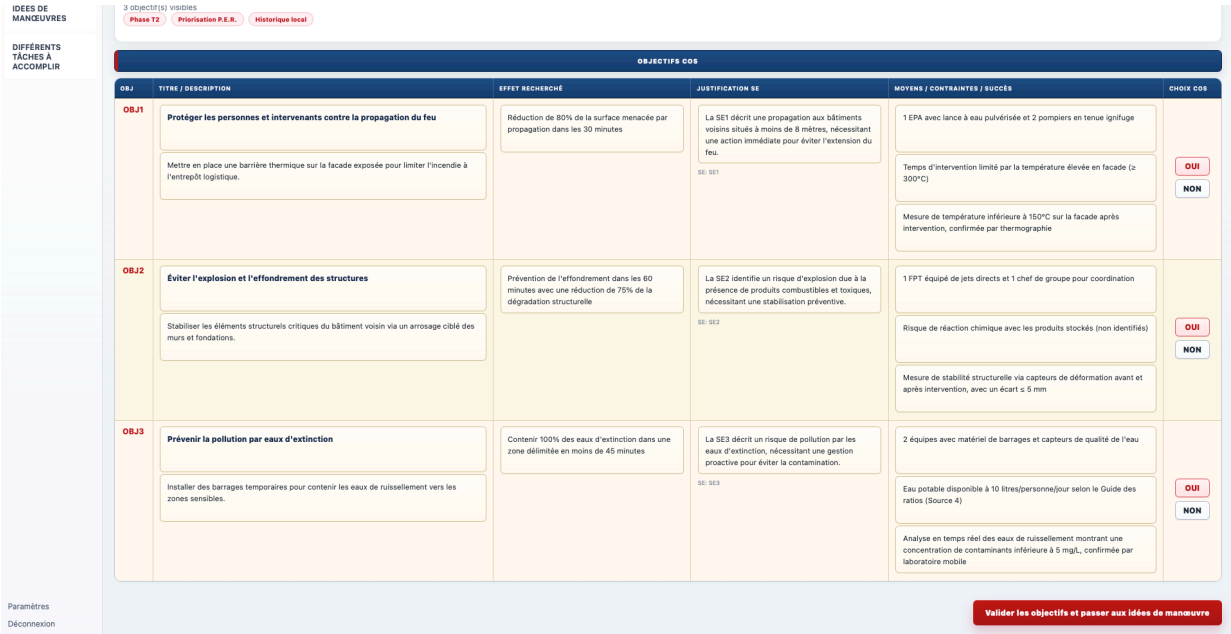



**Figure 22**: Proposed COS objectives with their expected effects, justifications, constraints, success criteria, and validation controls.

The explicit "Yes" and "No" controls allow to retain or reject each objective. Only retained objectives by the COS are propagated to the next generation stage. The AI therefore prepares structured proposals without automatically converting them into operational decisions.

### 5.1.5 Comparison and selection of maneuver ideas

For every retained objective, the third AI-supported stage generates several maneuver alternatives. Figure 23 presents each alternative through its operational description, advantages, disadvantages, costs, risks, and effectiveness criteria.

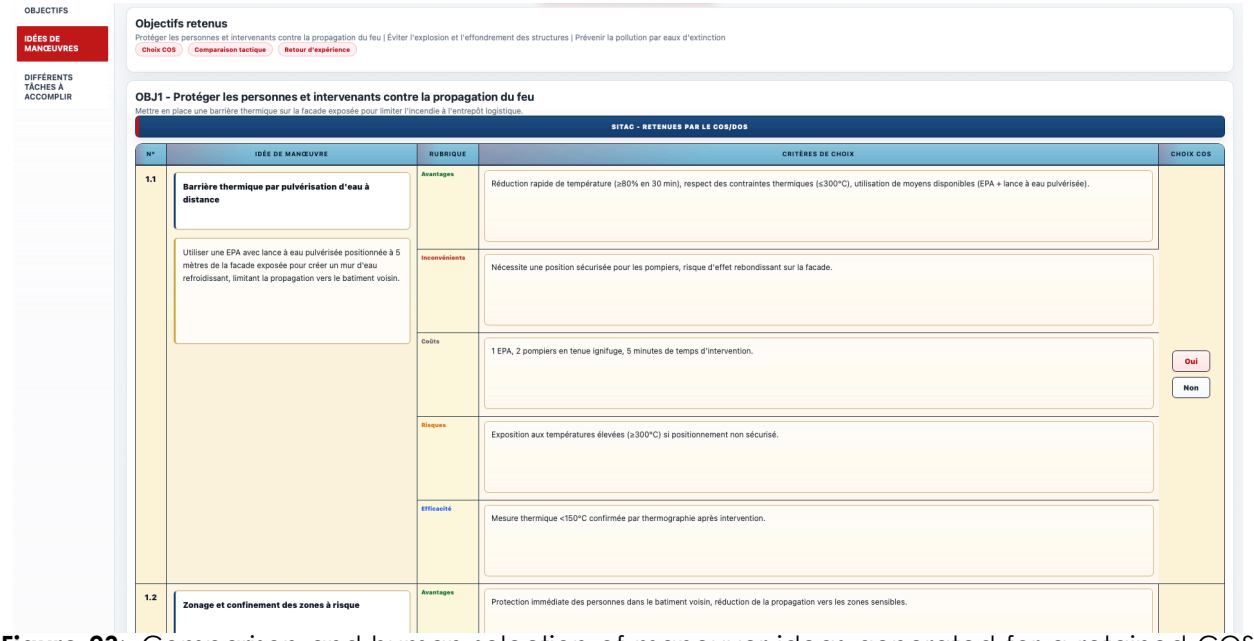



**Figure 23**: Comparison and human selection of maneuver ideas generated for a retained COS objective.

The page supports comparison rather than automatically recommending one maneuver. It also preserves the relationship between the possible situation, the associated objective, and each proposed response. The COS can examine the trade-offs and explicitly retain or reject each alternative.

### 5.1.6 Generation of detailed tasks

The final AI-supported stage converts each retained maneuver into different tasks to accomplish. As shown in Figure 24, every task contains an identifier, description, responsible actor, location, timing, required staff, required material, and intended effect.

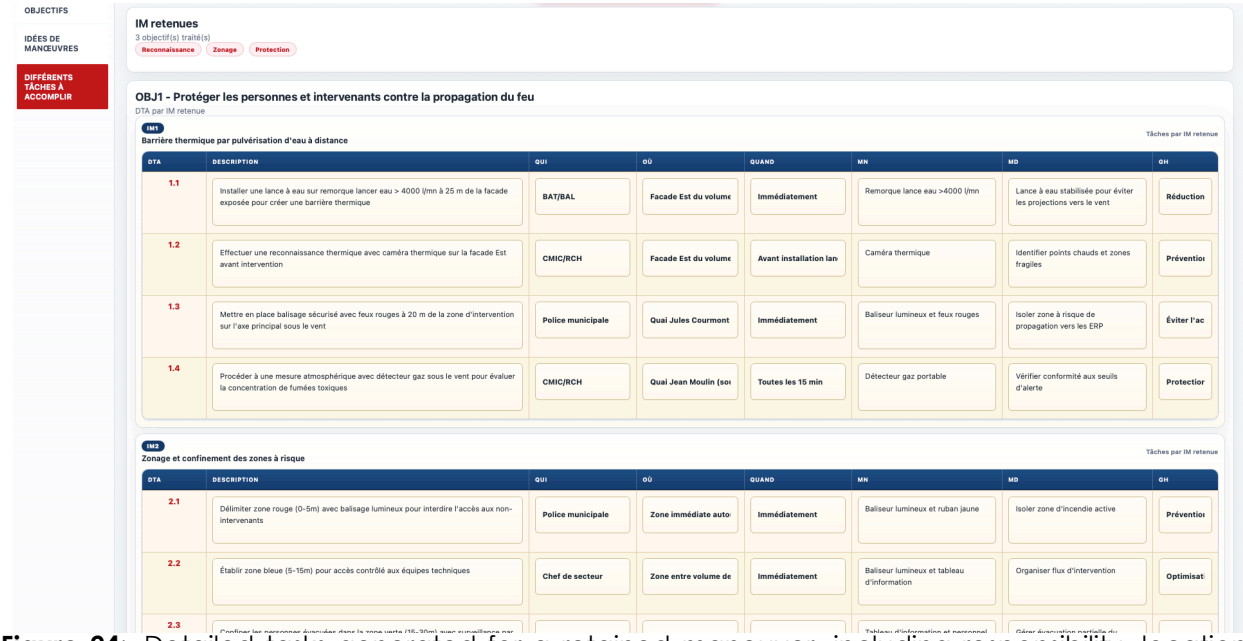



**Figure 24**: Detailed tasks generated for a retained maneuver, including responsibility, location, timing, personnel, material, and intended effect.

The DTA page demonstrates that Anticip-42 can maintain the relationship between a possible situation, a COS objective, a retained maneuver, and its proposed execution tasks. However, structural completeness does not guarantee operational validity. The responsible actor, location, timing, means, and risk treatment must still be reviewed by qualified officer.

### 5.1.7 Implementation findings

The implementation established that the complete workflow could be executed through a shared FastAPI backend while preserving validated information across the React pages. The language model could be changed without modifying the workflow, and each generation stage could retrieve a different combination of definitions, doctrine, operational guides, equipment information, ratios, and risk documents.

The implementation also showed the value of combining spatial and tabular contextual views and of making failures visible. Invalid model responses, unavailable services, and incomplete external results generated explicit errors or warnings rather than being presented as valid operational information. Together, these findings demonstrate that the operational requirements and system models could be transformed into an executable, modular research artifact.

## 5.2 Preliminary Evaluation of Local Language Models

The model evaluation was designed as a preliminary technical benchmark rather than a certification of operational quality. Six locally deployed models were tested on an industrial warehouse-fire scenario:

- Qwen3 30B
- Mistral Small 3.1
- Qwen2.5 32B
- Qwen2.5 72B
- Llama 3.1 70B
- GPT-OSS 20B

Each model received the same incident information, selected documentary collection, RAG mechanism, prompts, workflow stages, and expected response schemas. The evaluation measured proposal counts, required fields, structural quality, automatic repair attempts, detected issues, task completeness, DTA executability, RAG coverage, and total runtime.

The deterministic overall score assigned 30% to situations, 20% to objectives, 20% to maneuver ideas, and 30% to DTA. This score measured minimum structure and completeness; it did not validate tactical correctness.

The results were saved by the backend and could be reopened from the React dashboard.

## 5.3 Industrial Warehouse-Fire Results

Table 9 presents the principal deterministic results. Runtime is reported in minutes. The DTA execution index represents the proportion of tasks containing an action, responsible actor, location, timing, means, and expected effect.

Table 9: Automatic model comparison for the industrial warehouse-fire scenario.

| Model | Time | Overall | SE | Obj. | IM | DTA | Tasks | Exec. |
|---|---|---|---|---|---|---|---|---|
| Qwen3 30B | 37.16 | 1.00 | 1.00 | 1.00 | 0.97 | 1.00 | 36 | 1.00 |
| Mistral Small 3.1 | 46.31 | 1.00 | 1.00 | 1.00 | 1.00 | 0.92 | 36 | 0.50 |
| Qwen2.5 32B | 81.41 | 1.00 | 1.00 | 1.00 | 1.00 | 0.94 | 36 | 0.67 |
| Qwen2.5 72B | 222.50 | 1.00 | 1.00 | 1.00 | 1.00 | 0.96 | 36 | 0.67 |
| Llama 3.1 70B | 103.98 | 0.97 | 1.00 | 1.00 | 1.00 | 0.82 | 14 | 0.07 |
| GPT-OSS 20B | 17.81 | 0.93 | 0.81 | 1.00 | 0.80 | 1.00 | 4 | 1.00 |

*Note:* Time is reported in minutes. "Tasks" is the total number of generated DTA tasks. "Exec." is the automatic DTA execution index.

Four models obtained an automatic overall score of 1.00. The DTA results and runtimes nevertheless revealed substantial differences. Qwen3 30B completed the expected workflow in

37.16 minutes and generated 36 tasks, all of which satisfied the automatic executability criteria. Mistral Small 3.1 also generated 36 tasks, but only 50% satisfied all executability criteria. Qwen2.5 32B and Qwen2.5 72B produced 36 tasks with execution indexes of 0.67.

Model size did not produce a proportional improvement. Qwen2.5 72B required 222.50 minutes, approximately six times the Qwen3 30B runtime, without obtaining a higher overall score. Llama 3.1 70B generated only 14 tasks and obtained an execution index of 0.07. GPT-OSS 20B was the fastest model at 17.81 minutes, but it produced only one possible situation, one objective, one maneuver idea, and four tasks. Its execution index of 1.00 therefore reflected the completeness of a limited output rather than complete workflow coverage.

The complete workflow required between approximately 18 and 223 minutes on the available hardware. This result confirms that model selection and infrastructure capacity will directly affect whether the system can provide support within an acceptable operational time.

## 5.4 LLM-as-Judge Assessment

The deterministic evaluation could verify structure, proposal counts, required fields, and selected quality rules, but it could not fully assess operational coherence, realism, specificity, or risk anticipation. A complementary LLM-as-judge assessment was therefore conducted using Qwen3 30B as a fixed judge.

The judge evaluated the complete output produced by each candidate model without receiving the candidate name, previous ranking, automatic score, or detected quality issues. The rubric covered operational relevance, tactical coherence, realism, specificity, risk anticipation, DTA executability, documentary grounding, and hallucination control.

The combined score assigned 40% to the deterministic evaluation, 35% to the semantic judge score, 10% to DTA executability, 10% to reliability, and 5% to runtime efficiency. Because Qwen3 30B was also one of the candidate models, the assessment must be interpreted as complementary automated evidence rather than independent professional validation.

Table 10 presents the resulting ranking.

Table 10: Combined deterministic and LLM-as-judge evaluation for the warehouse-fire scenario.

| Rank | Model | Combined | Semantic | DTA exec. | Time |
|---|---|---|---|---|---|
| 1 | Qwen3 30B | 0.912 | 0.819 | 1.00 | 37.16 min |
| 2 | Qwen2.5 32B | 0.872 | 0.837 | 0.67 | 81.41 min |
| 3 | Qwen2.5 72B | 0.864 | 0.837 | 0.67 | 222.50 min |
| 4 | Mistral Small 3.1 | 0.851 | 0.800 | 0.50 | 46.31 min |
| 5 | GPT-OSS 20B | 0.826 | 0.675 | 1.00 | 17.81 min |
| 6 | Llama 3.1 70B | 0.682 | 0.656 | 0.07 | 103.98 min |

Qwen3 30B obtained the highest combined score. Its principal strengths were workflow

completeness, DTA executability, reliability, and a substantially shorter runtime than the larger Qwen2.5 models. Qwen2.5 32B and Qwen2.5 72B obtained slightly higher semantic scores, but their lower execution indexes and longer runtimes reduced their combined rankings.

The semantic assessment also identified limitations that were not captured by the deterministic score. These included potentially unavailable equipment, incomplete treatment of pollution or collapse risks, overly strict performance criteria, optimistic success conditions, and insufficient task coverage. Models with similar structural scores therefore did not provide equivalent semantic quality or operational realism.

This result supports the use of complementary evaluation methods. Deterministic checks can identify missing fields and incomplete structures, while semantic assessment can detect broader coherence and realism problems. Neither method replaces evaluation by SDIS professionals.

## 5.5 Methane-Release Scenario Results

A second evaluation considered a methane release involving underground networks and a methane pool. While the first evaluation focused on an industrial fire, this test focused on a chemical-risk emergency. Qwen3 30B, Qwen2.5 32B, and Qwen2.5 72B were evaluated to determine whether Anticip-42 could apply the same RAG pipeline, generation stages, and structured workflow to a different type of operational scenario. This provided an initial indication that the proof of concept was not limited to warehouse fires.

Table 11: Automatic comparison for the methane-release scenario.

| Model | Time | Overall | SE | IM | Tasks | Exec. |
|---|---|---|---|---|---|---|
| Qwen3 30B | 16.44 min | 1.00 | 0.87 | 0.97 | 20 | 1.00 |
| Qwen2.5 32B | 75.91 min | 1.00 | 1.00 | 1.00 | 36 | 0.67 |
| Qwen2.5 72B | 208.15 min | 1.00 | 0.96 | 1.00 | 36 | 1.00 |

All three models generated three possible situations, three objectives, and nine maneuver ideas. No automatic repair was required, and no quality issue was detected by the deterministic checks.

Qwen3 30B completed the workflow in 16.44 minutes and produced 20 executable tasks. Qwen2.5 32B produced 36 tasks, but its execution index was 0.67. Qwen2.5 72B produced 36 executable tasks but required 208.15 minutes.

## 5.6 Answers to the Research Questions

This section provides direct answers to the four supporting research questions and the main research question.

### 5.6.1 Answer to Research Question 1

SQ1 raised the following question: *What requirements shape the design of a proactive intervention system for emergency response?*

The requirements were derived from the user-story backlog, operational documents, discussions with the SDIS 42 supervisor, and exercise observations. Table 12 converts the initial user stories into system requirements.

Table 12: Requirements shaping the design of Anticip-42.

| ID | Requirement | Required system behavior | Backlog basis |
|---|---|---|---|
| R1 | Operational information management | Allow operational information, orders, resources, constraints, and field reports to be entered, reviewed, and updated. | US3 |
| R2 | Progressive anticipation | Generate possible situations, objectives, maneuver ideas, and DTA as four connected stages rather than as one complete plan. | US1, US2, US7, US8 |
| R3 | Human decision control | Keep every AI output editable and require human selection or validation before it can be used in the following stage. | US1, US2, US7, US8 |
| R4 | Context and cartography | Display the incident location, nearby stakes, water points, infrastructure, risks, and other relevant operational information on a map and in tables. | US10 |
| R5 | Documentary and experience support | Retrieve relevant local documents and allow knowledge from previous interventions to support later operations. | US4 |
| R6 | Data-source integration | Reuse information from available SDIS sources and allow documents or images to be added to the operation. | US5, US6 |
| R7 | Continuity and traceability | Preserve entered information, selected proposals, corrections, retrieved document references, and the current operation state. | US3, US4 |
| R8 | Output communication | Allow generated tables and operational information to be exported or printed. | US9 |
| R9 | Robustness and data control | Detect invalid model responses, show unavailable contextual data, and maintain local control over operational documents and model execution. | Cross-cutting |

The answer to SQ1 is that Anticip-42 requires more than an AI text generator. It requires a structured and persistent operational workflow, contextual and documentary support, visible error handling, and explicit human control over every proposal produced by the AI.

### 5.6.2 Answer to Research Question 2

SQ2 raised the following question: *How can system modeling be used to translate the identified requirements into a comprehensive architectural specification?*

The findings show that one diagram is not sufficient. A combination of UML and BPMN models is needed because each model answers a different part of the specification problem. Their individual contributions are summarized in Table 13.

Table 13: Contribution of each model to the system specification.

| Model | Question answered | Resulting specification |
|---|---|---|
| UML use-case diagram | Who interacts with Anticip-42 and for what purpose? | Defines the system boundary, operational actors, and functions available to each actor. |
| UML activity diagram | Who performs each activity and where are decisions made? | Defines the distribution of work, activity order, decision branches, corrections, and validation points. |
| UML sequence diagram | What information is exchanged and in what order? | Defines the chronological exchanges used to request, return, correct, and validate the outputs of each stage. |
| Business-level BPMN | How does the complete operational process work? | Defines the end-to-end process from the initial order to the final decision, including actors, information flows, and COS validation loops. |
| Technical-level BPMN | Which functions are required to support the operational process? | Details data entry, storage, mapping, generation, correction, transfer, error handling, and export functions within the same process. |

The models complement one another. The business BPMN preserves the operational view, while the technical BPMN adds the functions required to support it. The UML diagrams specify the system boundary, responsibilities, activity flow, and information exchanged between participants.

The direct answer to SQ2 is that requirements can be translated into an architectural specification by assigning each modeling question to an appropriate UML or BPMN view and then checking that the views describe the same actors, stages, information, and validation rules.

### 5.6.3 Answer to Research Question 3

SQ3 raised the following question: *How can a dynamic human-in-the-loop AI system be designed to enable proactive intervention while maintaining effective collaboration between human operators and AI components?* The implemented workflow assigns a limited role to the AI at every stage and keeps the operational decisions under human control. The division of responsibilities is summarized in Table 14.

Table 14: Human and AI responsibilities in the progressive workflow.

| Stage | AI contribution | Human responsibility | Control rule |
|---|---|---|---|
| Operational context | Structures the available information and retrieves relevant documentary and contextual data. | Enters, verifies, corrects, and completes the operational information. | Generation begins only from information accepted by the user. |
| Possible situations | Proposes several possible developments of the incident. | Reviews the alternatives, corrects them, and retains the relevant situations. | Unvalidated situations cannot be used to generate objectives. |
| Objectives | Proposes operational objectives based on the validated situations. | Examines and modifies the proposals; the COS retains decision-making authority. | Only validated objectives continue to the maneuver stage. |
| Maneuver ideas | Produces alternative maneuvers and supporting arguments. | Compares, modifies, accepts, or rejects the alternatives. | The AI cannot select the operational maneuver. |
| DTA | Breaks selected maneuvers into structured proposed tasks. | Checks their relevance, resources, location, timing, and executability. | Tasks remain proposals until human validation. |
| Updated situation | Generates a revised proposal when requested. | Decides when changed information requires correction or regeneration. | The system does not modify the operational plan automatically. |

The workflow is dynamic because its information can be updated and proposals can be generated again when the situation changes. However, this does not mean that the system monitors or changes the operation autonomously. The officer decides when an update should affect the workflow.

The evaluation also showed why human review is necessary. A model response can be structurally complete while still containing unrealistic resources, incomplete risk treatment, weak success criteria, or tasks that are difficult to execute. Structured outputs, document retrieval, and automatic checks can support the review, but they cannot establish tactical correctness.

The answer to SQ3 is that effective human–AI collaboration requires a progressive cycle of generation, review, correction, and validation. The AI prepares alternatives, while the anticipation officer and the COS retain control over what is accepted and passed to the following stage.

### 5.6.4 Answer to Research Question 4

SQ4 raised the following question: *What services and system interfaces are required to implement the proposed proactive intervention system and support interoperability among its components?* The proof of concept required a user interface, a shared backend, local AI and retrieval services, persistent storage, and adapters for external contextual information. Table 15 presents the role and interface of each service group.

Table 15: Services and interfaces required by Anticip-42.

| Service | Interface | Role | Interoperability requirement |
|---|---|---|---|
| React interface | HTTP requests and JSON responses | Provides operational forms, maps, proposal tables, correction controls, and validation actions. | Must remain separate from AI, retrieval, and external-service implementations. |
| FastAPI backend | Shared REST API | Checks requests, coordinates the workflow, calls the required services, and returns structured responses. | Provides a common access point for the frontend and backend services. |
| AI-generation service | `/v2/se`, `/v2/objectifs`, `/v2/im`, and `/v2/dta` | Runs the four generation stages through locally deployed Ollama models. | Uses common JSON structures so that models can be changed without redesigning the interface. |
| Document-retrieval service | Stage-specific RAG requests | Retrieves relevant passages and returns their document references with the generated response. | Keeps retrieval separate from generation and preserves source information. |
| Storage service | PostgreSQL, pgvector, and `/operations` | Stores indexed document content, embeddings, and saved operation records. | Allows the database location to change without changing the user interface. |
| Contextual-data services | `/geocode`, `/meteo`, and `/enjeux` | Provide location, weather, nearby stakes, infrastructure, and industrial-risk information. | Require timeouts, alternative providers, partial results, and visible unavailable values. |
| Availability service | `/health` | Reports whether the backend and required supporting services are available. | Prevents a failed service from appearing as a valid empty response. |

Interoperability depends on more than connecting the services. The exchanged JSON must follow a known structure, model responses must be checked, and service failures must be communicated clearly. Weather retrieval therefore uses `wttr.in` when Open-Meteo fails, while nearby-stake retrieval can use three Overpass servers. Géorisques and the Base Adresse Nationale provide independent additional information and do not stop the remaining services when they are unavailable.

The answer to SQ4 is that Anticip-42 requires separate but connected interfaces for user interaction, workflow coordination, AI inference, document retrieval, storage, and external context. Shared JSON contracts, validation, configurable connections, fallback mechanisms, and visible error handling allow these components to work together without creating one tightly coupled application.

### 5.6.5 Answer to the Main Research Question

The main research question raised the following question: *How can a proactive intervention system for firefighters be designed and implemented using human-in-the-loop AI to support operational decision-making during emergency response?*

Table 16 combines the answers obtained from the four supporting research questions.

Table 16: Synthesis of the answer to the main research question.

| Design dimension | Answer provided by the study | Main boundary |
|---|---|---|
| Operational grounding | Begin with the operational workflow, actors, information needs, user stories, and validation responsibilities. | The available requirements do not represent every SDIS process or information source. |
| System specification | Use complementary UML and BPMN models to define the system boundary, workflow, exchanges, functions, and human-validation points. | The models describe the proposed system but do not prove operational effectiveness. |
| Human–AI collaboration | Divide generation into possible situations, objectives, maneuver ideas, and DTA, with human review and validation between stages. | AI outputs remain proposals and may contain tactically unsuitable information. |
| Technical implementation | Connect React, FastAPI, local Ollama models, RAG, PostgreSQL, pgvector, and contextual APIs through explicit interfaces. | The proof of concept was not deployed on the final SDIS infrastructure. |
| Reliability and control | Validate model responses, preserve operation state, show document sources, use fallback providers, and display service failures. | These controls reduce technical failures but do not guarantee tactical correctness or service availability. |

The answer to the main research question is that a proactive intervention system can be designed as a progressive and modular human-in-the-loop workflow. The AI organizes informa-

tion, retrieves relevant documents, and prepares structured alternatives. Operational users review, correct, select, and validate these alternatives before they can influence the following stage.

The proof of concept demonstrates the technical feasibility of connecting this workflow to local language models, document retrieval, persistent storage, and external contextual services. Under the tested conditions, Qwen3 30B provided the strongest measured balance between completeness, task executability, semantic quality, reliability, and runtime. This result supports further testing but does not establish a definitive model for operational deployment.

Anticip-42 should therefore be considered a decision-support prototype rather than an autonomous decision-maker. Its outputs remain proposals requiring qualified human judgment, and further evaluation with authorized SDIS data, target infrastructure, professional users, and realistic operational conditions is required.

# 6 DISCUSSION

This chapter interprets the findings presented in the previous chapter and considers their significance beyond the technical implementation of Anticip-42. It first revisits the research questions and evaluates the extent to which they were answered. It then discusses the contributions and implications of the study for operational organizations and future research. Finally, it examines the main threats to the validity of the findings and the measures used to reduce their effects.

Anticip-42 remains a proof of concept. The discussion must therefore distinguish between demonstrating that the proposed approach is technically feasible and demonstrating that it is operationally effective. The study provides evidence for the first claim, but operational effectiveness would require evaluation on the final SDIS infrastructure, with the complete authorized documentary corpus, qualified operational users, and a wider range of realistic emergency scenarios.

The previous chapter presented direct answers to the research questions. This section discusses how well each question was answered, how the findings relate to previous research, and what can reasonably be concluded within the boundaries of the proof of concept.

## 6.1 Revisiting the Research Questions

The Results and Findings chapter provided direct answers to the research questions. This section interprets those answers in relation to the studies reviewed earlier.

### 6.1.1 Research Question 1

Research Question 1 examined the requirements that shape the design of a proactive intervention system for emergency response. The study found that these requirements must begin with the operational process rather than with the capabilities of a language model.

This finding agrees with research showing that emergency information should be selected according to the user's role, current task, and operational objective [43], [44]. It is also consistent with emergency-system architectures that begin with operational activities and user requirements before defining technical services [41], [42].

Research on situational awareness generally focuses on selecting and organizing information into a common operational picture [53], [54]. Research on digital SITAC tools similarly examines how operational information can be represented in a form familiar to firefighters [56]. Anticip-42 uses the same general principle but applies it to a wider progression: operational information is used to prepare possible situations, objectives, maneuver ideas, and DTA.

The findings therefore confirm the importance of operationally grounded and human-centered requirements. The context-specific addition is a requirements specification for the anticipation

function, including information continuity, editable AI proposals, human validation, documentary support, cartography, traceability, and visible failure handling.

This contribution remains limited to the investigated context. The requirements were developed with SDIS 42, a limited number of stakeholder interactions, and a feasibility document corpus. They provide a basis for the proof of concept but cannot yet be treated as a complete specification for every SDIS organization or emergency category.

### 6.1.2 Research Question 2

Research Question 2 examined how system modeling could translate the identified requirements into a comprehensive architectural specification. The findings showed that several complementary views were required because the operational process, system boundary, responsibilities, interactions, and technical functions could not be represented clearly in one diagram.

This finding is consistent with previous emergency-system research in which user requirements, operational scenarios, and process models guide the design of the supporting architecture [41], [42], [55]. Crisis-modeling studies also show the value of representing actors, information, and events before implementing the technical system [54].

The present study does not introduce a new modeling language. Its contribution is the application of established UML and BPMN views to the specific anticipation process investigated at SDIS 42. The use-case diagram defined the system boundary, the activity diagram represented the distribution of work, and the sequence diagram showed the order of exchanges. The business BPMN represented the high-level operational process, while the technical BPMN detailed the functions required to support the same activities.

The models also preserved the requirements identified through SQ1. In particular, they maintained the separation between AI-generated proposals and decisions made by operational officers. This traceability was important because a technically complete model would not have been suitable if it transferred operational authority to the system.

RQ2 therefore resulted primarily in a technical contribution: a connected system-modeling and architectural specification for Anticip-42. The study confirms the value of complementary models but does not claim that this combination is a new general modeling method. Broader review by operational and technical stakeholders would still be required before using the models as a definitive deployment specification.

### 6.1.3 Research Question 3

Research Question 3 examined how a dynamic human-in-the-loop AI system could support proactive intervention while preserving effective collaboration between operational users and AI

components.

Previous generative systems have shown that language models can prepare candidate emergency plans or provide information extracted from operational manuals [61], [62]. Disaster-ResponseGPT also allows a generated plan to be revised through dialogue. These studies support the use of generative AI for proposal preparation, but they do not implement the complete progression from possible situations to objectives, maneuver ideas, and DTA.

The preservation of human decision authority is also consistent with systems such as IRIS-AI, where analytical and optimization components support resource recommendations while the incident commander retains the final decision [47]. More generally, the design agrees with research emphasizing that users must be able to understand, inspect, and correct the behavior of automated systems [38], [39].

Anticip-42 adds a context-specific stage-by-stage approach. Human control is not placed only at the end of the process. Each AI-generated result can be reviewed, modified, rejected, or generated again before it is passed to the following stage. This makes the relationship between the operational context, possible situations, objectives, maneuver ideas, and DTA visible throughout the workflow.

The evaluation also qualifies the expected value of generative AI. Some outputs satisfied the required structure while still containing unrealistic resources, incomplete risk treatment, optimistic success criteria, or insufficiently detailed tasks. This supports research showing that fluent or well-structured language-model responses may still contain unsupported or incorrect information [33].

The scientific contribution associated with SQ3 is therefore not the claim that human oversight is generally necessary, since this is already established in the literature. The contribution is the application of continuous human validation to the specific progressive anticipation sequence examined in this study. Its operational value has not yet been demonstrated under real command-post conditions.

### 6.1.4 Research Question 4

Research Question 4 examined the services and interfaces required to implement Anticip-42 and support interoperability among its components.

Related emergency-response architectures commonly separate user interaction, contextual information, analytical services, and decision support [46], [47]. Proposed SDIS dashboards also combine information from operational systems, weather services, geographic data, and sensors [66]. These studies support the use of modular components rather than one tightly connected application.

Anticip-42 confirms this need for separation. The proof of concept uses React for the op-

erational interface and FastAPI as the shared backend. Local language models are accessed through Ollama, documents are retrieved through the RAG service, and PostgreSQL with pgvector supports storage. External adapters provide weather, geocoding, nearby stakes, and industrial-risk information.

The study also showed that interoperability cannot be reduced to successful data exchange. Language models may return invalid structures, and external services may be slow or unavailable. Shared JSON structures, response validation, timeouts, fallback providers, partial results, and visible error messages are therefore part of the interoperability design.

This finding complements the related architectures by providing an executable example that connects local AI, document retrieval, persistent operation state, and external contextual services within the progressive anticipation workflow. The migration from Streamlit to React also provided practical evidence that the interface could change while the principal backend services remained available through the same separation of responsibilities.

The contribution associated with SQ4 is exploratory technical evidence from an integrated local proof of concept. It does not establish interoperability with NexSIS or other live SDIS systems. NexSIS and the other French initiatives reviewed earlier address alert processing, reporting, doctrine access, mapping, or operational information [63], [64], [65]. Anticip-42 should therefore be viewed as a complementary anticipation concept rather than a replacement for these systems.

### 6.1.5 Main Research Question

The main research question examined how a proactive intervention system for firefighters could be designed and implemented using human-in-the-loop AI to support operational decision-making during emergency response.

The related literature addresses separate parts of this problem. Fire-behavior models predict the physical development of a fire [49], [51], [52]. Situational-awareness systems organize operational information [43], [53]. Intervention analytics and resource-allocation methods estimate demand, duration, escalation, or resource needs [47], [57], [58], [59], [60]. Generative systems prepare candidate plans or provide access to procedural knowledge [61], [62].

Anticip-42 does not replace these specialized approaches. Fire-behavior predictions, resource estimates, maps, or existing operational systems could provide useful inputs to the proposed workflow. The purpose of Anticip-42 is different: it organizes available information and supports the progressive preparation of possible situations, objectives, maneuver ideas, and DTA while keeping the responsible officers in control of every transition.

Among the publicly documented systems reviewed in this study, no evaluated system was identified that combined this complete French anticipation sequence with local language-model

execution, document retrieval, contextual services, persistent operation state, and explicit human validation between every generation stage. This comparison identifies the specific position of Anticip-42 without claiming that it is the first emergency-response AI system or that it is more effective than the related systems.

The findings support three limited scientific contributions. First, they confirm that operational roles, information exchanges, constraints, and validation responsibilities must be considered alongside technical requirements. Second, they provide a context-specific stage-by-stage approach to human–AI collaboration. Third, they provide exploratory evidence that local language models, document retrieval, storage, and contextual APIs can be connected within one workflow, while also showing that this integration does not guarantee reliable operational output.

The study also produced four technical contributions that correspond directly to the supporting research questions: a requirements specification, a system-modeling and architectural specification, a progressive human-in-the-loop workflow, and an integrated proof-of-concept implementation. These artifacts explain how the proposed system was designed and implemented.

The main research question was therefore answered at the level appropriate for a design-science proof of concept. Anticip-42 demonstrates that a progressive human-in-the-loop anticipation workflow can be implemented through modular local AI, retrieval, storage, and contextual services. It does not demonstrate that the system improves operational decisions during real interventions. Professional evaluation, a broader authorized corpus, representative infrastructure, and realistic operational testing remain necessary.

## 6.2 Contributions and Implications

This section presents the main contributions of the study and discusses their implications for practice and future research.

### 6.2.1 Contributions of the Study

The contributions follow the progression of the four research questions. They include the technical artifacts produced during the development of Anticip-42 and the context-specific knowledge obtained from their evaluation. Together, the implemented artifacts and their evaluation provide evidence of technical feasibility but do not establish operational effectiveness.

Among the studies reviewed, no evaluated system was identified that combined the complete anticipation sequence examined in this project: operational information, possible situations, COS objectives, maneuver ideas, and DTA. This positions Anticip-42 as a complementary approach to existing work on fire prediction, situational awareness, resource allocation, and generative emergency planning.

**Technical contributions.** The study produced four technical contributions corresponding directly to the supporting research questions:

1. **A requirements specification for Anticip-42.**

   The specification identifies the functional and non-functional requirements that shape the design of Anticip-42. It covers operational information, contextual-data needs, traceability, and the responsibilities retained by human operators. This contribution corresponds to SQ1.

2. **A system-modeling and architectural specification.**

   The UML and BPMN models translate the requirements into a specification of the operational workflow, system boundary, actors, activities, system components, information exchanges, validation points, and data structures. Together, these models form the architectural specification of Anticip-42. This contribution corresponds to SQ2.

3. **A progressive human-in-the-loop AI workflow.**

   The workflow separates possible situations, objectives, maneuver ideas, and DTA into successive generation stages. Each proposal can be reviewed, modified, rejected, or validated. Information accepted by an officer is preserved and used during the following stages. This contribution corresponds to SQ3.

4. **An integrated proof-of-concept implementation.**

   The implementation connects the React interface, FastAPI backend, local language models, document retrieval, persistent storage, and external contextual services through defined interfaces. This contribution corresponds to SQ4.

**Scientific contributions.** The scientific contribution is exploratory and design-oriented. The findings confirm, add to, and qualify specific observations from the existing literature:

1. **Confirmation of the importance of operational requirements.**

   The findings confirm that emergency decision-support systems cannot be designed from technical capabilities alone. Operational roles, information exchanges, time constraints, and validation responsibilities must also shape the design. This is consistent with existing human-centered emergency-system research [41], [42], [43], [44].

2. **A context-specific stage-by-stage approach to human–AI collaboration.**

   Existing research already establishes the importance of human control in automated decision support [38], [39]. This study adds a context-specific application of that principle by placing human review and validation between possible situations, objectives, maneuver ideas, and DTA rather than only at the end of the generation process.

3. **Exploratory evidence from an integrated local prototype.**

   The proof of concept shows that local language models, document retrieval, persistent storage, and external contextual services can be connected within one progressive workflow. However, the evaluation qualifies this result: successful technical integration does not guarantee tactically correct or operationally reliable output. Its usefulness remains dependent on response quality, documentary relevance, execution time, service availability, and professional human review.

The model comparison, automatic checks, LLM-as-judge assessment, and runtime measurements support these contributions but are not presented as separate scientific contributions. They provide preliminary evidence about the limitations of the prototype and inform the implications for future research and infrastructure planning.These contributions remain limited to a proof of concept developed in the context of SDIS 42. Further evaluation with qualified personnel, an authorized operational corpus, representative infrastructure, and realistic emergency conditions is required before conclusions can be made about operational effectiveness or safety.

### 6.2.2 Implications for Business and Practice

For SDIS organizations, the study provides a concrete basis for discussing how AI could support operational anticipation without disrupting the established chain of command. The workflow preserves the authority of the anticipation officer and the COS while using AI to organize information, retrieve documentary context, and prepare alternatives for examination.

The familiar structure of the interface is also practically important. Introducing AI through representations based on existing command-post supports may reduce the learning effort and resistance associated with a completely new tool. However, interface familiarity should not be interpreted as evidence that users will automatically trust or correctly use the system. Training, operational guidance, and exercises would still be required.

The architecture provides a foundation for a staged deployment strategy. A future implementation could place the React frontend, FastAPI backend, database, RAG services, and local models on separate virtual machines. Because the components communicate through configurable URLs and structured interfaces, they can be moved or scaled independently, subject to appropriate network, authentication, and security configuration.

The runtime results provide an initial baseline for infrastructure planning. They indicate that larger models can require substantially greater processing time without producing proportionally better outputs. Before selecting production hardware or a production model, SDIS 42 should therefore evaluate stage-level latency, model-loading time, memory and accelerator use, concurrent requests, and recovery from component failures. Acceptable response times should be defined

with operational users rather than inferred solely from technical benchmarks.

The documentary corpus will also influence deployment requirements. The proof of concept used a limited feasibility corpus rather than the complete authorized SDIS knowledge base. Future scaling should organize the documents by operational domain, such as chemical risks, natural hazards, floods, firefighter doctrine, and intervention feedback. Incremental indexing by folder or document category would make corpus updates easier to control and would support evaluation of retrieval quality for each operational domain.

External contextual services represent another practical concern. Mapping, weather, geocoding, and nearby-feature services improved the operational picture, but their response time and availability were outside the direct control of the application. A production architecture should consider caching, approved alternative providers, service monitoring, and clearly defined degraded modes. The absence of an external response must never be presented as confirmation that no relevant hazard or resource exists.

Security and governance are equally important. Local model deployment and local document storage provide greater control over sensitive material, but they do not remove the need for access control, network segmentation, audit logs, data-retention rules, model-version tracking, and authorization of the indexed documents. Every generated proposal should remain attributable to its input data, retrieved sources, model version, and human validation history.

The principal practical implication is that Anticip-42 should be scaled gradually and under clearly defined technical and operational controls. Production readiness depends on acceptable reliability, response time, documentary grounding, usability, security, and governance. Until these conditions have been evaluated with qualified officers, the system should remain a decision-support proof of concept rather than an operational tool.

### 6.2.3 Implications for Research

The findings suggest that research on AI for emergency response should examine more than language-model performance. The complete operational workflow must also be considered, including the information provided to the system, the responsibilities of each officer, the validation points, and the way information moves between stages.

Future research should further evaluate the progressive workflow used in Anticip-42. Dividing the process into possible situations, objectives, maneuver ideas, and detailed tasks provides several opportunities for officers to examine and correct generated content. Comparative studies are still needed to determine whether this approach provides better control than generating a complete operational proposal in a single step.

The results also show that one evaluation score is not enough to judge a model. Some mod-

els produced the expected tables and fields but still contained unrealistic resources, incomplete risk treatment, or insufficient task coverage. Structural checks should therefore be combined with an assessment of the meaning and operational relevance of the generated content.

The LLM-as-judge evaluation helped identify problems that were not detected by the automatic checks. However, an AI judge cannot replace evaluation by qualified SDIS professionals, especially when the judge is also one of the models being compared. The results of this study should therefore be treated as preliminary evidence rather than proof of operational quality.

Finally, the study shows that RAG can provide useful documentary context and make the retrieved sources visible to the officer. However, retrieving a document does not guarantee that the model interprets it correctly. Research on AI-assisted emergency response must therefore examine both the quality of the retrieved documents and the quality of the proposals generated from them.

## 6.3 Validity Threats

The findings must be interpreted in relation to several validity threats. Following commonly used categories, this section discusses construct, internal, external, and conclusion validity, together with the measures used to address each threat.

### 6.3.1 Construct Validity

Construct validity concerns whether the concepts and measures used in the study accurately represented the investigated problem. In this study, concepts such as operational relevance, executability, human control, and proactive support are complex and cannot be completely represented by software metrics.

The deterministic evaluation primarily measured structure, field completion, proposal counts, and predefined quality conditions. A high score did not necessarily indicate tactical correctness, sufficient risk treatment, or operational usefulness. Similarly, the DTA execution index measured whether a task contained an action, responsible actor, location, timing, means, and expected effect. It did not determine whether the task was safe or appropriate for the actual incident.

Several actions were taken to reduce this threat. Operational concepts were derived from doctrine, ENSOSP material, project documentation, command-post observations, and stakeholder explanations. Requirements, models, interface elements, and evaluation criteria were kept traceable to these sources. Deterministic evaluation was complemented by an LLM-as-judge assessment and by formative feedback during the design iterations.

Nevertheless, these measures cannot replace systematic assessment by qualified SDIS professionals. The reported scores must therefore be interpreted as indicators of structural and preliminary semantic quality, not as measures of operational validity.

### 6.3.2 Internal Validity

Internal validity concerns whether the observed results can reasonably be attributed to the factors examined in the study. Differences among model outputs may have been influenced by model architecture, quantization, generation variability, prompt interpretation, retrieval results, or temporary system conditions. Runtime may also have been affected by model loading, hardware availability, background processes, and external service latency.

To reduce these effects, the models were tested through the same progressive pipeline, structured output contracts, scenarios, and principal evaluation criteria. The deterministic checks were applied consistently, and the LLM judge did not receive the candidate model's name, previous score, or ranking. Requirements, design decisions, implementation components, and evaluation criteria were also connected through explicit traceability.

A remaining threat is that Qwen3 30B was both a candidate model and the LLM judge. Although candidate identities were hidden from the judge, the model may still favor outputs resembling its own style or reasoning patterns. The judge results were therefore treated only as complementary evidence and were not presented as independent professional validation.

The tests also did not isolate every infrastructure variable or perform enough repeated runs to measure generation variance statistically. The reported runtime differences should consequently be understood as preliminary measurements under the available test conditions.

### 6.3.3 External Validity

External validity concerns whether the findings can be generalized beyond the conditions of the study. Anticip-42 was developed in the context of SDIS 42, with a limited number of stakeholder interactions and a feasibility document corpus. It was not deployed in a live command post or evaluated during an actual emergency intervention.

The industrial warehouse-fire and methane-release scenarios represented two different technical risk contexts. Reusing the same architecture and workflow for both scenarios provided initial evidence of adaptability. However, these tests cannot demonstrate applicability to all fires, natural hazards, transport incidents, mass-casualty events, or other emergency categories.

The models were also evaluated on the available development hardware rather than the final SDIS infrastructure. The measured runtimes cannot therefore be generalized directly to a production deployment. Performance may change with different processors, accelerators, memory, model quantization, document volumes, network conditions, or concurrent users.

The use of recognized doctrine, familiar command-post structures, modular interfaces, and a second scenario helped reduce this threat. Nevertheless, stronger external validity will require multi-scenario testing, evaluation with the complete authorized corpus, deployment on represen-

tative infrastructure, and participation by a broader group of officers from SDIS 42 and potentially other SDIS organizations.

### 6.3.4 Conclusion Validity

Conclusion validity concerns whether the conclusions are adequately supported by the available evidence. The study compared six models for the warehouse-fire scenario and three models for the methane-release scenario. This provided useful comparative evidence, but the number of scenarios, executions, and professional evaluations remained limited.

The combined model ranking also depended on predefined weights assigned to deterministic quality, semantic assessment, executability, reliability, and runtime. Different weights could change the final ranking. In addition, the absence of detected deterministic issues did not prove that an output was tactically complete or safe.

To reduce this threat, the study did not rely on one measure alone. Requirements evidence, design artifacts, technical observations, deterministic checks, semantic assessment, scenario testing, runtime measurements, and formative stakeholder feedback were considered together. The conclusions were also deliberately limited to technical feasibility, initial model comparison, and preliminary infrastructure planning.

Accordingly, the study does not claim that Anticip-42 improves operational outcomes or that Qwen3 30B is the definitive production model. It concludes that the proposed architecture and human-in-the-loop workflow are feasible, that model and infrastructure choices materially affect system performance, and that further professional and operational evaluation is necessary before deployment.

# 7 CONCLUSIONS

This study investigated how a proactive intervention system for firefighters could be designed and implemented using human-in-the-loop AI. The work was carried out in the context of the anticipation function at SDIS 42 and resulted in Anticip-42, an integrated proof of concept that supports the preparation of possible situations, COS objectives, maneuver ideas, and detailed tasks.

The project began with an examination of the existing operational process, actors, information exchanges, command-post supports, and validation responsibilities. These findings were translated into requirements, user stories, BPMN models, UML diagrams, and a technical architecture. A Streamlit prototype was first used to test the feasibility of the AI workflow. The final proof of concept was then implemented with a React and TypeScript interface, a FastAPI backend, local language models, a RAG pipeline, PostgreSQL and pgvector storage, and external contextual-information services.

## 7.1 Fulfillment of the Aim and Objectives

The main aim of the study was to design and implement a proactive human-in-the-loop decision-support system for emergency response. This aim was fulfilled at the proof-of-concept level.

The first objective was to identify the requirements that shape the design of the system. This was achieved by examining the anticipation process, operational roles, information needs, existing command-post supports, data sources, and human-validation requirements. The resulting specification included functional, operational, data, security, traceability, and interoperability requirements.

The second objective was to translate these requirements into a comprehensive architectural specification. Business and technical BPMN models represented the operational process and its system-supported activities. UML use-case, activity, and sequence diagrams defined the system boundary, actor responsibilities, activity flow, information exchanges, decision points, and validation loops.

The third objective was to design a dynamic human-in-the-loop workflow. This was achieved through a progressive process in which the AI prepares structured proposals and the officers review, correct, retain, reject, or regenerate them. The operation state preserves validated information between stages, while the anticipation officer and the COS retain their operational responsibilities and decision-making authority.

The fourth objective was to define and implement the services and interfaces required for

interoperability. The proof of concept connected the frontend, backend, local models, document retrieval, persistent storage, and contextual services through explicit interfaces and structured data contracts. The implementation also included validation, fallback behavior, partial-result handling, and visible errors.

The study therefore fulfilled its objectives by progressing from operational requirements to system models, architecture, implementation, and preliminary evaluation. However, fulfilling these objectives does not mean that Anticip-42 is ready for operational deployment. The work establishes technical feasibility and provides a basis for further evaluation.

## 7.2 Major Findings

The first major finding is that AI support should follow the existing anticipation process rather than replace it with a single automated recommendation. Dividing generation into possible situations, objectives, maneuver ideas, and detailed tasks makes the progression easier to examine and creates several opportunities for human correction and validation. The second finding is that familiar operational representations are important. The interface uses tables and markers based on those observed at the command post. This allows officers to work with information organized in a form that is closer to their existing practices.

The third finding is that structured output does not guarantee operational quality. Several language models produced the expected tables and fields while still including unrealistic resources, incomplete risk treatment, optimistic success criteria, or insufficient task coverage. Deterministic validation, RAG, and LLM-as-judge assessment can support review, but they cannot replace the judgment of qualified SDIS officers. The fourth finding is that interoperability requires more than connecting components through an API. Shared data contracts, response validation, normalization, persistent storage, timeouts, fallback providers, partial results, and visible errors are also necessary. This was particularly important for language-model responses and public contextual APIs, which could be incomplete, slow, or temporarily unavailable.

The model evaluation also showed that a larger model does not automatically provide a better operational trade-off. For the warehouse-fire scenario, Qwen3 30B achieved the strongest combined balance among workflow completeness, task executability, semantic quality, reliability, and runtime. The methane-release test also showed that the same architecture and generation stages could be reused for a different technical-risk scenario. These results remain preliminary and do not identify a definitive production model.

Finally, the measured runtimes confirmed that infrastructure and model selection will affect the practical use of the system. The results provide an initial reference for capacity planning, but they were obtained on development hardware with a limited document corpus and without con-

current users.

## 7.3 Recommendations

The first recommendation is to preserve human authority at every stage. AI-generated content should remain editable and should never be treated as an automatic operational order. The anticipation officer and the COS must retain responsibility for interpreting the situation and validating the proposals.

The second recommendation is to continue development through controlled stages. The next stage should be a technical pilot on infrastructure representative of the intended SDIS environment. This pilot should measure stage-level response time, model-loading time, memory and accelerator use, concurrent requests, and recovery from service failures.

The third recommendation is to evaluate the system with qualified operational users. Scenario-based exercises should examine whether the interface is understandable, whether the generated proposals are useful, how much correction they require, and whether the workflow supports rather than delays the work of the anticipation officer.

The fourth recommendation concerns the documentary corpus. Authorized SDIS documents should be organized by operational domain, such as chemical risks, natural hazards, floods, operational doctrine, equipment, and intervention feedback. Document versions, sources, dates, and authorization status should be preserved during indexing. Retrieval quality should be evaluated separately for each domain.

The fifth recommendation is to strengthen security and governance before any operational deployment. The system will require server-side authentication, role-based access control, protected configuration values, network segmentation, audit logs, document-access rules, model-version tracking, and data-retention policies. The current proof-of-concept authentication should not be treated as a production security mechanism.

Finally, the contextual-information services should be monitored and supported by clearly defined degraded operating modes. A missing response from an external provider must be displayed as unavailable information, not interpreted as evidence that no hazard, resource, or exposed element exists.

## 7.4 Future Work

The findings define an agenda for moving from the current proof of concept toward a system that can be evaluated under representative operational conditions. This agenda concerns not only further technical development, but also professional validation, infrastructure, documentary governance, system security, and the conditions required for responsible operational use. Fu-

ture work should first evaluate Anticip-42 with the complete authorized SDIS document corpus. This evaluation should measure whether the correct documents are retrieved for each generation stage and whether the generated proposals remain consistent with their sources. It should also examine how the system handles outdated, conflicting, or differently prioritized documents.The model comparison should be repeated on the target infrastructure and across a wider range of emergency scenarios. Repeated runs would help measure generation variability, while tests with concurrent users would provide a more realistic estimate of the required computing resources. Any future model selection should consider response time, coverage, executability, reliability, and operational review rather than model size alone.

The evaluation process should also include independent review by qualified SDIS professionals. Multiple professional evaluators and, where useful, independent AI judges could be used to compare tactical relevance, risk coverage, documentary grounding, and task executability. This would provide stronger evidence than automated scores alone. User-centered evaluation should measure the effect of the system on workload, understanding, verification behavior, trust, and decision time. It should also examine whether repeated validation helps officers identify unsuitable proposals or creates additional cognitive effort during time-sensitive situations.

Future versions could extend the system to additional information sources, including authorized SDIS operational systems, images from the field, drone or satellite imagery, and structured feedback from previous interventions. Saved operation records could support return-of-experience analysis, subject to appropriate authorization, anonymization, and governance. The technical architecture could also be strengthened through production containerization, centralized monitoring, structured logging, secure service configuration, backup procedures, and resilient deployment across separate infrastructure. These improvements should be introduced only after the required security, data-protection, and operational responsibilities have been clearly defined.

## 7.5 Closing Statement

This study began with the objective of moving towards proactive emergency response. The findings clarify what this objective means in the context of AI-assisted operational anticipation. Proactive support should not be understood as allowing an AI system to predict an emergency and make decisions autonomously. It should instead help officers organize changing information, consider possible developments, examine operational alternatives, and prepare for what may happen next.

Anticip-42 provides an initial implementation of this approach. It connects operational requirements, system models, local language models, document retrieval, contextual information, and persistent operation state within a progressive human-in-the-loop workflow. Possible situations,

objectives, maneuver ideas, and DTA are prepared separately, while qualified officers remain responsible for reviewing, correcting, selecting, and validating the proposed content.

The next generation of Anticip-42 must move beyond technical feasibility. Its development agenda should focus on evaluation with qualified SDIS personnel, use of the complete authorized documentary corpus, deployment on representative infrastructure, integration with approved operational systems, and stronger security, traceability, and degraded-mode management. These steps are necessary to determine whether the proposed workflow is useful, usable, reliable, and safe under realistic conditions.

Anticip-42 is therefore not the completion of the move towards proactive emergency response. It is a first step that identifies the architecture, human responsibilities, and evaluation conditions required to continue that move without transferring operational authority to AI.

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# A APPENDIX

This appendix presents the complete process models developed during the study. The business and technical BPMN models are each divided into three overlapping parts to keep their activities and information flows readable. The complete activity and sequence diagrams are also divided into three consecutive parts because their original dimensions make the text difficult to read on a single page. Cropped extracts were included in Chapter 4, while the following pages preserve the complete content of the models.

The UML use-case, activity, and sequence diagrams were produced during the exploratory modeling phase of Iteration 1. They supported the definition of the system boundary, actor responsibilities, activity flow, validation decisions, and interactions between the operational officers and the anticipation system. The business and technical BPMN models were subsequently finalized during Iteration 2 using the requirements and modeling findings established during Iteration 1.

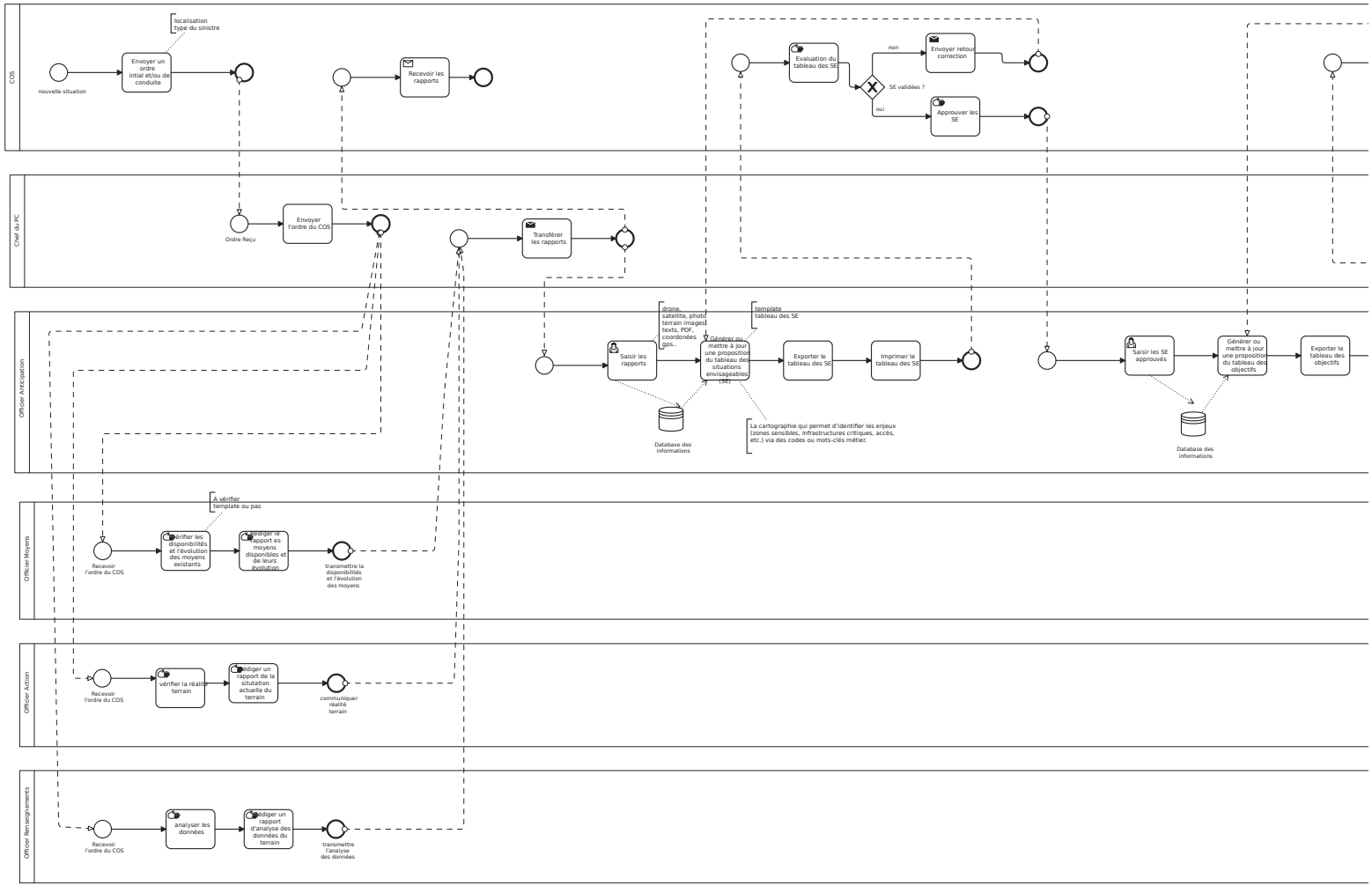



**Figure 25**: Complete business-level BPMN model of the operational anticipation process, Part 1 of 3.

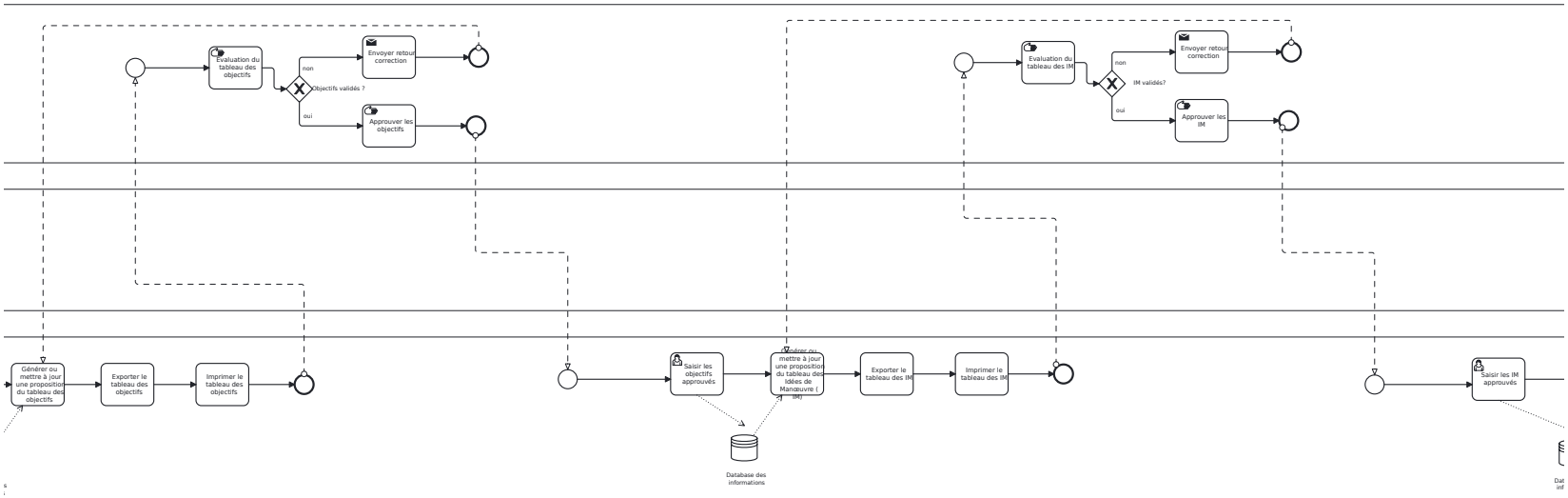



**Figure 26**: Complete business-level BPMN model of the operational anticipation process, Part 2 of 3.

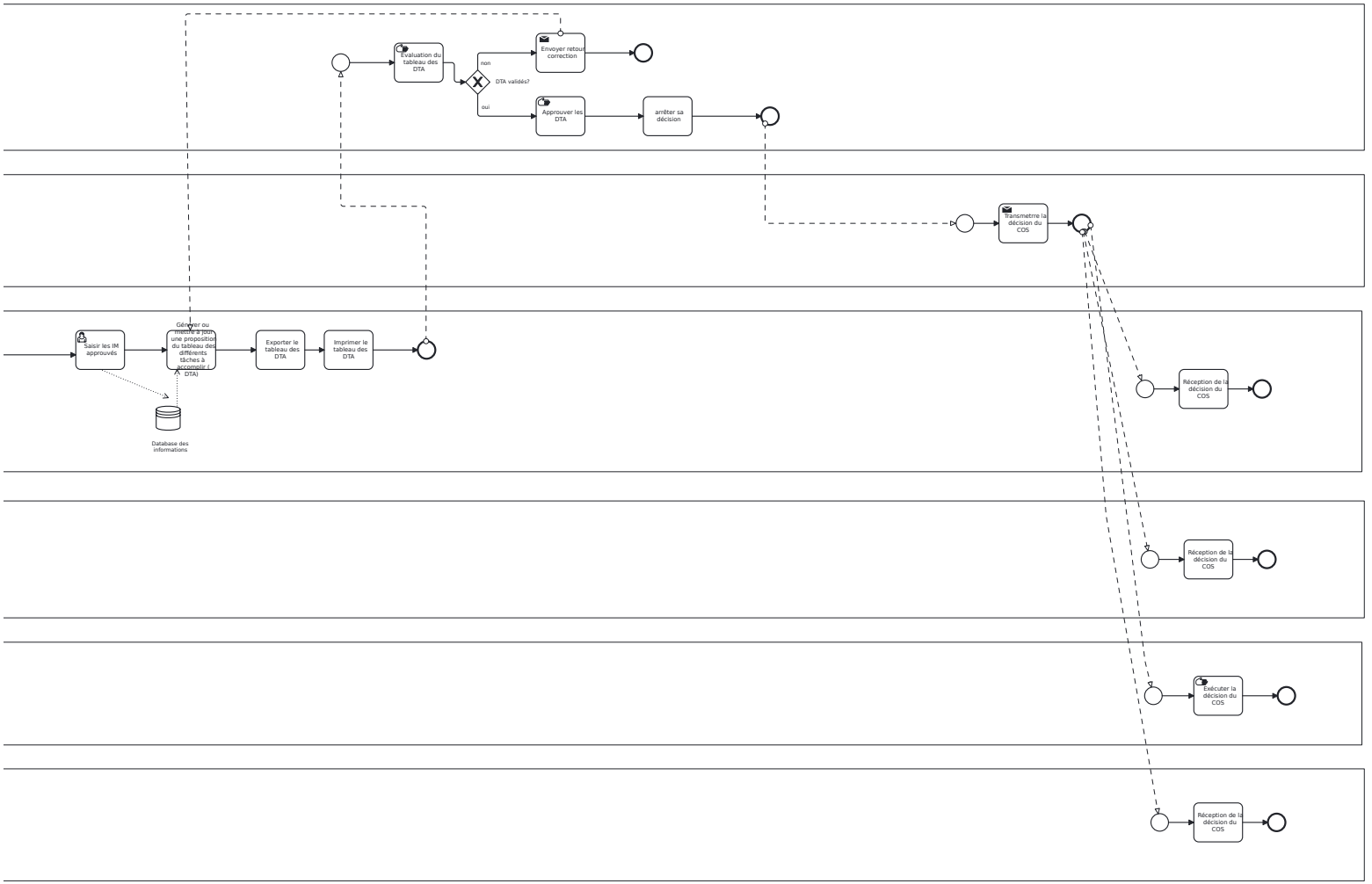



**Figure 27**: Complete business-level BPMN model of the operational anticipation process, Part 3 of 3.

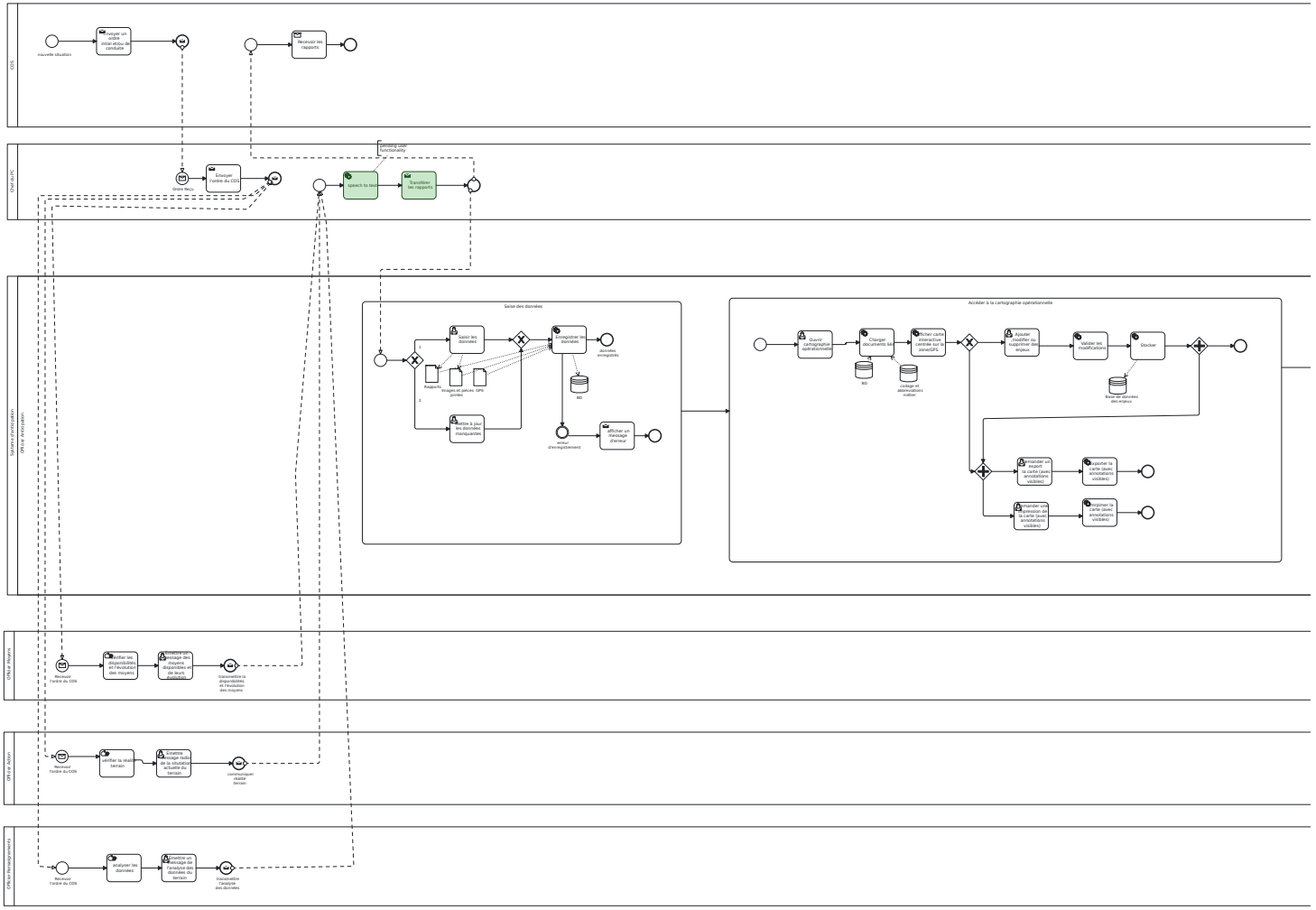



**Figure 28**: Complete technical BPMN model of the proposed anticipation system, Part 1 of 3.

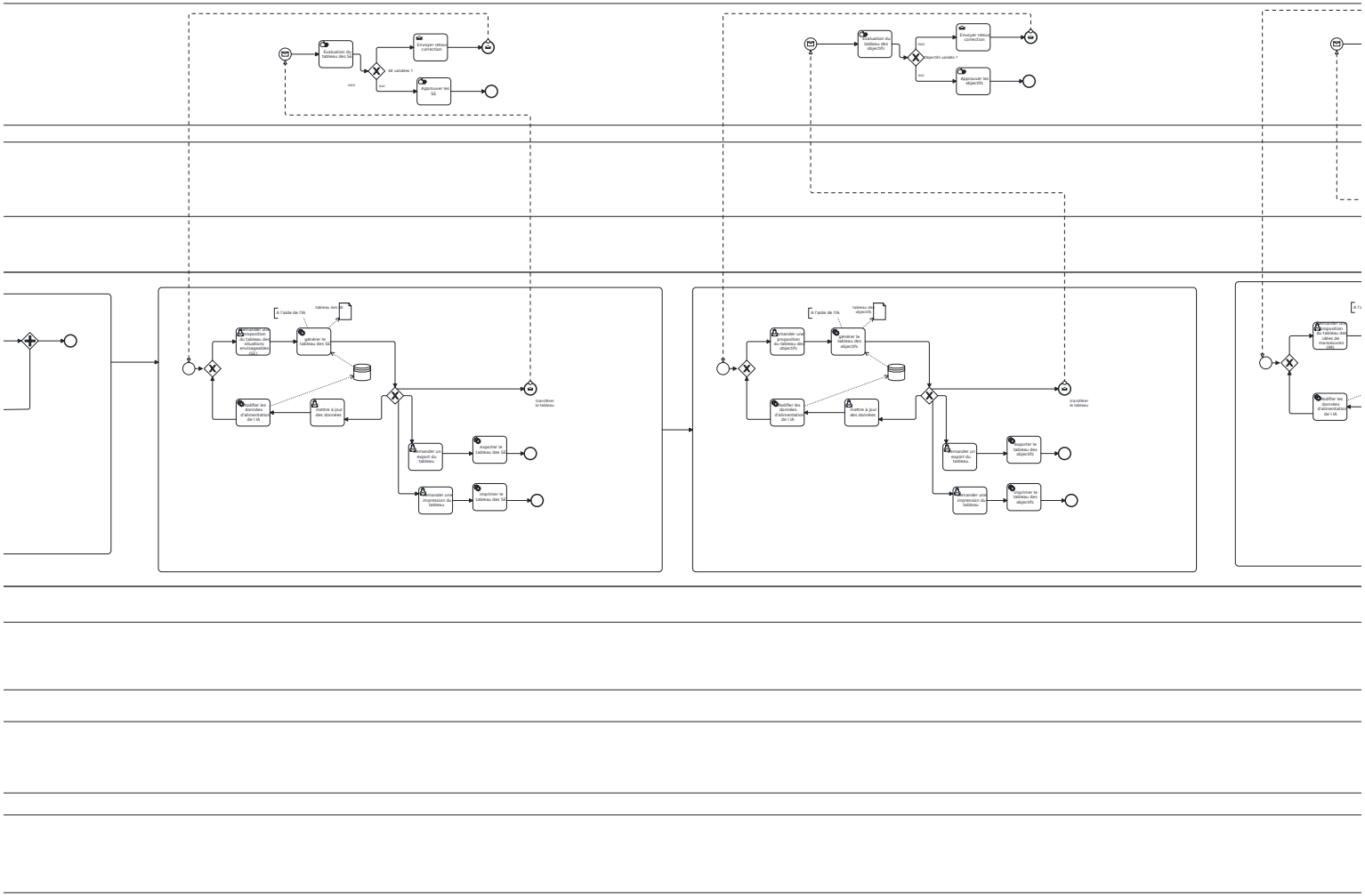


**Figure 29**: Complete technical BPMN model of the proposed anticipation system, Part 2 of 3.

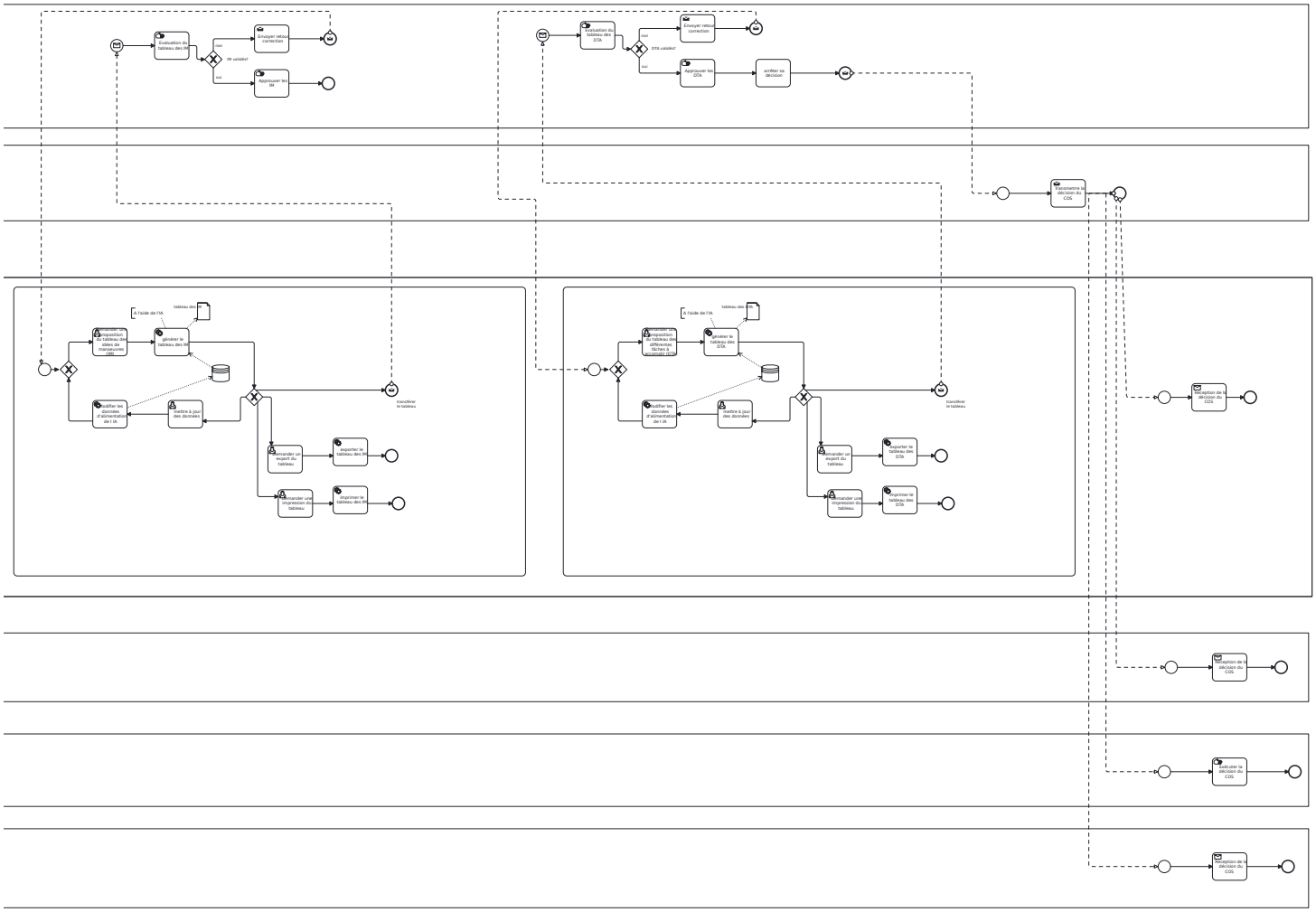


**Figure 30**: Complete technical BPMN model of the proposed anticipation system, Part 3 of 3. Some represented capabilities remain planned extensions rather than implemented proof-of-concept functions.

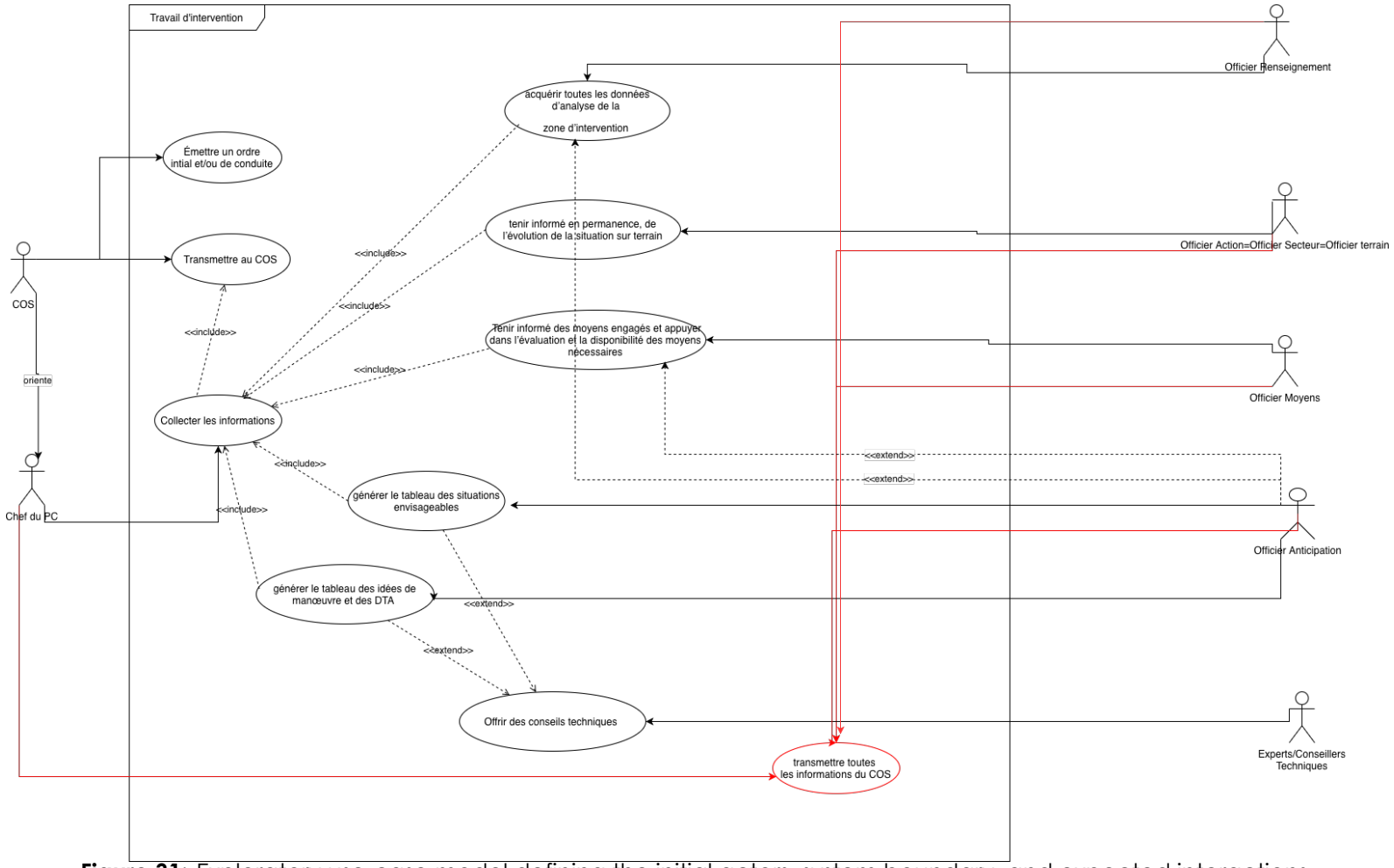



**Figure 31**: Exploratory use-case model defining the initial actors, system boundary, and expected interactions.

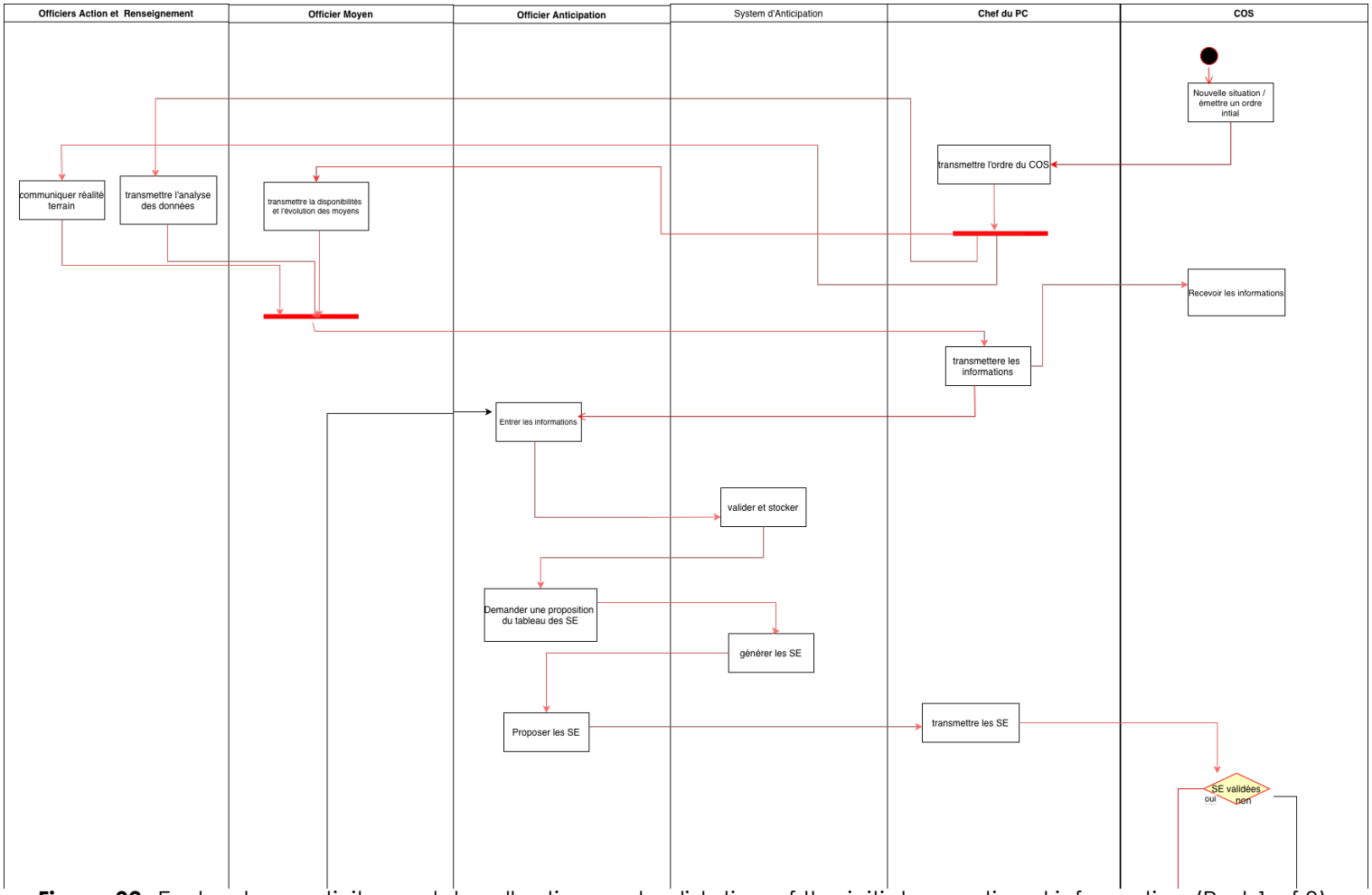



**Figure 32**: Exploratory activity model: collection and validation of the initial operational information (Part 1 of 3).

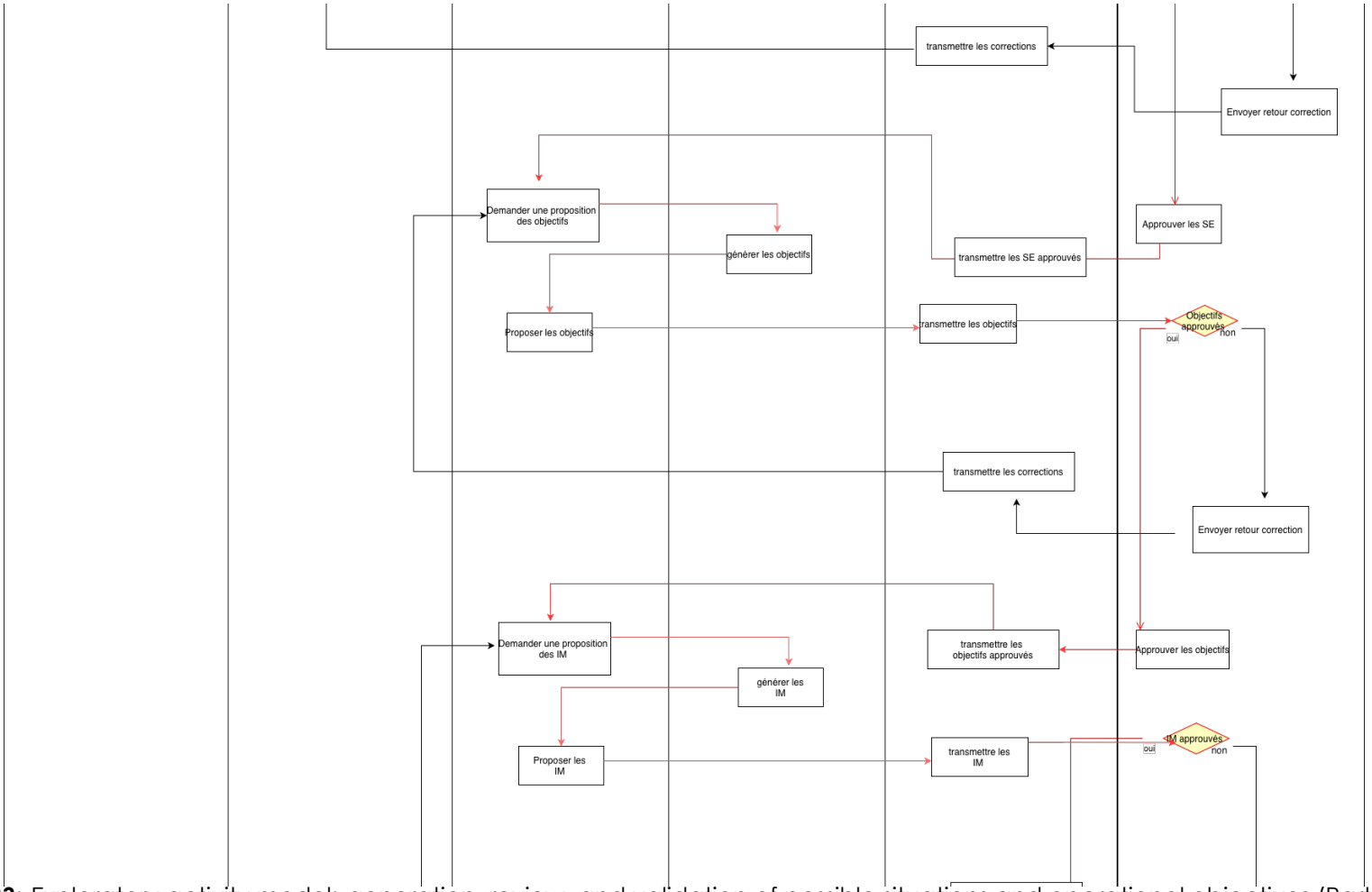



**Figure 33**: Exploratory activity model: generation, review, and validation of possible situations and operational objectives (Part 2 of 3).

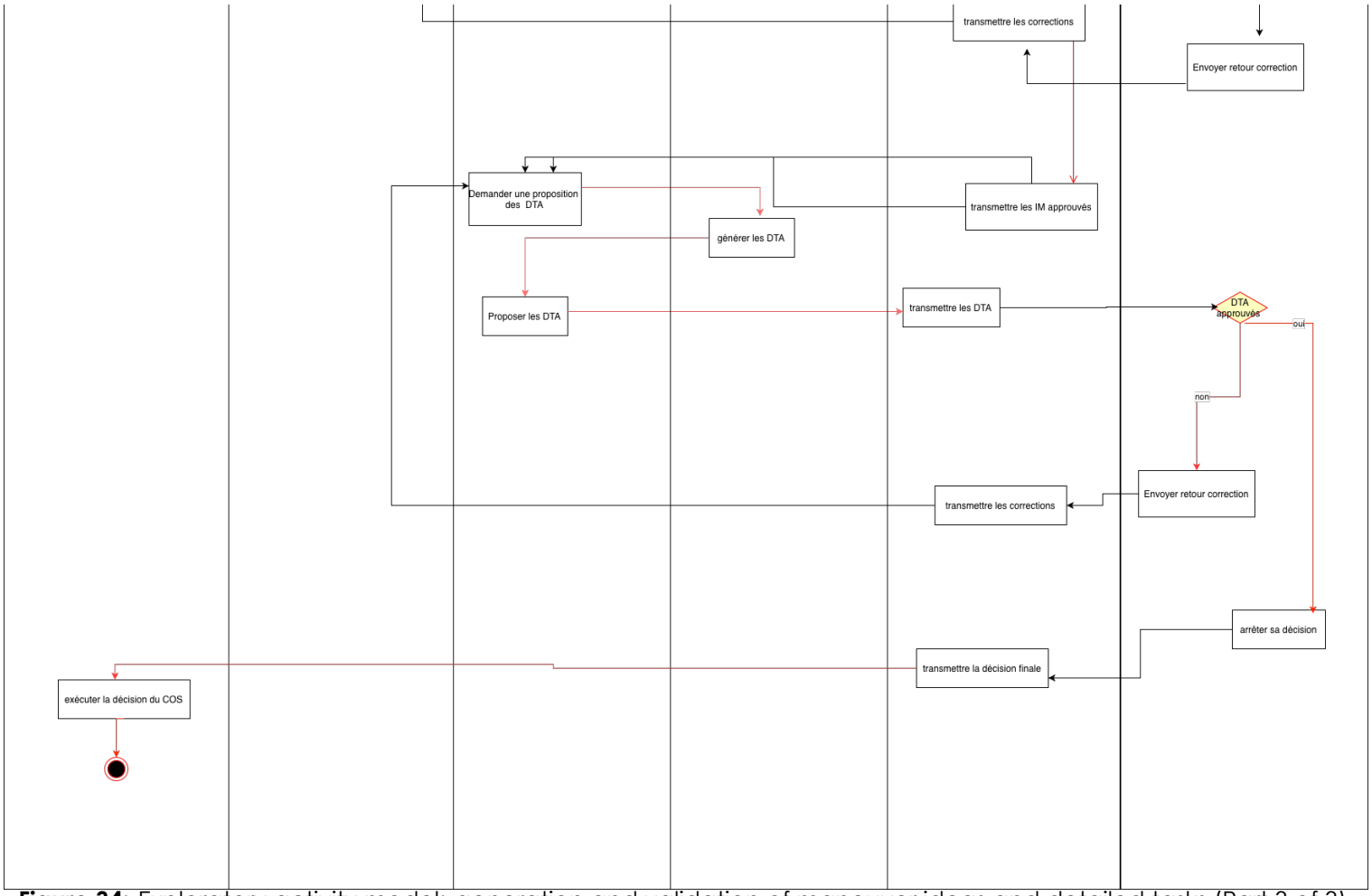



**Figure 34**: Exploratory activity model: generation and validation of maneuver ideas and detailed tasks (Part 3 of 3).

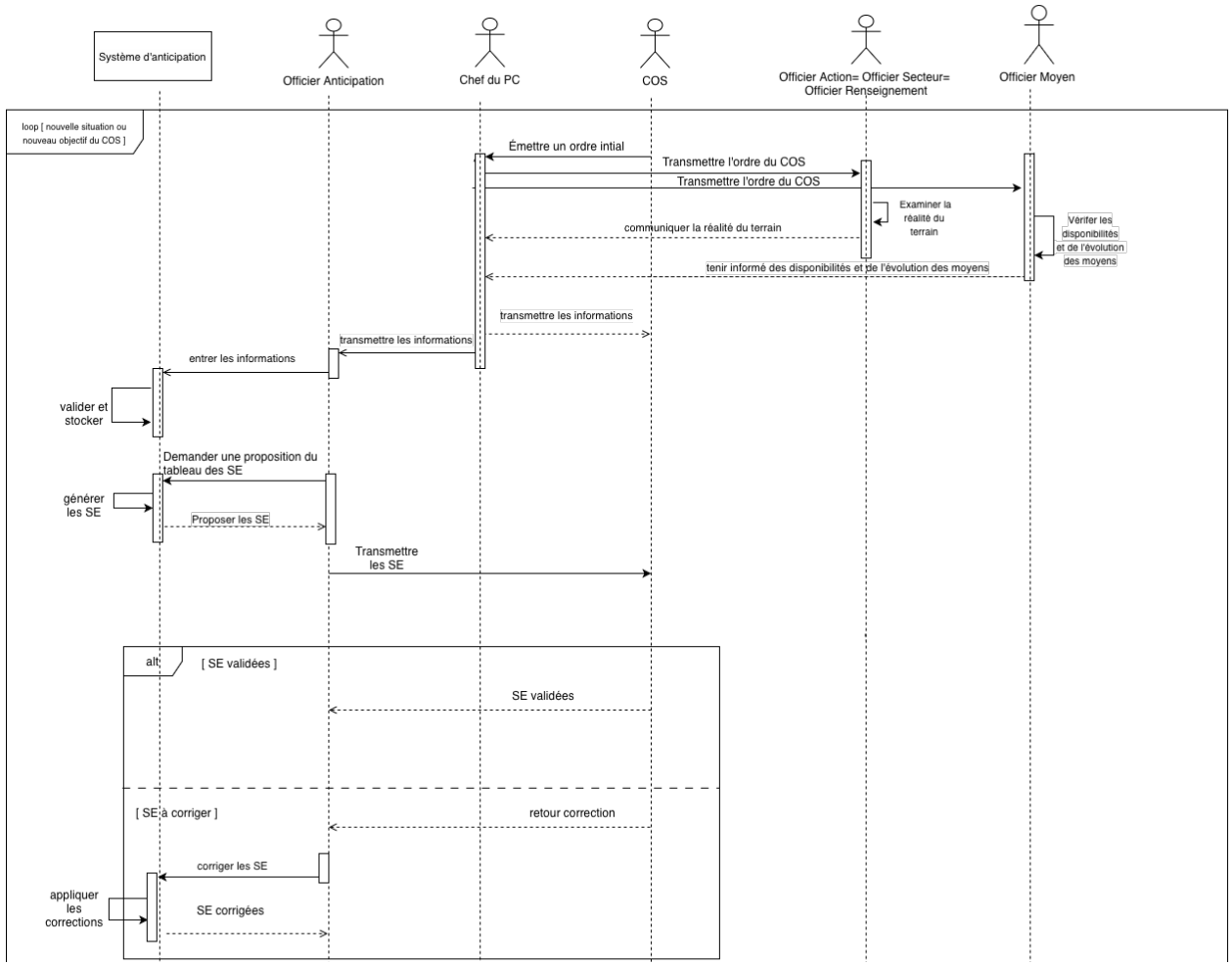



**Figure 35**: Exploratory sequence model: transmission of the initial operational information and generation of possible situations (Part 1 of 3).

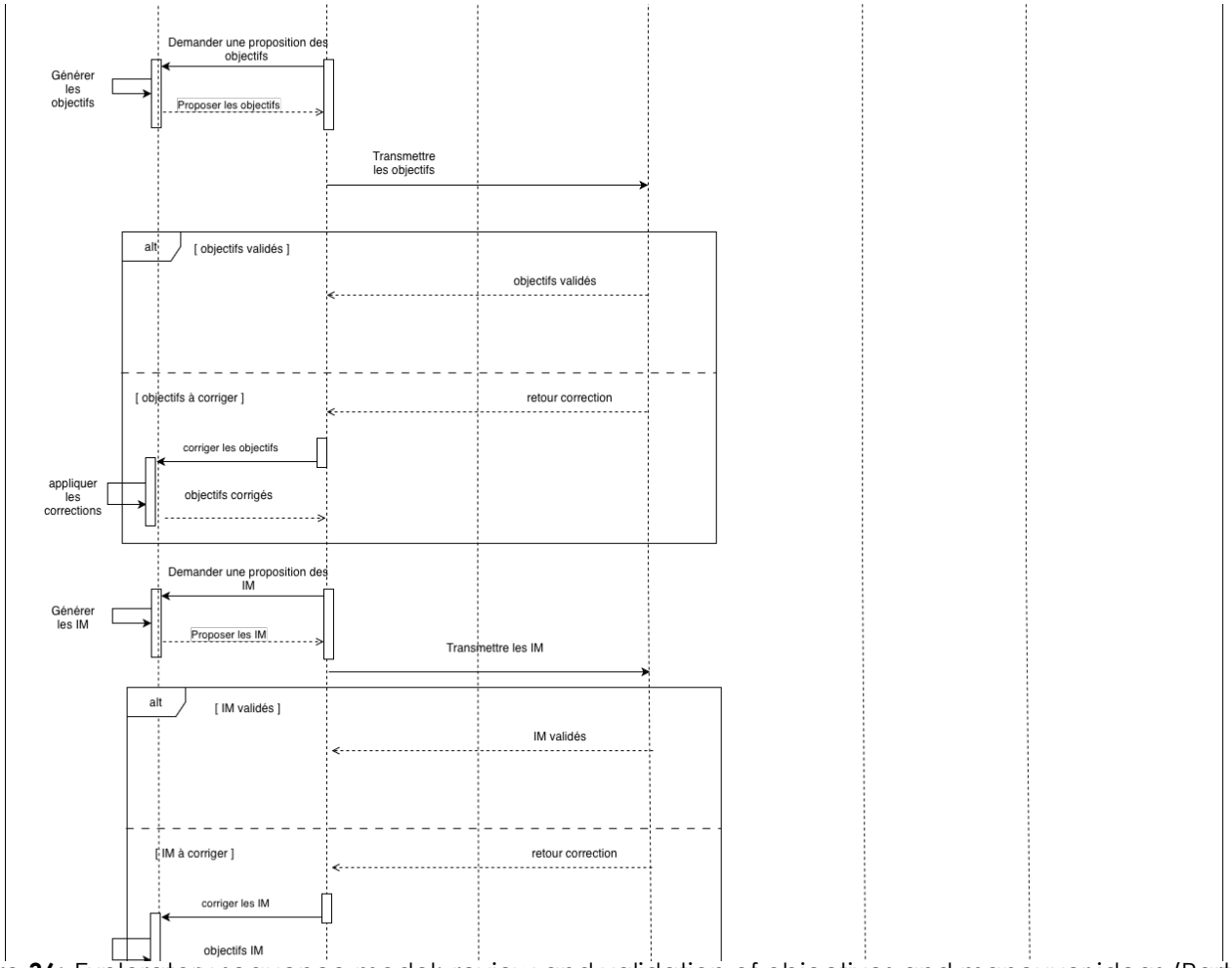



**Figure 36**: Exploratory sequence model: review and validation of objectives and maneuver ideas (Part 2 of 3).

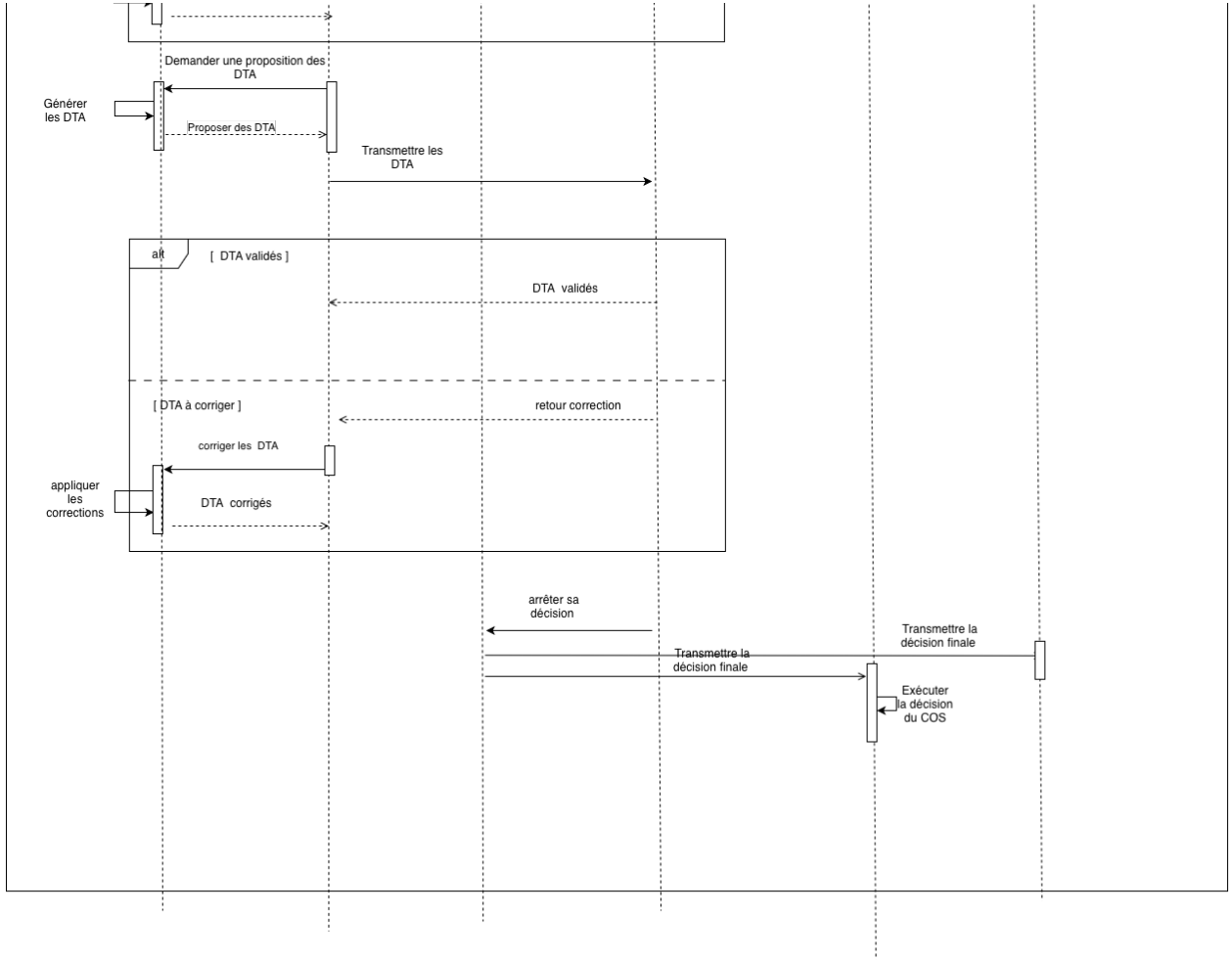



**Figure 37**: Exploratory sequence model: generation, correction, and validation of detailed tasks and transmission of the final decision (Part 3 of 3).